

CLASS NOTES

Review

GDP definition: the market value of all newly produced final goods and services within a nation during some period of time.

- **Market value** means that we measure goods and services at their market prices, so is, in principle, measure by doing this: $GDP = p_1q_1 + p_2q_2 + p_3q_3 \dots p_nq_n$ where p_n = market price of good n and q_n is the quantity of good n produced. This allows us to put all goods and services into a common unit (the dollar unit in the case of the US), and thus allows us to aggregate. However, it also means we have to make sure we do not confuse price changes for quantity changes when measuring GDP.
- **Newly produced**. This is self-evident. We do not count transactions involving goods produced in *previous* periods (for example, used cars). If were to do this would mean double counting, which we want to avoid.
- **Final good** vs. intermediate good. Only include final goods in GDP.
Intermediate good- a good used up in the production of another good during the same period that it itself was produced. We exclude these goods to avoid double counting. If flour, for example, is currently produced, but not sold (i.e., not used up in the production of bread), then it is considered a final good. If it is sold *and* used to make bread, then it is an intermediate good and therefore not counted in GDP.
- **Within a nation**. This implies we count exports but not imports in GDP.

Components of GDP.

$GDP = C + I + G + NX$. Roughly \$21.4 Trillion for US in 2019. Large drop in first half of 2020.

- Consumption (C) = durable consumer goods + non-durable consumer goods + services.
- Investment (I) = fixed investment (purchase of newly produced capital goods) + inventory investment (changes in the level of inventories). *Notice that this definition of investment does NOT include financial transactions.* Financial transactions more or less correspond to the transfer of ownership/claims to investment goods, not the production of investment goods. Hence, we use the word Investment in this class, we are NOT using the same way you hear it in the news or in finance classes.
- Government purchases of goods and services (G). This includes *all* levels of government. Does NOT include transfer payments (for example, does not include such things as food stamps). Transfer payments do not represent payments to individuals in exchange for goods and services currently produced.
- Net exports (NX) = exports minus imports

Problems with GDP as a measure of “true” production

- a. It excludes household production because it is difficult to measure this non-market production. But since household production represents current production we would like to count this in GDP. But, as a practical matter, this can be difficult to measure.
- b. Illegal production. Again, it is difficult to measure these types of non-market transactions. Some current production may not be reported (to avoid paying taxes) and some current production may be illegal (bootleg DVDs).
- c. Environmental goods such as clean air or clean water are difficult to measure because there is no formal market transaction involved. People value these goods just as they value cars, food, shelter, etc., so we would ideally like to include this in GDP.
- d. there are some other problems but the above are the main ones.

Accounting identities

Total production of newly produced goods $\equiv$ total spending on production $\equiv$ total income

To understand the logic of the first part of this identity, we assume that goods not sold to consumers are “purchased” by the firm from itself. For example, if 100 widgets are newly produced with 90 of these units sold to consumers, the other 10 “unsold” units are said to be purchased by the firm from itself (it is referred to as inventory investment and shows up in the “I” component of GDP). To understand the last part of this identity, note that every dollar of spending must generate a dollar worth of income for someone (even if the income is taken in the form of, say, an “unsold” widget).

Price Level review

The price level is a measure of the average money price of final goods and services. It is measured by a price index. Price index is a formula that tells us how to take the *weighted average*.

Use of price index

Inflation in year $t = 100\% * [\text{Price index}_t - \text{Price index}_{t-1}] / \text{Price index}_{t-1}$.

Year	Price Index	Inflation rate in year t
0	100	.
1	110	$(110-100)/100 = 0.10$ or 10%
2	121	$(121-110)/110 = 0.10$ or 10%
3	133	$(133-121)/121 = 0.10$ or 10% (approx.)

To convert a nominal variable (that is, a variable *not* adjusted for price level changes) into a real variable (variable that is adjusted for price level changes), you do the following:

$\text{Real variable}_t = [\text{Nominal variable}_t] / [\text{price index}_t/100]$

Year	nominal income	price index	real income
0	\$80,000	100	$80,000/[100/100] = 80,000$
1	\$88,000	110	$88,000/[110/100] = 80,000$

tables to be updated with handout in class 2023

In reading about GDP, notice that Real GDP = [Nominal GDP]/[price index/100].

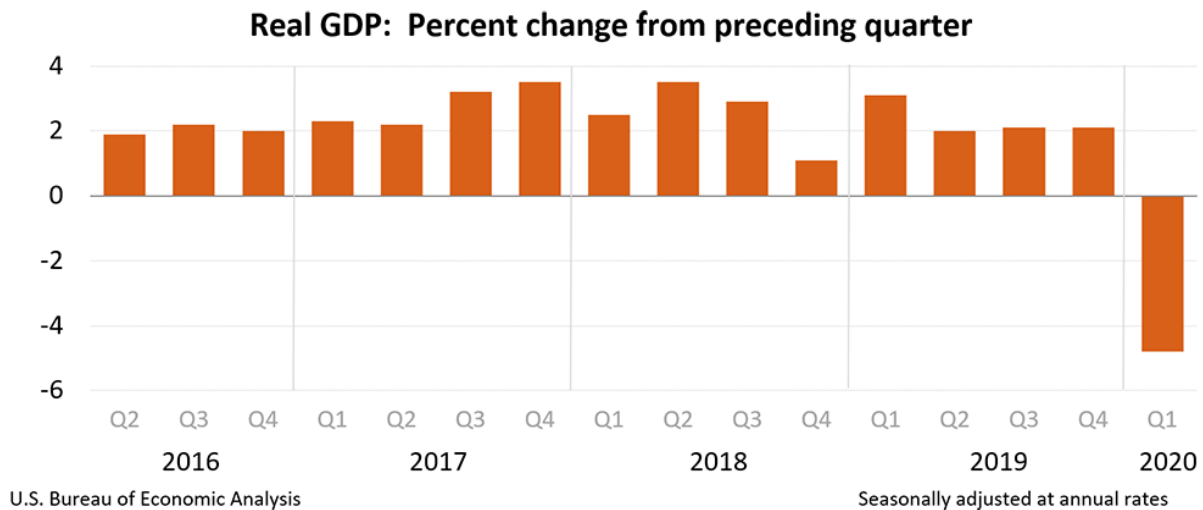
[Quarterly data at seasonally adjusted annual rates]												
Year or quarter	Gross domestic product	Personal consumption expenditures (C)			Gross private domestic investment (I)							
		Total	Goods	Services	Total	Fixed investment					Residential	Change in private inventories
						Total	Nonresidential					
							Total	Structures	Equipment	Intellectual property products		
		Billions of dollars										
2004.	12,213.7	8,212.7	2,902.0	5,310.6	2,281.3	2,217.2	1,467.4	307.7	721.9	437.8	749.8	64.1
2005.	13,036.6	8,747.1	3,082.9	5,664.2	2,534.7	2,477.2	1,621.0	353.0	794.9	473.1	856.2	57.5
2006.	13,814.6	9,260.3	3,239.7	6,020.7	2,701.0	2,632.0	1,793.8	425.2	862.3	506.3	838.2	69.0
2007.	14,451.9	9,706.4	3,367.0	6,339.4	2,673.0	2,639.1	1,948.6	510.3	893.4	544.8	690.5	34.0
2008.	14,712.8	9,976.3	3,363.2	6,613.1	2,477.6	2,506.9	1,990.9	571.1	845.4	574.4	516.0	-29.2
2009.	14,448.9	9,842.2	3,180.0	6,662.2	1,929.7	2,080.4	1,690.4	455.8	670.3	564.4	390.0	-150.8
2010.	14,992.1	10,185.8	3,317.8	6,868.0	2,165.5	2,111.6	1,735.0	379.8	777.0	578.2	376.6	53.9
2011.	15,542.6	10,641.1	3,518.1	7,123.0	2,332.6	2,286.3	1,907.5	404.5	881.3	621.7	378.8	46.3
2012.	16,197.0	11,006.8	3,637.7	7,369.1	2,621.8	2,550.5	2,118.5	479.4	983.4	655.7	432.0	71.2
2013.	16,784.9	11,317.2	3,730.0	7,587.2	2,826.0	2,721.5	2,211.5	492.5	1,027.0	691.9	510.0	104.5
2014.	17,527.3	11,822.8	3,863.0	7,959.8	3,044.2	2,960.2	2,400.1	577.6	1,091.9	730.5	560.2	84.0
2015.	18,224.8	12,284.3	3,920.3	8,363.9	3,223.1	3,091.2	2,457.4	572.6	1,121.5	763.3	633.8	131.9
2016.	18,715.0	12,748.5	3,995.9	8,752.6	3,178.7	3,151.6	2,453.1	545.8	1,093.6	813.8	698.5	27.1
2017.	19,519.4	13,312.1	4,165.0	9,147.0	3,370.7	3,340.5	2,584.7	586.8	1,143.7	854.2	755.7	30.2
2018.	20,580.2	13,998.7	4,364.8	9,633.9	3,628.3	3,573.6	2,786.9	633.2	1,222.6	931.1	786.7	54.7
2019 ^p .	21,429.0	14,563.9	4,508.6	10,055.2	3,742.8	3,676.1	2,878.7	625.8	1,240.9	1,012.0	797.4	66.8
		68%				17%						

TABLE B-3. Gross domestic product, 2004–2019— <i>Continued</i>								
[Quarterly data at seasonally adjusted annual rates]								
Year or quarter	Net exports of goods and services (NX)			Government consumption expenditures and gross investment (G)				
	Net exports	Exports	Imports	Total	Federal			State and local
					Total	National defense	Nondefense	
	Billions of dollars							
2004.	-619.1	1,177.6	1,796.7	2,338.9	891.7	569.9	321.9	1,447.1
2005.	-721.2	1,305.2	2,026.4	2,476.0	947.5	609.4	338.0	1,528.5
2006.	-770.9	1,472.6	2,243.5	2,624.2	1,000.7	640.8	359.9	1,623.5
2007.	-718.4	1,660.9	2,379.3	2,790.8	1,050.5	679.3	371.2	1,740.3
2008.	-723.1	1,837.1	2,560.1	2,982.0	1,150.6	750.3	400.2	1,831.4
2009.	-396.5	1,582.0	1,978.4	3,073.5	1,218.2	787.6	430.6	1,855.3
2010.	-513.9	1,846.3	2,360.2	3,154.6	1,297.9	828.0	469.9	1,856.7
2011.	-579.5	2,103.0	2,682.5	3,148.4	1,298.9	834.0	465.0	1,849.4
2012.	-568.6	2,191.3	2,759.9	3,137.0	1,286.5	814.2	472.4	1,850.5
2013.	-490.8	2,273.4	2,764.2	3,132.4	1,226.6	764.2	462.4	1,905.8
2014.	-507.7	2,371.7	2,879.4	3,168.0	1,215.0	743.4	471.6	1,953.0
2015.	-519.8	2,266.8	2,786.6	3,237.3	1,221.5	730.1	491.4	2,015.7
2016.	-518.8	2,220.6	2,739.4	3,306.7	1,234.1	728.4	505.7	2,072.6
2017.	-575.3	2,356.7	2,932.1	3,412.0	1,269.3	746.2	523.1	2,142.7
2018.	-638.2	2,510.3	3,148.5	3,591.5	1,347.3	793.6	553.7	2,244.2
2019 ^P .	-632.0	2,503.8	3,135.7	3,754.3	1,423.4	846.6	576.8	2,330.8
	-2.9%			17.5%				

Gross Domestic Product, 1st Quarter 2020

Real gross domestic product (GDP) decreased at an annual rate of 4.8 percent in the first quarter of 2020 (see chart), according to the "advance" estimate released by the Bureau of Economic Analysis (<https://www.bea.gov/>). In the fourth quarter of 2019, real GDP increased 2.1 percent.

The GDP estimate released today is based on source data that are incomplete or subject to further revision by the source agency (see "Source Data for the Advance Estimate" on page 2). The "second" estimate for the first quarter, based on more complete data, will be released on May 28, 2020.



Coronavirus (COVID-19) Impact on the Advance First-Quarter 2020 GDP Estimate

The decline in first quarter GDP was, in part, due to the response to the spread of COVID-19, as governments issued "stay-at-home" orders in March. This led to rapid changes in demand, as businesses and schools switched to remote work or canceled operations, and consumers canceled, restricted, or redirected their spending. The full economic effects of the COVID-19 pandemic cannot be quantified in the GDP estimate for the first quarter of 2020 because the impacts are generally embedded in source data and cannot be separately identified.

The decrease in **real GDP** in the first quarter reflected negative contributions from personal consumption expenditures, nonresidential fixed investment, exports, and private inventory investment that were partly offset by positive contributions from residential fixed investment, federal government spending, and state and local government spending. The decrease in Consumption reflected decreases in services, led by health care, and goods, led by motor vehicles and parts. The decrease in nonresidential fixed investment primarily reflected a decrease in equipment, led by transportation equipment. The decrease in exports primarily reflected a decrease in services, led by travel.

Current-dollar GDP (nominal GDP) decreased 3.5 percent, or \$191.2 billion, in the first quarter to a level of \$21.54 trillion. In the fourth quarter, GDP increased 3.5 percent, or \$186.6 billion

The **price index for gross domestic purchases** increased 1.6 percent in the first quarter, compared with an increase of 1.4 percent in the fourth quarter. **The PCE price index (consumer price index)** increased 1.3 percent, compared with an increase of 1.4 percent. Excluding food and energy prices, the PCE price index increased 1.8 percent, compared with an increase of 1.3 percent.

See <https://www.bea.gov/> for details and updates on the economy.

Present value and future value

Tools needed: PV, FV, Law of one price, S&D, opportunity costs, etc.

Future Value and Present Value

Basic terms and symbols V_0 , V_n , i , V . Symbols defined.

V_0 = Present Value (some dollar amount received, paid, deposited, etc. today)

V_1 = Future Value in one year (some dollar amount received, paid, deposited, etc. one year from today)

V_n = Future Value in n years (some dollar amount received, paid, deposited, etc. n years from today)

i = annual rate of interest (it is assumed that interest is compounded annually)

n = number of periods (we select years to be consistent with interest rate, but of course would could convert all of this to, say, monthly terms).

Future Value (FV) of a single current dollar amount.

a. Derivation of FV formula. Start with a very simple example: You deposit \$100 today at an interest rate of 10%. How much do you accumulate at the end of one year (assuming interest is compounded annually)?

Set up: principal (deposit) + interest income = principal + (principal)(i) = amount at end of the year

$\$100 + (\$100)(.10) = \$110$. In terms of our notation

$V_0 + (V_0)(i) = V_1 \rightarrow$ factor out V_0 to get $V_0(1+i) = V_1 \rightarrow \$100(1.1) = \$110$.

How much do you accumulate at end of two years?

Time: 0-----1-----2
 V_0 $V_0(1+i) = V_1$ $V_1(1+i) = V_2$

Since $V_1(1+i) = V_2$ then by substitution we can write $V_0(1+i)(1+i) = V_2$ which can be simplified to $V_0(1+i)^2 = V_2$

$\rightarrow 100(1.1)^2 = \121 .

We can keep repeating this story for 3 years, for 4 years, etc. We can generalize for n years and get our first important formula **$V_0(1+i)^n = V_n$** . This is the future value formula for a *one time* deposit today.

Very important: this formula has four terms in it--- V_0 , V_n , i , and n . So it can be used to solve for any problem with one these terms being the unknown and the other three as the "givens".

b. Examples.

Example 1: How much do you accumulate at end of 10 years at 6% if you deposit \$100 today?

Givens: **$V_0 = \$100$, $i = 6\%$, $n = 10$ years**. The unknown is V_{10} .

$V_0(1+i)^n = V_n \rightarrow \$100(1.06)^{10} = V_{10}$. We can use **table A1.1** to find the value of $(1+i)^n$. Find $i=6\%$ column and $n = 10$ years row. Should see 1.79. Plugging this in gives us $\$100(1.79) = \179 . [Footnote: A rule-of-thumb, the rule of 72. You can find how many years it takes to double your money by dividing the interest rate into 72. So $72/6 = 12$. This tells us it takes approximately 12 years to double your money. Check table A1.1 to confirm this.]

Example 2: How much do I have to deposit *today* to accumulate \$10,000 in 5 years at 4%?

Givens: **$V_5 = \$10,000$, $i = 4\%$, $n = 5$ years**. The unknown is V_0 . $V_0(1+i)^n = V_n \rightarrow V_0(1.04)^5 = \$10,000$.

Can use table A1.1 to find $(1.04)^5$. $V_0 = 10,000/1.22 = \$8,197$ (rounding).

Example 3: How many years will it take to accumulate \$10,000 if I deposit \$5000 today at 10%?

Givens: $V_0 = \$5000$, $i = 10\%$, $V_n = 10,000$. Unknown is n .

$V_0 (1+i)^n = V_n \rightarrow (\$5000)(1.10)^n = \$10,000$. Can use table A1.1. Important: in using the table to solve for either i or n we first need to isolate the term with these variables in it:

$(1+i)^n = (V_n/V_0) = (1.10)^n = \$10,000 / \$5000 = 2$. Now refer to table A1.1, find the $i = 10\%$ column and read down this column until find the number closest to 2. See what n value this corresponds to. That is your answer. The answer is roughly 7 years.

Example 4: If I deposit \$5000 today and 7 years later it accumulates to \$10,000, what is the annual return on this deposit? NOTE: In this case, the variable i in this formula is interpreted as the annual return on the "investment" (also called the internal rate of return, more on this concept in later lectures). Givens: $V_0 = \$5000$, $n = 7$ years, $V_7 = \$10,000$. Unknown is i . $V_0 (1+i)^n = V_n \rightarrow (\$5000)(1+i)^7 = \$10,000$. Can use table A1.1. However, as explained above, in using the table to solve for either i or n we first need to isolate the term with these variables in it:

$(1+i)^n = (V_n/V_0) = (1+i)^7 = \$10,000 / \$5000 = 2$. Now refer to table A1.1, find the $n=7$ row and read across this row until find the number closest to 2. See what i value this corresponds to. That is your answer. The answer is approx. 10%.

Table A1.1: $V_n = V_0(1+i)^n$. Future Value formula. Used to find future value of a single current dollar amount.

Future value interest factor of \$1 per period at i% for n periods, FVIF(i,n).																				
Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	1.010	1.020	1.030	1.040	1.050	1.060	1.070	1.080	1.090	1.100	1.110	1.120	1.130	1.140	1.150	1.160	1.170	1.180	1.190	1.200
2	1.020	1.040	1.061	1.082	1.103	1.124	1.145	1.166	1.188	1.210	1.232	1.254	1.277	1.300	1.323	1.346	1.369	1.392	1.416	1.440
3	1.030	1.061	1.093	1.125	1.158	1.191	1.225	1.260	1.295	1.331	1.368	1.405	1.443	1.482	1.521	1.561	1.602	1.643	1.685	1.728
4	1.041	1.082	1.126	1.170	1.216	1.262	1.311	1.360	1.412	1.464	1.518	1.574	1.630	1.689	1.749	1.811	1.874	1.939	2.005	2.074
5	1.051	1.104	1.159	1.217	1.276	1.338	1.403	1.469	1.539	1.611	1.685	1.762	1.842	1.925	2.011	2.100	2.192	2.288	2.386	2.488
6	1.062	1.126	1.194	1.265	1.340	1.419	1.501	1.587	1.677	1.772	1.870	1.974	2.082	2.195	2.313	2.436	2.565	2.700	2.840	2.986
7	1.072	1.149	1.230	1.316	1.407	1.504	1.606	1.714	1.828	1.949	2.076	2.211	2.353	2.502	2.660	2.826	3.001	3.185	3.379	3.583
8	1.083	1.172	1.267	1.369	1.477	1.594	1.718	1.851	1.993	2.144	2.305	2.476	2.658	2.853	3.059	3.278	3.511	3.759	4.021	4.300
9	1.094	1.195	1.305	1.423	1.551	1.689	1.838	1.999	2.172	2.358	2.558	2.773	3.004	3.252	3.518	3.803	4.108	4.435	4.785	5.160
10	1.105	1.219	1.344	1.480	1.629	1.791	1.967	2.159	2.367	2.594	2.839	3.106	3.395	3.707	4.046	4.411	4.807	5.234	5.695	6.192
11	1.116	1.243	1.384	1.539	1.710	1.898	2.105	2.332	2.580	2.853	3.152	3.479	3.836	4.226	4.652	5.117	5.624	6.176	6.777	7.430
12	1.127	1.268	1.426	1.601	1.796	2.012	2.252	2.518	2.813	3.138	3.498	3.896	4.335	4.818	5.350	5.936	6.580	7.288	8.064	8.916
13	1.138	1.294	1.469	1.665	1.886	2.133	2.410	2.720	3.066	3.452	3.883	4.363	4.898	5.492	6.153	6.886	7.699	8.599	9.596	10.699
14	1.149	1.319	1.513	1.732	1.980	2.261	2.579	2.937	3.342	3.797	4.310	4.887	5.535	6.261	7.076	7.988	9.007	10.147	11.420	12.839
15	1.161	1.346	1.558	1.801	2.079	2.397	2.759	3.172	3.642	4.177	4.785	5.474	6.254	7.138	8.137	9.266	10.539	11.974	13.590	15.407
16	1.173	1.373	1.605	1.873	2.183	2.540	2.952	3.426	3.970	4.595	5.311	6.130	7.067	8.137	9.358	10.748	12.330	14.129	16.172	18.488
17	1.184	1.400	1.653	1.948	2.292	2.693	3.159	3.700	4.328	5.054	5.895	6.866	7.986	9.276	10.761	12.468	14.426	16.672	19.244	22.186
18	1.196	1.428	1.702	2.026	2.407	2.854	3.380	3.996	4.717	5.560	6.544	7.690	9.024	10.575	12.375	14.463	16.879	19.673	22.901	26.623
19	1.208	1.457	1.754	2.107	2.527	3.026	3.617	4.316	5.142	6.116	7.263	8.613	10.197	12.056	14.232	16.777	19.748	23.214	27.252	31.948
20	1.220	1.486	1.806	2.191	2.653	3.207	3.870	4.661	5.604	6.727	8.062	9.646	11.523	13.743	16.367	19.461	23.106	27.393	32.429	38.338
25	1.282	1.641	2.094	2.666	3.386	4.292	5.427	6.848	8.623	10.835	13.585	17.000	21.231	26.462	32.919	40.874	50.658	62.669	77.388	95.396
30	1.348	1.811	2.427	3.243	4.322	5.743	7.612	10.063	13.268	17.449	22.892	29.960	39.116	50.950	66.212	85.850	111.065	143.371	184.675	237.376
35	1.417	2.000	2.814	3.946	5.516	7.686	10.677	14.785	20.414	28.102	38.575	52.800	72.069	98.100	133.176	180.314	243.503	327.997	440.701	590.668
40	1.489	2.208	3.262	4.801	7.040	10.286	14.974	21.725	31.409	45.259	65.001	93.051	132.782	188.884	267.864	378.721	533.869	750.378	1,051.668	1,469.772
50	1.645	2.692	4.384	7.107	11.467	18.420	29.457	46.902	74.358	117.391	184.565	289.002	450.736	700.233	1,083.657	1,670.704	2,566.215	3,927.357	5,988.914	9,100.438

Future Accumulate Value (FAV) of an equal annual series of dollar amounts (first in series one year from now).

FAV of a series of equal annual dollar amounts, first in series is one year from now. Future Accumulated Value (FAV) of a series of equal annual dollar amounts. Long way to solve. How to derive a "short cut". Very useful for retirement/pension problems.

Derivation of formula. Suppose we deposit V each year for 5 years (first deposit one year from today). How much will this accumulate to at end of 5 years at interest i ?

$$V(1+i)^4 + V(1+i)^3 + V(1+i)^2 + V(1+i) + V = \text{Future Accumulated Value (FAV)}.$$

Factoring out V : $V[(1+i)^4 + (1+i)^3 + (1+i)^2 + (1+i) + 1] = \text{Future Accumulated Value (FAV)}$.

Generalizing to n years of deposits (first one year hence): $\text{FAV} = V[(1+i)^{n-1} + (1+i)^{n-2} + \dots + (1+i)^2 + (1+i) + 1]$

The term in brackets collapses to: **$\text{FAV} = V \{ [(1+i)^n - 1] / i \}$** with the value in the braces available from a table.

Example 1: If we make equal annual deposits of \$10,000 per year (first deposit one year hence), then how much will we accumulate at the end of 20 years at 6%?

Givens: **$V = \$10,000$, $i = 6\%$, $n = 20$ years.** The unknown is FAV.

$\text{FAV} = V \{ [(1+i)^n - 1] / i \} \rightarrow \text{FAV} = 10,000 \{ [(1.06)^{20} - 1] / 0.06 \}$ with the term in braces found in table A1.2.

$$\text{FAV} = 10,000 \{ 36.79 \} = \$367,900.$$

Example 2: How many years will it take to accumulate \$120,000 at 6% with make equal annual deposits of \$10,000 per year (first deposit one year hence)?

Givens: **$V = \$10,000$, $i = 6\%$, $\text{FAV} = \$120,000$.** The unknown is n .

$$\text{FAV} = V \{ [(1+i)^n - 1] / i \} \rightarrow 120,000 = 10,000 \{ [(1.06)^n - 1] / 0.06 \}$$

Can use table A1.2. However, just as explained in the previous examples, in using the table to solve for either i or n we first need to isolate the term with these variables in it:

$$\text{FAV}/V = \{ [(1+i)^n - 1] / i \} \rightarrow 120,000/10,000 = \{ [(1.06)^n - 1] / 0.06 \} \rightarrow 12 = \{ [(1.06)^n - 1] / 0.06 \}$$

Now refer to table A1.2, find the $i = 6\%$ column and read down this column until find the number closest to 12. See what n value this corresponds to. That is your answer. Between 9 and 10 years.

Table A1.2 $FAV = V \{ [(1+i)^n - 1] / i \}$ Future accumulated value for a series of equal e).

Future value interest factor of an ordinary annuity of \$1 per period at i% for n periods, FVIFA(i,n).																				
Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2	2.010	2.020	2.030	2.040	2.050	2.060	2.070	2.080	2.090	2.100	2.110	2.120	2.130	2.140	2.150	2.160	2.170	2.180	2.190	2.200
3	3.030	3.060	3.091	3.122	3.153	3.184	3.215	3.246	3.278	3.310	3.342	3.374	3.407	3.440	3.473	3.506	3.539	3.572	3.606	3.640
4	4.060	4.122	4.184	4.246	4.310	4.375	4.440	4.506	4.573	4.641	4.710	4.779	4.850	4.921	4.993	5.066	5.141	5.215	5.291	5.368
5	5.101	5.204	5.309	5.416	5.526	5.637	5.751	5.867	5.985	6.105	6.228	6.353	6.480	6.610	6.742	6.877	7.014	7.154	7.297	7.442
6	6.152	6.308	6.468	6.633	6.802	6.975	7.153	7.336	7.523	7.716	7.913	8.115	8.323	8.536	8.754	8.977	9.207	9.442	9.683	9.930
7	7.214	7.434	7.662	7.898	8.142	8.394	8.654	8.923	9.200	9.487	9.783	10.089	10.405	10.730	11.067	11.414	11.772	12.142	12.523	12.916
8	8.286	8.583	8.892	9.214	9.549	9.897	10.260	10.637	11.028	11.436	11.859	12.300	12.757	13.233	13.727	14.240	14.773	15.327	15.902	16.499
9	9.369	9.755	10.159	10.583	11.027	11.491	11.978	12.488	13.021	13.579	14.164	14.776	15.416	16.085	16.786	17.519	18.285	19.086	19.923	20.799
10	10.462	10.950	11.464	12.006	12.578	13.181	13.816	14.487	15.193	15.937	16.722	17.549	18.420	19.337	20.304	21.321	22.393	23.521	24.709	25.959
11	11.567	12.169	12.808	13.486	14.207	14.972	15.784	16.645	17.560	18.531	19.561	20.655	21.814	23.045	24.349	25.733	27.200	28.755	30.404	32.150
12	12.683	13.412	14.192	15.026	15.917	16.870	17.888	18.977	20.141	21.384	22.713	24.133	25.650	27.271	29.002	30.850	32.824	34.931	37.180	39.581
13	13.809	14.680	15.618	16.627	17.713	18.882	20.141	21.495	22.953	24.523	26.212	28.029	29.985	32.089	34.352	36.786	39.404	42.219	45.244	48.497
14	14.947	15.974	17.086	18.292	19.599	21.015	22.550	24.215	26.019	27.975	30.095	32.393	34.883	37.581	40.505	43.672	47.103	50.818	54.841	59.196
15	16.097	17.293	18.599	20.024	21.579	23.276	25.129	27.152	29.361	31.772	34.405	37.280	40.417	43.842	47.580	51.660	56.110	60.965	66.261	72.035
16	17.258	18.639	20.157	21.825	23.657	25.673	27.888	30.324	33.003	35.950	39.190	42.753	46.672	50.980	55.717	60.925	66.649	72.939	79.850	87.442
17	18.430	20.012	21.762	23.698	25.840	28.213	30.840	33.750	36.974	40.545	44.501	48.884	53.739	59.118	65.075	71.673	78.979	87.068	96.022	105.93
18	19.615	21.412	23.414	25.645	28.132	30.906	33.999	37.450	41.301	45.599	50.396	55.750	61.725	68.394	75.836	84.141	93.406	103.74	115.27	128.12
19	20.811	22.841	25.117	27.671	30.539	33.760	37.379	41.446	46.018	51.159	56.939	63.440	70.749	78.969	88.212	98.603	110.28	123.41	138.17	154.74
20	22.019	24.297	26.870	29.778	33.066	36.786	40.995	45.762	51.160	57.275	64.203	72.052	80.947	91.025	102.44	115.38	130.03	146.63	165.42	186.69
25	28.243	32.030	36.459	41.646	47.727	54.865	63.249	73.106	84.701	98.347	114.41	133.33	155.62	181.87	212.79	249.21	292.10	342.60	402.04	471.98
30	34.785	40.568	47.575	56.085	66.439	79.058	94.461	113.28	136.31	164.49	199.02	241.33	293.20	356.79	434.75	530.31	647.44	790.95	966.71	1,181.9
35	41.660	49.994	60.462	73.652	90.320	111.43	138.24	172.32	215.71	271.02	341.59	431.66	546.68	693.57	881.17	1,120.7	1,426.5	1,816.7	2,314.2	2,948.3
40	48.886	60.402	75.401	95.026	120.80	154.76	199.64	259.06	337.88	442.59	581.83	767.09	1,013.7	1,342.0	1,779.1	2,360.8	3,134.5	4,163.2	5,529.8	7,343.9
50	64.463	84.579	112.80	152.67	209.35	290.34	406.53	573.77	815.08	1,163.9	1,668.8	2,400.0	3,459.5	4,994.5	7,217.7	10,436	15,090	21,813	31,515	45,497

PV of single future dollar amount

a. Derivation of formula. This simply involves a rearrangement of $V_0(1+i)^n = V_n$. So we now get the famous present value formula: $V_0 = V_n / (1+i)^n$. Aside from a purely mechanical, plug-in-the-numbers use of this formula, what exactly does concept of present value mean?

Point number 1: In Economics and Finance, we frequently need to make comparisons of dollars received at different points in time. But just as apples and oranges are not the same goods and therefore cannot be directly added together, neither are dollars received *at different points in time* the same good. For example, assuming a positive interest rate, someone would (presumably) not be indifferent receiving \$100 now versus receiving \$100 ten years from now (obviously, if you picked the \$100 now you could always save it and ten year later get more \$100, hence the logic why someone would not be indifferent between these choices). Both receipts are for \$100 but the fact that someone is rationally not indifferent between these options implies that are not the same thing. If we are to compare dollars at different points in time to make rational comparisons, we need to put everything into the same unit. That common unit is the present value unit (it is true we could use some future value date as a common unit but that is typically not done and is not as intuitive). For example, if I ask you to find the PV of \$100 received one year from now at 10%, we would use $V_0 = V_n / (1+i)^n$ to answer this question.

$V_0 = 100 / (1.1) = \$90.91$ (rounding). But how are we interpret this number? It could be said that receiving \$90.91 today is equivalent of receiving \$100 one year later at 10% since \$90.91 could always be deposited today at 10% and accumulate to \$100 a year later. Thus, I am indifferent between receiving \$90.91 today versus \$100 one year later if the interest rate is 10% (I am ignoring uncertainty at this point in the discussion, so if you choose \$100 a year later you are guaranteed to receive it). Although this seems clear enough, how would you respond to this comment: "My preferences are such that I only care about today, so I would not be indifferent between these two options—I would prefer \$90.91 option since I get it NOW. I don't want to wait." The response this comment is addressed in point number two below.

Table A1.3. $V_0 = V_n / (1+i)^n$. Present Value of a one-time future dollar amount.

Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065
16	0.853	0.728	0.623	0.534	0.458	0.394	0.339	0.292	0.252	0.218	0.188	0.163	0.141	0.123	0.107	0.093	0.081	0.071	0.062	0.054
17	0.844	0.714	0.605	0.513	0.436	0.371	0.317	0.270	0.231	0.198	0.170	0.146	0.125	0.108	0.093	0.080	0.069	0.060	0.052	0.045
18	0.836	0.700	0.587	0.494	0.416	0.350	0.296	0.250	0.212	0.180	0.153	0.130	0.111	0.095	0.081	0.069	0.059	0.051	0.044	0.038
19	0.828	0.686	0.570	0.475	0.396	0.331	0.277	0.232	0.194	0.164	0.138	0.116	0.098	0.083	0.070	0.060	0.051	0.043	0.037	0.031
20	0.820	0.673	0.554	0.456	0.377	0.312	0.258	0.215	0.178	0.149	0.124	0.104	0.087	0.073	0.061	0.051	0.043	0.037	0.031	0.026
25	0.780	0.610	0.478	0.375	0.295	0.233	0.184	0.146	0.116	0.092	0.074	0.059	0.047	0.038	0.030	0.024	0.020	0.016	0.013	0.010
30	0.742	0.552	0.412	0.308	0.231	0.174	0.131	0.099	0.075	0.057	0.044	0.033	0.026	0.020	0.015	0.012	0.009	0.007	0.005	0.004
35	0.706	0.500	0.355	0.253	0.181	0.130	0.094	0.068	0.049	0.036	0.026	0.019	0.014	0.010	0.008	0.006	0.004	0.003	0.002	0.002
40	0.672	0.453	0.307	0.208	0.142	0.097	0.067	0.046	0.032	0.022	0.015	0.011	0.008	0.005	0.004	0.003	0.002	0.001	0.001	0.001
50	0.608	0.372	0.228	0.141	0.087	0.054	0.034	0.021	0.013	0.009	0.005	0.003	0.002	0.001	0.001	0.001	0.000	0.000	0.000	0.000

Point number 2: This brings us to something known as the **Separation Theorem**. Assume the same options above *AND* assume the interest of 10% is the rate that borrowing and lending can take place at. Then we can say that REGARDLESS of your time preferences, you would still be indifferent between the two options at 10%! To see why: Suppose I choose the first option of \$90.91 today. What does this allow me to do in terms of consumption spending patterns? If I so wish, I can go out and spend this amount now, or I can save it all, or do some combination of these. If I save it all, get this one year later $\$90.91(1.1) = \100 . Thus, I have \$100 to spend in the future just as I would have if I had selected option two. Suppose I instead choose the second option of receiving \$100 one year later. What does this allow me to do? If I so wish, I can go out and spend this amount one year later, or I can borrow against all of it, or do some combination of these. How much could I borrow today? Let B equal the most you can borrow against \$100 at 10%: $B = \$100/1.1 = \90.91 . With \$90.91 cash in hand today, I could go out and spend that amount today (just like option one!). One year later I would then run up a debt of $\$90.91(1.1) = \text{Debt} = \100 which I can pay off my income of \$100. We have exactly the same set of spending (consumption) choices regardless of which option I choose! This is a very important principle. It means we only need to worry about PV comparisons and do not need to know individual preferences as to whether the person “lives for today” or is “very patient, only cares about future”.

This story can be translated to budget lines, with the line given by $C_0 + C_1/(1+i) = E_0 + E_1/(1+i)$, where C_0 = consumption spending today, C_1 = consumption spending one year from today, E_0 = your endowment of income today (this would be \$90.91 if select option 1 but zero if select option two) and E_1 = endowment of income one year later (this would be zero if select option 1 but \$100 if select option two). The left-hand side is the PV of your consumption spending and the right-hand side is the PV of your income stream. It therefore has a simple interpretation—the PV of consumption spending is equal to the PV of your income stream.

With option one : $C_0 + C_1/(1+i) = 90.91 + 0 = \90.91

With option two: $C_0 + C_1/(1+i) = 0 + 100/(1.1) = \90.91 . So the PV of the income stream is the same. Any combination of C_0 and C_1 which has a PV of \$90.91 is achievable under either option with some sort of borrowing or lending behavior. One more example: Suppose I desire a spending pattern of \$40.91 today and \$55 one year later. How can I achieve this if I choose option one? Spend \$40.91 out my \$90.91, leaving me with \$50 to save. This will accumulate to \$55 at 10%. How about option two? Borrow \$40.91 today, spend it, and then out of \$100 pay off my debt of $\$40.91(1.1) = \45 one year later. After paying off my debt, I am left with $\$100 - \$45 = \$55$.

Example of PV formula: Suppose a consumer is willing to pay up to \$10 for a lightbulb that lasts one year. They are indifferent between buying two consecutive one year lightbulbs (purchase one today, purchase another one a year later) or buying today a two-year lightbulb. Either way, the consumer has the services of a lightbulb over a two year period. Suppose it cost the seller \$5 to make either kind of lightbulb. First question is this: At 10%, what is the maximum price a seller charge for a two year lightbulb today? Second question: What is the most profitable strategy for a lightbulb seller? Answer 1: A two year lightbulb can be sold for $10 + 10/1.1 = \$19.09$ today.

Answer 2: Profit from sale of two-year lightbulb today = $19.09 - 5 = \$14.09$. Profit in present value terms from sale of two consecutive one-year lightbulbs = $(10 + 10/1.1) - (5 + 5/1.1) = 19.09 - 9.55 = \9.54 .

PV of an annuity

By far, the most interesting problems in economics and in finance involve the present value of a series of future dollar amounts (stocks, bonds, real estate, firms' investments, etc.)

a. If we want to find the present value of a series of future dollar amounts, the "long way" to set it up is: $V_0 = V_1/(1+i) + V_2/(1+i)^2 + V_3/(1+i)^3 + \dots + V_n/(1+i)^n$. If the annual dollar amounts differ, then there are no short cuts so this could be a tedious calculation without a calculator. However, suppose we introduce a simplification. Specifically, suppose we assume equal annual payments so that $V_1 = V_2 = V_3 = \dots = V_n = V$ where V is called the annuity amount (i.e., the equal annual payment). Then by substitution: $V_0 = V/(1+i) + V/(1+i)^2 + V/(1+i)^3 + \dots + V/(1+i)^n$.

Factor out V to now get: $V_0 = V [1/(1+i) + 1/(1+i)^2 + 1/(1+i)^3 + \dots + 1/(1+i)^n]$. The term in brackets collapses into this: $V_0 = V [(1+i)^n - 1] / [(i)(1+i)^n]$ for which we have a table, Table A1.4. This is the PV of an annuity formula (subject to the first payment in the series being one year from today).

Example 1: How much would you be willing to pay *today* for the right to receive (ignore uncertainty) \$100 per year for five years (first in the series one year from today) at 10%? NOTE: This simple question is at the very core of the discussion on what determines any asset price!! There are additional factors of course that can complicate the problem, but this is the first step in understanding asset pricing.

Givens: **$V = \$100$ per year, $i = 10\%$, $n = 5$ years.** The unknown is V_0 , which in this case can be interpreted as the asset price today.

$V_0 = V [(1+i)^n - 1] / [(i)(1+i)^n] \rightarrow V_0 = (100) [(1.1)^5 - 1] / [(0.10)(1.1)^5]$ with the term in brackets found in table A1.4. (3.79 in this case).

$V_0 = (100) [3.79] = \$379$ (rounding). Based on our discussion above on the Separation Theorem, in what sense is receiving \$379 today (option A) the equivalent (and therefore a person is indifferent between we argue) of receiving a stream of payments of \$100 per year for five years (first in series one year later) at 10%?

To understand this, suppose you choose you choose option A (take \$379 right now). You would then be able to either spend it now or deposit this amount in the bank today and exactly withdraw \$100 per year for 5 years—same as if you had selected option B! (See table below to confirm this)

	1	2	3	4	5
beginning balance	379.0	316.9	248.6	173.4	90.8
interest over the year (10% annual rate)	37.9	31.7	24.9	17.3	9.1
end of year balance before withdrawal	416.9	348.6	273.4	190.8	99.9
net of withdrawal of \$100	316.9	248.59	173.449	90.794	0

If you had selected option B, then you could borrow at most up to \$379 today, go out and spend it on consumption if you so choose, and have just enough to pay off your debt in 5 years by making annual payments of \$100. The table above would then be interpreted as:

	1	2	3	4	5
beginning debt	379.0	316.9	248.6	173.4	90.8
interest on debt (10% annual rate)	37.9	31.7	24.9	17.3	9.1
end of year debt owe bank before payment	416.9	348.6	273.4	190.8	99.9
debt you owe the bank after making payment of 100	316.9	248.59	173.449	90.794	0

In other words, options A and B provide you with the exact same consumption choices!

Example two: Suppose you take out a loan today of \$100,000 and desire to repay it in 12 equal annual payments with an interest rate of 6% charged on the loan.

Givens: **$V_0 = \$100,000$, $i = 6\%$, $n = 12$ years.** The unknown is V .

$V_0 = V [(1+i)^n - 1] / [(i)(1+i)^n] \rightarrow \$100,000 = (V) [(1.06)^{12} - 1] / [(0.06)(1.06)^{12}]$ with the term in brackets found in table A1.4 (8.38 roughly in this case).

$V = \$100,000 / [8.38] = \$11,933$ (approximately).

A variation on this question. Supposed you are charged an upfront fee for this loan (“points”, “finance fee”; NOTE: A point is a fee equal to 1% of the loan amount. A 30-year, \$600,000 mortgage might have a rate of 4% but come with a charge of 1 point, or \$6,000). So the annual loan payment is calculated based on the \$100,000 loan but, say, you have to pay \$10,000 immediately out this loan to the bank in the form of a finance fee. What rate are you really paying on the part of the loan you actually get to use (the annual effective rate)?

$V_0 = V [(1+i)^n - 1] / [(i)(1+i)^n] \rightarrow [\$100,000 - 10,000] = \$90,000 = (11,933) [(1+i^*)^{12} - 1] / [(i^*)(1+i^*)^{12}]$

Can use table A1.4. However, as explained in the previous examples, in using the table to solve for either i or n we first need to isolate the term with these variables in it:

$V_0 / V = [(1+i)^n - 1] / [(i)(1+i)^n] \rightarrow (90,000/11,933) = 7.54 = [(1+i^*)^{12} - 1] / [(i^*)(1+i^*)^{12}]$

Now refer to table A1.4, find the $n = 12$ row and read down this column until you find the number closest to 7.54. See what i value this corresponds to. That is your answer. Answer is approx. 8%.

Example three: Your firm has purchased a machine today for \$24,000. The machine generates a net benefit stream of \$3,000 per year for 17 years. It then falls apart and has no scrap value. That is the rate of return on this investment in the machine (the “internal rate of return” as it is usually called)?

Givens: **$V_0 = \$24,000$, $V = \$3,000$, and $n = 17$ years.** The unknown is i which is interpreted as the rate of return on this investment in this case. $V_0 = V [(1+i)^n - 1] / [(i)(1+i)^n] \rightarrow \$24,000 = (\$3,000) [(1+i)^{17} - 1] / [(i)(1+i)^{17}]$

Can use table A1.4. Once again we have a problem where the unknown is either i or n and so once again we proceed by first isolating the term with these variables in it:

$V_0 / V = [(1+i)^n - 1] / [(i)(1+i)^n] \rightarrow 24,000/3,000 = 8 = [(1+i^*)^{17} - 1] / [(i^*)(1+i^*)^{17}]$

Now refer to table A1.4, find the $n = 17$ row and read down this column until you find the number closest to 8. See what i value this corresponds to. That is your answer. Answer is roughly $i^* = 10\%$.

Notice this would be a good investment if the best you could get on your \$24,000 in an ALTERNATIVE investment of similar risk is some rate less than 10%.

Table A1.4. $V_0 = V [(1+i)^n - 1] / [(i)(1+i)^n]$ Present value of an equal annual series of dollar amounts (first in the series one year from now).

Present value interest factor of an (ordinary) annuity of \$1 per period at i% for n periods, PVIFA(i,n).																				
Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675
16	14.718	13.578	12.561	11.652	10.838	10.106	9.447	8.851	8.313	7.824	7.379	6.974	6.604	6.265	5.954	5.668	5.405	5.162	4.938	4.730
17	15.562	14.292	13.166	12.166	11.274	10.477	9.763	9.122	8.544	8.022	7.549	7.120	6.729	6.373	6.047	5.749	5.475	5.222	4.990	4.775
18	16.398	14.992	13.754	12.659	11.690	10.828	10.059	9.372	8.756	8.201	7.702	7.250	6.840	6.467	6.128	5.818	5.534	5.273	5.033	4.812
19	17.226	15.678	14.324	13.134	12.085	11.158	10.336	9.604	8.950	8.365	7.839	7.366	6.938	6.550	6.198	5.877	5.584	5.316	5.070	4.843
20	18.046	16.351	14.877	13.590	12.462	11.470	10.594	9.818	9.129	8.514	7.963	7.469	7.025	6.623	6.259	5.929	5.628	5.353	5.101	4.870
25	22.023	19.523	17.413	15.622	14.094	12.783	11.654	10.675	9.823	9.077	8.422	7.843	7.330	6.873	6.464	6.097	5.766	5.467	5.195	4.948
30	25.808	22.396	19.600	17.292	15.372	13.765	12.409	11.258	10.274	9.427	8.694	8.055	7.496	7.003	6.566	6.177	5.829	5.517	5.235	4.979
35	29.409	24.999	21.487	18.665	16.374	14.498	12.948	11.655	10.567	9.644	8.855	8.176	7.586	7.070	6.617	6.215	5.858	5.539	5.251	4.992
40	32.835	27.355	23.115	19.793	17.159	15.046	13.332	11.925	10.757	9.779	8.951	8.244	7.634	7.105	6.642	6.233	5.871	5.548	5.258	4.997
50	39.196	31.424	25.730	21.482	18.256	15.762	13.801	12.233	10.962	9.915	9.042	8.304	7.675	7.133	6.661	6.246	5.880	5.554	5.262	4.999

SUMMARY OF FOUR FORMULAS AND some “TRICKS”

1. $V_0 (1+i)^n = V_n$. Future value of single current dollar amount. **Table A1.1.**
2. $FV = V \{ [(1+i)^n - 1] / i \}$ Future accumulated value for a series of equal annual dollar amounts (first in the series one year hence). **Table A1.2.**
3. $V_0 = V_n / (1+i)^n$. Present Value of a one-time future dollar amount. **Table A1.3.**
4. $V_0 = V [(1+i)^n - 1] / [(i)(1+i)^n]$ Present value of an equal annual series of dollar amounts (first in the series one year from now). **Table A1.4.**

TRICKS: Assuming the problem you are given only involves the use of just one formula (more on problems that use a combination later), then a few “tricks” to identify the nature of the problem. First, notice that only formulas 2 and 4 have the V variable. This variable is in dollars per year. So a tipoff is expressions such as “per year”, “every year”, “annually” → a signal that V is being referred to. Second, suppose you have spotted that it is some sort of annuity problem. How to distinguish between 2 and 4? Notice which variables the formulas share and the ONE that they do not share— V_0 (“current dollars”, “loan taken out today, buy machine today, current price of bond, etc.) shows up on formula #4 but NOT

in 2. For formula 2, we have the FAV term but not the V_0 term. Hence, again it should be easy to spot the difference.

Variations

1.a. Perpetuity formula.

Let us return to the PV of an annuity formula.

$$V_0 = V \left[\frac{(1+i)^n - 1}{(i)(1+i)^n} \right] = V \left\{ \frac{(1+i)^n}{(i)(1+i)^n} - \frac{1}{(i)(1+i)^n} \right\} \\ = V \left\{ \frac{1}{i} - \frac{1}{(i)(1+i)^n} \right\}$$

As n approaches infinity notice the last term in this expression approaches zero. Therefore, in the limit the formula becomes. $V_0 = V/i$. Needless to say, this is very convenient!

Even if n is not literally infinite (perpetual), it is a very useful approximation for a large number of periods. To see this, suppose $n = 60$. Refer to Table A1.4 to get the precise value term at an interest rate of 10%. You should see 9.9672. So if V was \$100, the PV would be \$996.72 (very close to \$1,000). Now use the perpetuity formula. $V_0 = V/i = 100/0.10 = \$1,000$. Almost the same. Close enough.

FOOTNOTE:

1.b. A variation on the perpetuity formula is the Gordon formula. Suppose the annual dollar amounts are not the same but do grow at the same constant growth rate of g . Then the new version of this perpetuity formula would be $V_0 = V/(i - g)$ subject to $i > g$. We will make use of this in a later on lecture on stock prices.

Problems involving a combination of formulas

Many real world problems are not as straightforward as the examples provided above.

Here is an example: Suppose you currently have \$100,000 in an account. You desire to know the equal annual dollar amount you could withdraw at 12% and still have \$70,000 left in the account at the end of 10 years. Let's identify the variables in terms of our symbols:

Givens: $V_0 = \$100,000$ (tipoff "currently have.."). $n = 10$ years, $i = 12\%$ and $V_{10} = \$70,000$ (some dollar amount, not a series, ten years from now). The unknown is V (\$ per year).

We know that the PV of the series of withdrawals PLUS PV the desired remaining balance in ten years must be equal to \$100,000. So set it up

$V_0 = V \left[\frac{(1+i)^n - 1}{(i)(1+i)^n} \right] + V_n / (1+i)^n$. As you can see, we are using formulas #3 and #4.

$100,000 = V \left[\frac{(1.12)^{10} - 1}{(.12)(1.12)^{10}} \right] + 70,000 / (1.12)^{10}$. Now just a simple matter of solving for V .

This type of problem is especially important for financial assets since, for example, a "bond" may pay annual interest payments (or monthly interest payments) plus some final payment called the maturity value.

Second example. When you retire you desire to have \$500,000 in your account. Will retire in 40 years and the interest rate is 4%. You currently have \$30,000 in your retirement account and you wish to know how much you will have to put into this account each year to achieve your goal (first of this equal annual deposits is one year from today). Givens: $V_{40} = 500,000$, $V_0 = 30,000$, $n = 40$ years, $i = 4\%$.

Unknown is V .

Set up: future value of current balance + FAV of equal annual deposits = $V_{40} = 500,000$. Using formulas one and two: $V_0 (1+i)^n + V \left[\frac{(1+i)^n - 1}{i} \right] = \$500,000 \rightarrow \$30,000(1.04)^{40} + V \left[\frac{(1.04)^{40} - 1}{0.04} \right] = \$500,000$. Now a simple matter to solve for V .

First in the series is more than one year from now

Suppose a project has annual benefit stream of \$2 million with the first in the series is 5 years from now. The benefit stream lasts 56 years (this the last payment is 60 years from now—not 61 !). Interest rate is 8%. *Long way to find PV:* $V_0 = V_5/(1+i)^5 + V_6/(1+i)^6 + V_7/(1+i)^7 \dots V_{60}/(1+i)^{60}$

$$= V/(1+i)^5 + V/(1+i)^6 + V/(1+i)^7 \dots V/(1+i)^{60}.$$

Unfortunately, our PV of annuity formula no longer directly works since the first in the series is 5 years from now.

However, we can still use it by doing the following:

	Timeline: 0	---	1	----	2	----	3	----	4	----	5	-----	6	-----		60
We want the PV of this											\$2		\$2			\$2

If we use this $V_0 = \$2 \text{ million } [(1.08)^{60} - 1] / [(0.08)(1.08)^{60}]$ we end of finding the PV of this stream:

	Timeline: 0	---	1	----	2	----	3	----	4	----	5	-----	6	-----		60
We get PV of this			\$2		\$2		\$2		\$2		\$2		\$2			\$2

But that is not what we want. We want to get rid of the first four years of payments which has a PV of $V_0 = \$2 \text{ million } [(1.08)^4 - 1] / [(0.08)(1.08)^4]$

The bottom line is that to end up with the PV stream we desire, you simply do this:

PV of \$2 million (first in series 5 years from now) = \$2 million $[(1.08)^{60} - 1] / [(0.08)(1.08)^{60}]$ MINUS \$2 million $[(1.08)^4 - 1] / [(0.08)(1.08)^4]$ = **\$2 million $[(1.08)^{60} - 1] / [(0.08)(1.08)^{60}]$ MINUS $[(1.08)^4 - 1] / [(0.08)(1.08)^4]$**

Bond Pricing/explaining “gaps” between rates questions

Definitions, zero-coupon bonds, coupon bonds, bond prices, relationship between i and bond prices, explaining “gaps” between bond yields, explaining co-movements in bond yields

Definitions and Terminology

Bonds are securities that represent a debt owed by the issuer (that is, the seller of bond—might a corporation or the government) to the investor buyer of the bond). Bonds are obligations buy the issuers to pay a **specified** amount by a given date, generally with periodic interest payments.

1. The **par, face or maturity value** of a bond is the amount the bond issuer must pay at maturity.
2. The **coupon rate** is the interest that issuer must pay, and this periodic interest payment is often called the coupon payment. THIS rate is usually fixed for the duration of the bond and does not fluctuate with market interest rates.
3. If default, bond holder has claim on assets of issuer.

Treasury Notes (1 to 10 years) and Bonds (10 to 30 original maturity). T-Bills less than one year. Treasury Inflation-protected securities (TIPS). Are adjusted for changes in inflation rate as measured by the CPI)

Bond Cash Flows, Prices, and Yield to Maturity

FV = face value (maturity value, par value)

Important: $CPN (\$ \text{ per period}) = (\text{Coupon Rate} * \text{Face Value}) / (\text{Number of Coupon Payments per Year})$

Zero-Coupon Bonds

1. Does not make coupon payments
2. Always sells at a discount (a price lower than face value), so they are also called pure discount bonds
3. Treasury Bills are U.S. government zero-coupon bonds with a maturity of up to one year.
4. Yield to Maturity (YTM) of a zero-coupon bond is the discount rate that sets the present value of the promised bond payments equal to the current market price of the bond (it is really just the IRR!)
5. It is typically quoted on an annual basis (APR = Annual Percentage rate)

NOTATION

P = price of bond, FV (or F) = the face value (maturity value, par value)

1. To find price: $P = FV / (1+i)$ where i is the appropriate discount rate.

2. To find the Yield to Maturity (YTM) $\rightarrow P = FV / (1+YTM)$

and to make it more general for maturity dates less than one year:

YTM = [(F-P)/P] * (365/d) where d is number of days until maturity. The last term is the adjustment to put it into APR.

EXAMPLE 1: Suppose that a one-year, risk-free, zero-coupon bond with a GIVENS:

F= \$100,000 face value, P =96,618, d = 365 days → solve for unknown YTM

has an initial price of \$96,618.36. $YTM = [(100,000-96,618)/96,618] * (365/365) = 0.035$ (3.5%)

EXAMPLE #2: If this was a *six-month* bond then:

GIVENS: F = 100, 000, P = 96,618, d = 180 days → solve for YTM

$YTM = [(100,000-96,618)/96,618] * (365/180) = 0.035$ (3.5%)* $(365/180) = .071$ (7.1%)

EXAMPLE #3: FV =100,000, P = 97,000 and YTM is 6%. How many days?

$0.06 = YTM = [(100,000-97,000)/97,000] * (365/d)$

$0.06 = 0.0309 * (365/d) \rightarrow 1.94 = 365/d \rightarrow d = 365/1.94 = 188$ days.

NOTICE: as the interest rate rises, the price of zero-coupon bond falls.

A lower P implies a higher YTM

Coupon Bonds

FV (or F) = face value (maturity value, par value)

CPN (\$ per period) = (Coupon Rate * Face Value) / (Number of Coupon Payments per Year)

a. Pay face value at maturity and b. Pay regular coupon interest payments

Two types of U.S. Treasury coupon securities are currently traded in financial markets:

Treasury Notes - U.S. Treasury coupon security with original maturities of 1–10 years

Treasury Bonds - U.S. Treasury coupon security with original maturities over 10 years

Return on a coupon bond comes from:

a. The difference between the purchase price and the principal value AND

b. Periodic coupon payments (that is, interest payments)

Coupon Bonds: Yield to Maturity

The YTM is the single discount rate that equates the present value of the bond's remaining cash flows to its current price—the IRR.

To compute the yield to maturity of a coupon bond, we need to know the coupon interest payments, and when they are paid.

Yield to Maturity of a Coupon Bond

$P = CPN/(1+ytm) + CPN/(1+ytm)^2 + CPN/(1+ytm)^3 + \dots + [CPN + FV]/(1+ytm)^n$

$P = (CPN) * [(1/(1+i)^n - 1) / [i * (1+i)^n]] + [Face Value / (1+i)^n]$.

[Note: in equilibrium the market i equals YTM]

#1.Example to find price

Example. The U.S. Treasury has just issued a ten-year (so 20 periods if semi-annual interest), \$1,000 bond with a 4% coupon rate and semi-annual coupon payments. What cash flows will you

receive if you hold the bond until maturity? The face value of this bond is \$1,000. Because this bond pays coupons **semiannually**, you will receive a coupon payment every six months of **CPN= \$1,000 * 4%/2 = \$20/six month pd.** Here is the timeline, based on a six-month period between coupon payments:



If discount rate is also 4% APR (2% semiannually)

$P = 20/(1.02) + 20/(1.02)^2 + \dots + 20 + 1000/(1.02)^{20} = \1000 . **Notice that when $i = c$, $P = FV$.**

c=coupon rate, FV = face value

Face Value (FV or F)	Coupon rate (c)	number of yrs. (yrs)	Frequency of payments per year (f)	Total number of payments (n)= yrs*f	CPN per pd. = (FV*c)/f	P	YTM (or i)	YTM/f
\$ 1,000	4%	10	2	20= 10*2	\$ 20	Find	4%	2.0%
\$ 1,000	8%	10	2	20	\$ 40	1034.74	7.00%	3.75%
\$ 1,000	8%	10	2	20	\$ 40	934.96	9%	4.5%

monthly = 12
quarterly = 4
annual = 1
semi-annual = 2

First four columns are the “givens”. You calculate the green highlighted columns.

Yellow highlights = values that you plug into equation.

Unknown, thing solved for, is either P or YTM

#2.Example to find YTM: Suppose a 10-year, \$1000 bond with an 8% coupon rate and **semiannual** coupons is trading for a price of \$1034.74. What is the bond’s yield to maturity? The yield to maturity is defined as the discount rate that equates the present value of the bond’s cash flows with its price.

- -Givens: n =10 years, P = 1034.71, FV = 1000, c = 8% (APR), paid semiannually.
- Conversions and other: 20 periods, $c/2 = 4\%$ and $0.04*1000 = \$40$ per semiannual period.
- Plug into formula:

$$\$1,034.74 = 40/(1+0.5YTM) + 40/(1+0.5YTM)^2 + \dots + 40 + 1000/(1+0.5YTM)^{20} \rightarrow$$

0.5YTM is 3.75%--> $YTM = 3.75\% \times 2 = 7.5\%$. NOTE: the solving of this equation gives us 3.75% or 0.5YTM = 3.75% (basically, letting $x = 0.5YTM$) so we need to convert to APR by doubling it.

That is, because the bond pays coupons **semiannually**, this yield solved for in the formula is for a six-month period. We convert it to an APR by multiplying by the number of coupon payments per year.

#3.Example when interest rate changes: If the bond's yield to maturity changes to **9% APR**, what will the bond's price be? (in equilibrium, YTM equals market rate)

$$PV = 40/(1+0.045) + 40/(1.045)^2 + \dots + 40+1000/(1+0.045)^{20} = \$934.96$$

Note that the YTM is higher than the coupon rate and the price is lower than the par value.

Interest rates have increased, so the higher return brings the bond's yield into line with the higher competitive rates being offered for similar risk and maturity elsewhere in the Market.

As i rises, P falls.

Terminology for bond pricing:

We say that a bond is selling at a discount when $P < FV$. This implies $YTM > c \rightarrow 9\% > c=8\%$

We say that a bond is selling at par when $P=FV$. This implies $YTM = c$ (example #1)

We say that a bond is selling at a premium when $P > FV$ (Example #2: $1034 > 1000$). This implies $YTM < c$ (example #2: $7.5\% < 8\%$)

However, the important lesson from all of this is that as interest rates increase, bond prices decreases. Simply a notion we see from doing PV calculations!

Note: Most issuers of coupon bonds choose a coupon rate so that the bonds will initially trade at, or very close to, par (that is, when first issued, can expect $c = i$). **After the issue date, the market price of a bond changes over time. Think of the concept of PV.**

Credit risk (default risk), bond ratings and yields

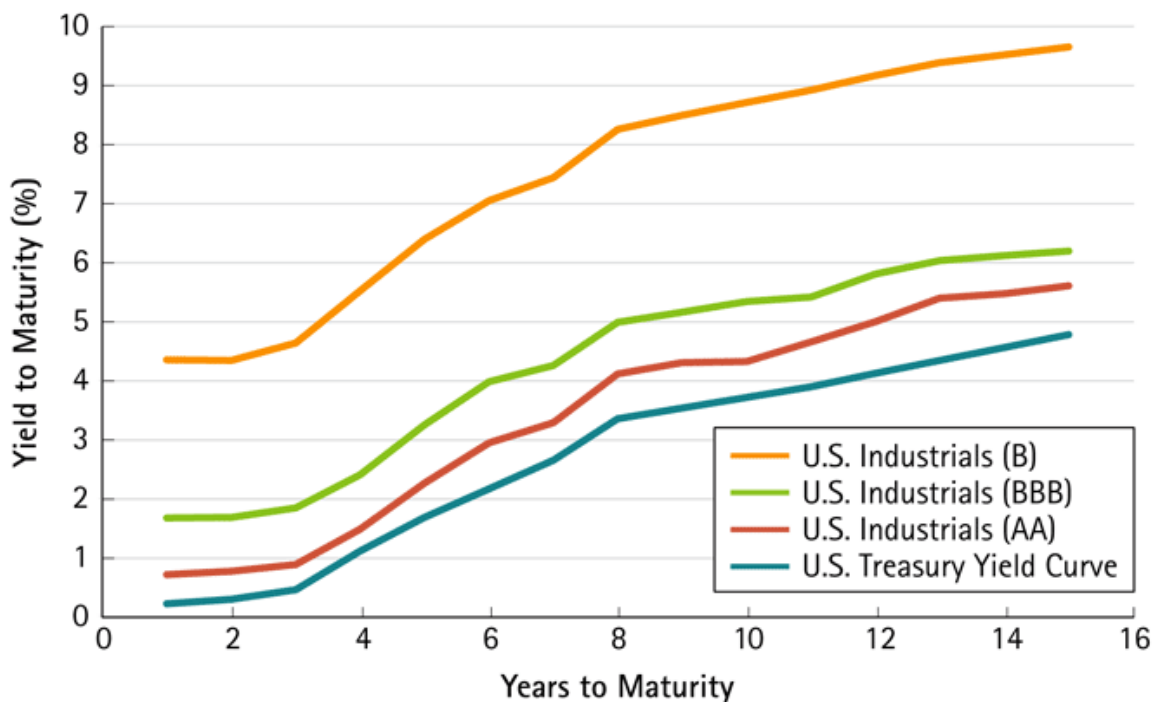
Moody's	Standard & Poor's	Number of Public Firms	Description (Moody's)
Investment Grade Debt			
Aaa	AAA	5	Judged to be of the best quality. They carry the smallest degree of investment risk and are generally referred to as "gilt edged."
Aa	AA	28	Judged to be of high quality by all standards. Together with the Aaa group, they constitute what are generally known as high-grade bonds.
A	A	164	Possess many favorable investment attributes and are considered as upper-medium-grade obligations. Factors giving security to principal and interest are considered adequate at present, but may not remain that way.
Baa	BBB	399	Are considered as medium-grade obligations (i.e., they are neither highly protected nor poorly secured).

Speculative Bonds ("Junk Bonds")

Ba	BB	318	Judged to have speculative elements; their future cannot be considered as well assured.
B	B	296	Generally lack characteristics of the desirable investment. Assurance of interest and principal payments over any long period of time may be small.
Caa	CCC	27	Are of poor standing. Such issues may be in default or there may be present elements of danger with respect to principal or interest.
Ca	CC	4	Are speculative to a high degree. Such issues are often in default or have other marked shortcomings.
C	C, D	8	Lowest-rated class of bonds, and issues so rated can be regarded as having extremely poor prospects of ever attaining any real investment standing.

Source: www.moodys.com and S&P Compustat.

Corporate Yield Curves for Various Bond ratings



Source: Bloomberg.

Several questions to answer here.

- What explains the gaps?
- Why do these rates move together?
- What is the relationship between yield to maturity and the yield? (this last question is not answered until later when we cover term structure of interest rates).

[Default risk

XYZ Industries needs to raise \$25 million to fund a new office complex. The company plans on issuing ten-year bonds with a face value of \$1000 and a coupon rate of 7.0% (annual payments). The following table summarizes the YTM for similar ten-year corporate bonds of various credit ratings:

Rating	AAA	AA	A	BBB	BB
YTM	6.70%	6.80%	7.00%	7.40%	8.00%

a. Find P for a given bond rating and YTM. If company receives an AAA rating, the price of bonds would be expected to be: $P = 70/(1.067) + 70/(1.067)^2 + \dots + 70 + 1000/(1+0.067)^{10} = \$1,021$

b. For a given price, find YTM and implied bond rating. Suppose that when these bonds were issued, XYZ received a price of \$972.42 for each bond. What is the likely rating that XYZ's bonds received?

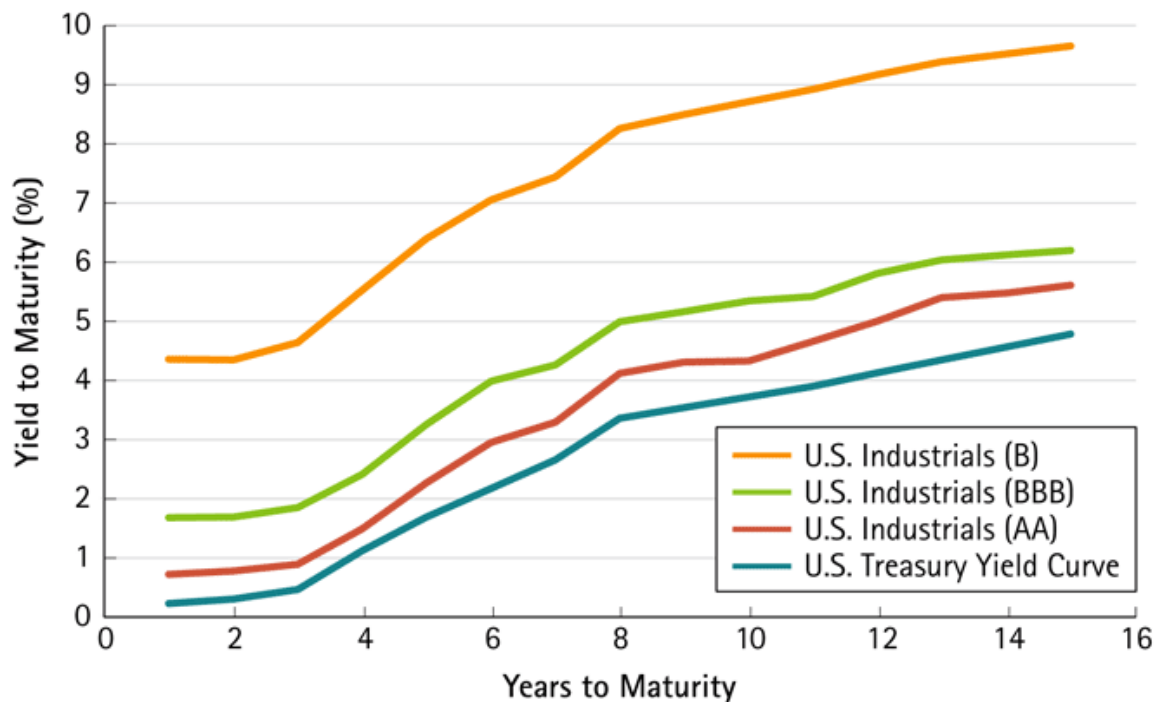
$$972.42 = 70/(1+YTM) + 70/(1+YTM)^2 + \dots + 70 + 1000/(1+YTM)^{10}$$

Solving for YTM gives us 7.4%. This would be in the BBB rating.]

Lesson: default risk (as measured by bond ratings) impacts yields and therefore bond prices (for given coupon rate and given face value). This helps explain the gaps between rates on the above chart. Will say more about this in a S&D format.

Explaining Gaps and Co-movements

Two things to notice in the chart below. Rates on different securities (a) tend to move together (go up and down together) and (b) the rates are not the same on these different securities (gaps)



Source: Bloomberg.

1. Why to do bond and other rates tend to move together?

We have seen from our discussion on the term structure of interest rates that there is a theoretical reason why short-term and long-term rates move together: namely, due to the **substitutability**. I can put my money into, say, Treasury Bills or Commercial Paper (a short-term, unsecured security typically issued by large well-known corporations). Just as the price of Coke and Pepsi tend to move together, likewise the yields on securities tend to move together.

2. What explains the different rates on various securities (that is, what explains the gaps)?

a. **Default risk or Credit risk** (see Moody's, Standard and Poor's, Fitch). Default risk or credit risk can explain much of the divergence (the gap). Bonds are given credit ratings by various credit rating service companies, such as Moody's and Standard and Poor's. As the above chart shows, bonds with higher credit ratings (and thus have lower credit risk) pay lower rates.

b. **Liquidity** (relative ease at which asset can be converted to cash). Discussed later in Asset Approach topic.

c. **Tax considerations** (example, municipal bonds) **Example of Tax considerations:** Municipal bonds are long-term securities issued by states, county, and local government entities typically to finance projects such as schools, roads, transportation projects, etc. They are not subject to federal taxation or state taxation for investors who live in the state that they were issued.

Suppose t = tax rate on interest income and suppose municipal bonds and similar bonds are both initially taxed at this rate. In equilibrium, we expect the after-tax returns to equalize: $i_{\text{other}}(1-t) = i_{\text{muni}}(1-t)$. Now suppose we remove the tax on munis. The immediate impact would be $i_{\text{other}}(1-t) < i_{\text{muni}}$. This would make holding municipal bonds more attractive and would increase the demand for those bonds (and decrease the demand for the other bonds). But this change in the pattern of demand would drive up municipal bond prices (and lower the price of the other bonds). This is equivalent to a decrease on the rate on munis and an increase in the rate on the other similar bonds. Thus, in the new equilibrium we should observe this $i_{\text{other}}(1-t) = i_{\text{muni}}$ or $(1-t) = (i_{\text{muni}} / i_{\text{other}})$. This is broadly consistent with what we observe. The historical evidence indicates that this ratio is around 70% (implying a 30% tax bracket). There is however a lot of variation around this average.

d. Other: **suppose government guarantees student loans**

Moody's	Standard & Poor's	Number of Public Firms	Description (Moody's)
Investment Grade Debt			
Aaa	AAA	5	Judged to be of the best quality. They carry the smallest degree of investment risk and are generally referred to as "gilt edged."
Aa	AA	28	Judged to be of high quality by all standards. Together with the Aaa group, they constitute what are generally known as high-grade bonds.
A	A	164	Possess many favorable investment attributes and are considered as upper-medium-grade obligations. Factors giving security to principal and interest are considered adequate at present, but may not remain that way.
Baa	BBB	399	Are considered as medium-grade obligations (i.e., they are neither highly protected nor poorly secured).

Speculative Bonds (“Junk Bonds”)

Ba	BB	318	Judged to have speculative elements; their future cannot be considered as well assured.
B	B	296	Generally lack characteristics of the desirable investment. Assurance of interest and principal payments over any long period of time may be small.
Caa	CCC	27	Are of poor standing. Such issues may be in default or there may be present elements of danger with respect to principal or interest.
Ca	CC	4	Are speculative to a high degree. Such issues are often in default or have other marked shortcomings.
C	C, D	8	Lowest-rated class of bonds, and issues so rated can be regarded as having extremely poor prospects of ever attaining any real investment standing.

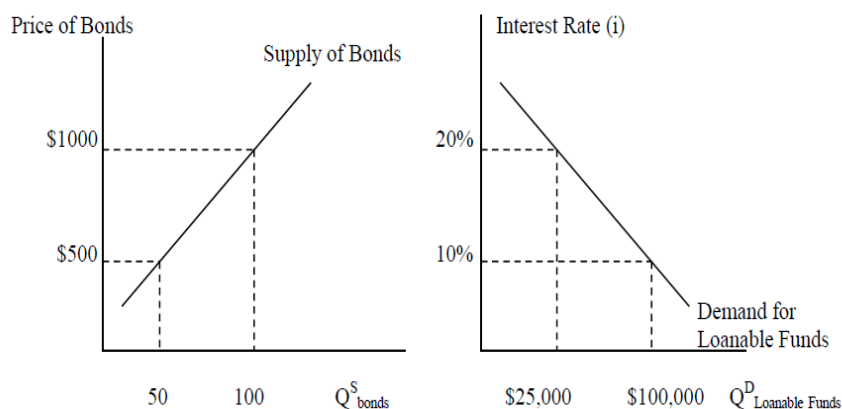
Source: www.moodys.com and S&P Compustat.

An important preliminary note before starting the S&D discussion on credit market interaction (rates go up and down together).

We sometimes tell our stories from the bond price “point of view” and other times from interest rate/yield to maturity point of view. It turns out they are one and the same.

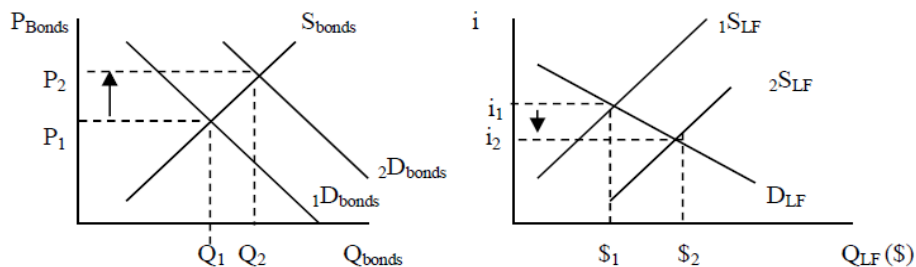
Assume that the price of a bond is given by the perpetuity formula: $V_0 = V/i \rightarrow P_{\text{bond}} = \text{annual interest payment}/i$.
Suppose annual interest payment is \$100.

i	Price of Bonds	Q of bonds supplied	Q of loanable funds demanded
10%	$\$100/0.10=\1000	100	$\$1000*100=\$100,000$
20%	$\$100/0.20=\500	50	$\$500*50=\$25,000$



It can be seen from above that the supply of bonds is the SAME as the demand for loanable funds—to indicate that someone wants to sell an IOU (bond) is equivalent to saying that this person wants to borrow. Likewise, the Demand for Bonds is equivalent to the supply of loanable funds—to desire to lend funds is the equivalent to the desire to buy IOUs.

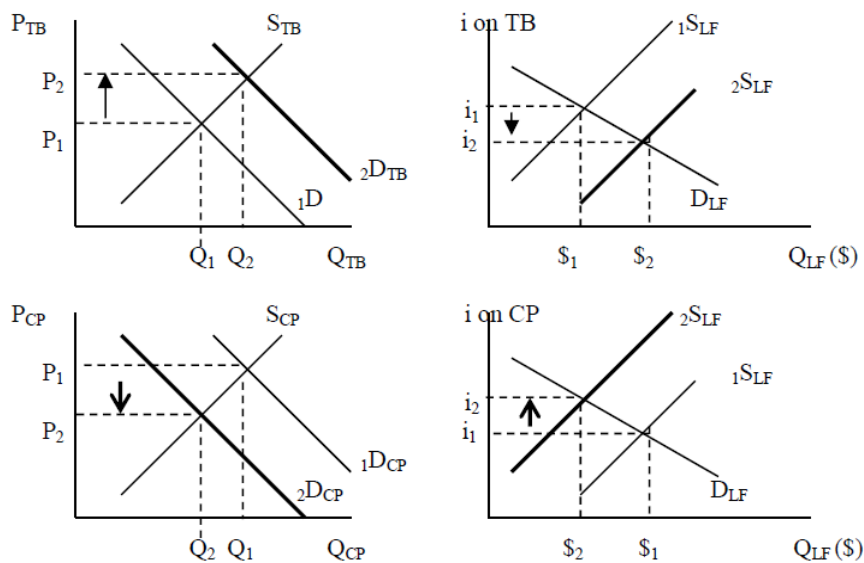
So, example, an increase in the demand for bonds (rightward shift) is the same as an increase in the supply of LF (rightward shift)



Using S & D to explain “gaps”

TB = Treasury Bills and CP = commercial paper (short-term, zero-coupon bonds issued by large well known corporations).

- Suppose initially TB and CP were roughly considered perfect substitutes—same default risk, same tax consideration, same liquidity, etc.
- Now there is reason to believe that CP has a higher default risk. This would lower the demand for CP, thereby lowering its price and raising its yield. It would also raise the demand for TB, thereby increasing its demand, raising its price and lowering its yield. This would explain the gap.



The exact same S&D analysis can be used to explain co-movements. Suppose each market is in equilibrium with $\text{rate on T-Bills} = (\text{CP rate} - \delta)$ to account for differences in default risk. Suppose the government issues more T-Bills (S of TB shifts out, D for LF in TB market shifts out) $\rightarrow i_{TB}$ rises. But this will mean TB become relatively more attractive to hold compared to CP at the initial value of i_{CP} . Thus, demand for CP falls, S of LF in CP falls (shifts in) causing i_{CP} to also rise! Thus, due to their substitutability then tend to move together.

Nominal and real rates/Fisher equation/Fisher effect

Nominal rate of interest, real interest rate and PV

Nominal rate of interest (i) does not adjust for inflation. It just looks at how many dollars you get back (if you are making a loan, for example) expressed in percentage terms without adjusting for changes in the price level over the life of loan. The real rate of interest (r) does adjust for inflation. It looks at the increase in purchasing power you get back expressed in percentage terms. Here is a loan example.

	Period 0	Period 1	%	%, general
an in nominal terms	\$100	\$110	$i=10\%$	$(V_1 - V_0)/V_0 = i$ Or $(V_1/V_0) = 1+i$
Price level (price of widgets) $P_0=\$1$		$P_1=\$1.1$	$\pi=10\%$	$(P_1 - P_0)/P_0 = \pi$ Or $(P_1/P_0) = 1+\pi$
Loan in real terms	$\$100/\$1=100$ Widgets	$\$110/1.1$ $=100$ widgets	$r=0\%$	$[(V_1/P_1) - (V_0/P_0)] / [V_0/P_0] = r$

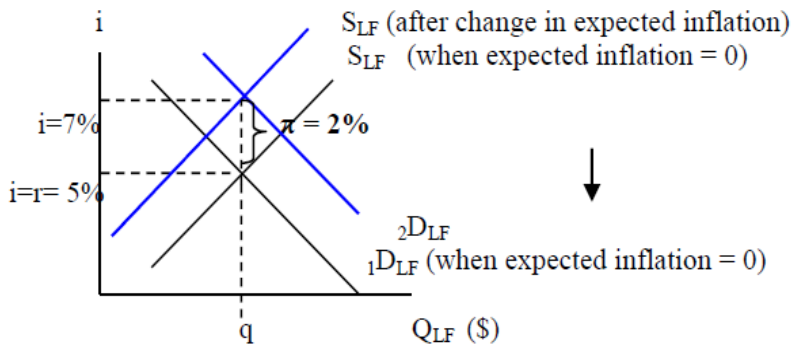
This last term--- $[(V_1/P_1) - (V_0/P_0)] / [V_0/P_0] = r$ --- can be rearranged as

$$[(V_1/V_0) * (P_0/P_1)] - 1 = r \rightarrow (1+i)*(1/(1+\pi)) = 1+r \rightarrow (1+i)/(1+\pi) = 1+r \rightarrow$$

$$(1+i) = (1+r)*(1+\pi)$$

$(1+i) = (1+r)*(1+\pi)$ is the Fisher Equation. To get an approximate version of this equation, $(1+i) = (1+r)*(1+\pi) = 1 + r + \pi + r*\pi$. The last term is usually considered “small” and therefore dropped to get this $1+i = 1 + r + \pi \rightarrow i = r + \pi$ which is the approximate version of the Fisher equation. If we interpret π as the *expected* inflation rate, then an implication of this equation is that an increase in the expected inflation rate will lead to an increase in the nominal rate of interest (this is call the Fisher Effect).

The **logic of the Fisher Effect** can be seen below in the graph. The original supply and demand for loanable funds assume lenders and borrowers do not expect any inflation. If they now come to expect 2% inflation, then since lenders were willing to lend at the q^{th} dollar at a 5% real return as indicated on the graph, to maintain that real rate would require at minimum a 7% nominal rate for the q^{th} dollar. This leads to vertical shift depicted below. Similar story for borrowers to get the shift below. Bottom line is that the nominal rate rises by the change in the expected inflation rate—this is the Fisher Effect. See graph below.



FOOTNOTE:

For the stories below, suppose we let $P_0 = 1$ (what the initial price level is somewhat arbitrary so we might as well pick an easy number).

1. Suppose there is a stream of payments of \$100 today and \$121 one year later. We desire to find the PV of this stream of payments. Assuming there is inflation, how do we deal with this?

The PV calculation expressly using the nominal rate of interest and *without* making any adjustment for inflation for the \$121 received in the future is given by **PV = 100 + 121/(1+i)**

But the PV calculation expressly making an adjustment for inflation by converting \$121 into real terms and using the real rate of interest to take PV is given by

$PV = 100 + (121/(1+\pi)) / (1+r)$ By substitution we can rewrite this as $100 + (121/(1+\pi)) / [(1+i)/(1+\pi)]$ since we know from above that $1+r = (1+i)/(1+\pi)$.

[Note: $(121/(1+\pi))$ is the \$121 expressly adjusted for inflation, that is, the expressed dollars are in purchasing power terms]

We can see that $PV = 100 + (121/(1+\pi)) * (1+\pi)/(1+i) = 100 + 121/(1+i)$. So we end up with the same answer either way. Important! If you convert future dollars into real terms (purchasing power terms) then you must use the real rate of interest to find the PV. If you do NOT convert future dollar amounts into real terms, that is, leave them in nominal terms then you must use the nominal rate of interest to find the PV. Intuition: If you do not expressly convert future dollars into real terms, but use the nominal rate of interest for PV calculations, then you are implicitly converting these future dollar streams into real terms since the nominal rate has built into the estimated future inflation rates (see Fisher equation again).

More generally,

$$PV = \sum_{t=1} [V(1+\pi)^t / (1+i)^t] = [\sum_{t=1} V(1+\pi)^t / ((1+r)^t (1+\pi)^t)] \text{ by substitution since } (1+i) = (1+r)(1+\pi)$$

But notice that the $(1+\pi)^t$ term cancels out, leaving us with this equation

$$V_0 = \sum_{t=1} V / (1+r)^t \text{ by where the } V \text{ can be thought of as the payment in real terms.}$$

NOTE: $V(1+\pi)^t$ = nominal value of payment stream at time t, assuming it grows at the same rate as the inflation rate. If it does grow at the same rate of inflation then in real terms it is constant over time at value of V. For example, $V_0 = \$10$, $V_1 = \$11$, $V_2 = \$12.1$ are in nominal terms. But if the inflation rate is 10% per year then in real terms the numbers are 10, $11/1.1=10$, $12.1/1.21=10$, all of which are 10.

Asset Approach and the Bond Market

S&D analysis (and its counterpart in the LF market), shift variables, applications.

Lecture Notes on the Asset Approach to the Bond Market

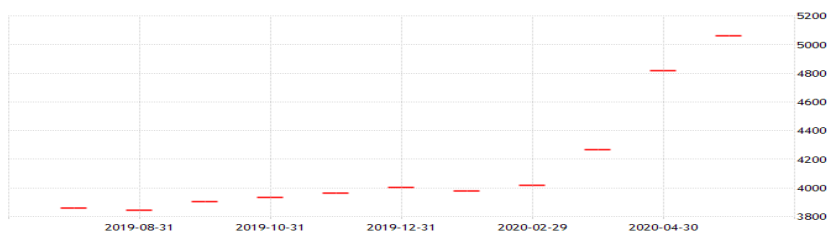
Definition of Asset Approach

This approach views the demand for bonds as part of broader decision for determining the optimal combination of assets to hold in one's portfolio or collection of assets. A person or firm's stock of wealth or assets consists of:

- Money (as defined by M1 for our purposes; M1 = Currency + Checking Accounts)
- Other financial assets (savings deposits, time deposits, bonds, stocks etc.)
- Non-financial assets (real estate for example)
- Human Capital (the skills, knowledge, talents we possess or have acquired through formal and informal education, training and experience).

Each asset has different characteristics with regard to **returns, risk and liquidity**, and hence there is a tradeoff. Cash, for example, has a high degree of liquidity but a low (zero) return. In principle, people will allocate their wealth across these different assets so that the marginal benefit of holding one more dollar in particular form is equal across assets. Liquidity refers to the ease and quickness at which one asset can be traded for another. Money (M1), by definition, is the asset commonly used to make payment or pay debts, and is therefore the most liquid of all assets.

M1 in billions of dollars



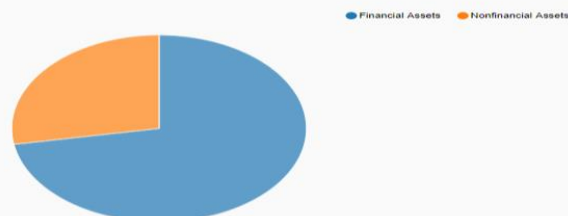
Note: the \$113 Trillion mentioned below does not include human capital

In 2018, U.S. households held over \$113 trillion in assets. For context, that is over five times as much as all the goods and services produced in the U.S. economy in a single year. If that amount were divided evenly across the U.S. population of 329 million, it would result in over \$345,000 for each person. For a family of three, that's over a million dollars in assets.

Almost three-quarters of aggregate household assets are in the form of financial assets—namely stocks and mutual funds, retirement accounts, and closely-held businesses. Real estate makes up the vast majority of nonfinancial assets.

Composition of Household Assets

2018



Z.1 Financial Accounts of the United States, First Quarter 2019, Table B.101.h Balance Sheet of Households, Federal Reserve Board.

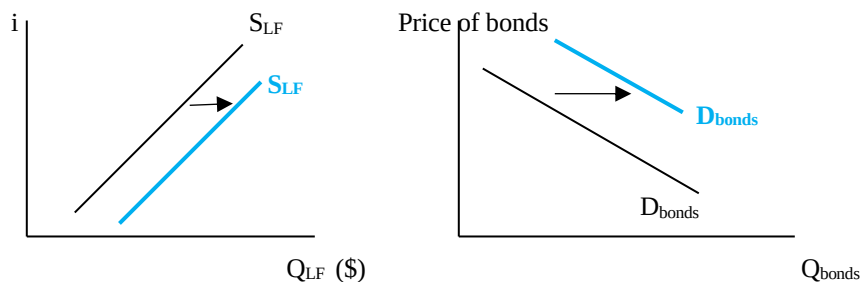
BROOKINGS

Bond Demand

Recall that the demand for bonds is the same as the supply of loanable funds (to indicate a willingness by an “IOU”, a bond, is the same as an indication of a willingness to lend funds).

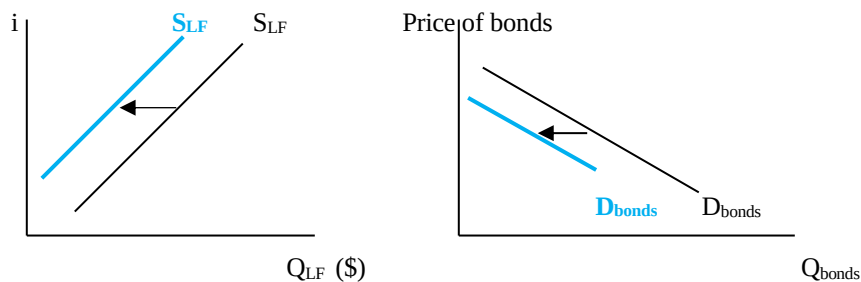
This asset approach model looks at four characteristics in determining bond demand:

i. A person or firm’s overall **wealth**. As wealth increases, it is generally the case that demand for each type of asset increases (each is a “normal” good to put in micro terms). For bonds, an increase in wealth means a rightward shift in the bond demand curve which is the same as an increase in the supply of loanable funds. See graphs below.



ii. **Expected future returns on one asset relative to another asset.** For bonds, a higher expected *FUTURE* return on bonds lowers its current period demand. An increase in higher expected *FUTURE* return can occur either because of higher expected future *real* return (r) or because of an increase expected inflation (π) [recall the Fisher Equation!]. This is counterintuitive, so here is an example. Suppose a bond that is currently issued payments annual interest of \$10 forever (hence, we can use the perpetuity formula). The interest rate is 10%, so using the perpetuity formula, we get a price of $V_0 = V/i = 10/0.10 = \$100$. Suppose now that new information becomes available that the interest rate next period will be 20%. This would therefore mean you expect the price of this bond to fall to $10/0.20 = \$50$. To avoid such an expected capital loss, people would reduce their demand for bonds today. (leftward shift in bond demand or leftward shift in supply of loanable funds).

Impact of an increase expected future returns on *current* demand for bonds:



iii. **Risk on one asset *relative* to another asset.** For bonds, an increase in the riskiness of bonds relative to other assets (perhaps because people come to believe the *stock* market has become less risky) would cause the demand for bonds to fall as people moved from bonds to, say, stocks or cash (leftward shift in bond demand/leftward shift in supply of loanable funds). See graphs above.

iv. **Liquidity on one asset *relative* to another asset.** For bonds, if some event lead to an increase in the liquidity of bonds relative to other assets (say, due to some development in institutional features of the bond market), this would increase the demand for bonds (since liquidity is a desirable feature in an asset). Rightward shift in bond demand curve, rightward shift in LF supply curve.

[note: it should be mentioned that there are other factors that shift the demand curve. Will be discussed as the need to do so arises.]

Supply of Bonds

Recall that the supply of bonds is the same as the demand for loanable funds (to indicate a willingness to sell an “IOU”, a bond, is the same as an indication of a desire to borrow funds).

This asset approach model looks at the following in determining bond supply:

i. **Expected Future Profitability of Investment (EFPI).** A change in the expected return on investment in new capital goods—that is, the internal rate of return-- (building, machines, tools etc.) would influence desired borrowing by firms to finance investment projects. To desire to borrow is equivalently a desire to sell bonds. So, for example, if EFPI increases, this would result in an increase the supply of bonds (rightward shift in bond supply curve/rightward shift in LF demand curve) as firms desire to borrow more funds.

ii. **Expected inflation.** Recall our discussion on the Fisher Effect. An increase in the expected inflation rate causes a shift in the demand for loanable funds curve to the right or vertically up (as previously explained). But a rightward shift in the LF demand curve is simply the same as a rightward shift in Bond supply curve. Another way to see this is as follows: At any GIVEN nominal rate, an increase in expected inflation rate implies a lower expected real rate of interest. This would increase the desire to borrow (i.e., to supply bonds).

iii. **Government Budget deficits ($G-T < 0$).** If the government spends more than it takes in from tax revenues, it will have to finance the difference by borrowing. The government borrows by selling bonds (for US, this means US Treasury Bills, Notes and Bonds). Thus, an increase in the government budget deficit implies a rightward shift in the bond supply curve.

In summary, the shift variables are:

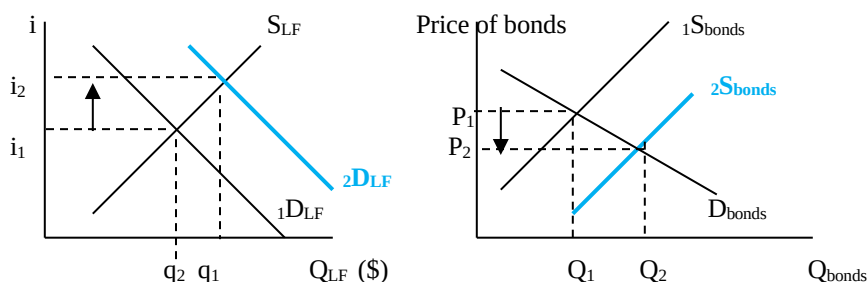
B^d (wealth, relative expected future returns, relative risk, and relative liquidity)

B^s (EFPI, expected inflation rate, government budget deficit)

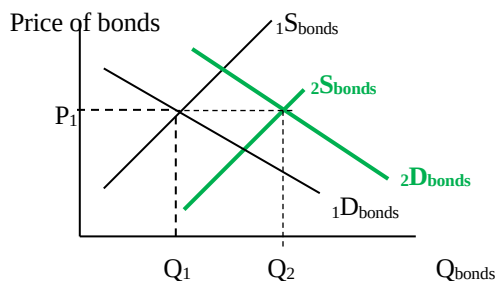
Why is the price of bonds not listed as a shift variable? Answer: a change in the price of bonds is represented as a movement along a given curve, not a shift.

Examples

1. Suppose EFPI increases. The increase in the EFPI means an increase in the supply of bonds (or, same thing, an increase in the demand for LF) as firms would find it more desirable to borrow to finance these better investment projects. The result is a decrease in the price of bonds or, same thing, an increase in the interest rate.



Example #2. Suppose the relative liquidity of a bond increases. At the same time suppose there is an increase in the government budget deficit. The increase in relative liquidity causes the demand for bonds to increase (rightward shift in bond demand curve). The increase in the budget deficit means a rightward shift on bond supply. By itself, a rightward shift in bond demand implies an increase in bond price (or from the LF market perspective, a decrease in the interest rate). By itself, the second shock causes a rightward shift in the bond supply curve which implies a decrease in the bond price (increase in interest rate). Since both happen at the same time, the impact on the bond price and interest rate is uncertain. However, equilibrium quantity of bonds definitely rises.



Example #3: Using example #2, what you have to assume to get a result that interest rates fall?

Falling interest rates mean rising bond prices. To get the result that bond prices rise, we have to assume that the shift in demand for bonds was stronger than the shift in the supply of bonds.

Example #4: Suppose you witness a sharp rise in bond prices. You also know that the EFPI rose (assume no other change in supply side shift variables). How would you possibly explain this outcome? We know that bond supply increased which—by itself—would mean a fall in bond prices. Thus, in order to generate an outcome in which bond prices rose we would have to assume something caused bond demand to rise and by a magnitude larger than the supply shift. We would then look at our list of candidates on the bond side to see what would generate this type of shift: increase in wealth, a lower than expected FUTURE return on bonds (equivalent to saying you expect future bond prices to rise), an increase in relative liquidity, a decrease in relative risk.

Term Structure of Interest Rates

Symbols

i_{ij} = annual yield on j^{th} -year bond purchased at time i . For example:

i_{01} = annual yield on 1-year bond purchased today

i_{02} = annual yield on 2-year bond purchased today.

i_{03} = annual yield on 3-year bond purchased today.

The i_{0n} are sometimes called the spot rates. These rates are observable.

Forward rates

i_{11} = expected annual yield on 1-year bond purchased one year from today (known as the forward rate)

i_{21} = expected annual yield on 1-year bond purchased two years from today.

i_{31} = expected annual yield on 1-year bond purchased three years from today.

These rates are not directly observable but can be inferred.

Expectations hypothesis: purchase a two-year bond today (call this option a) is a perfect substitute for purchasing two consecutive one-year bonds (call this option b) :

Option A: if invest \$1 in this option, after two years you would end up with

$$\$1(1+i_{02})(1+i_{02}) = \$1(1+i_{02})^2.$$

Option B: If invest \$1 in this option, after two years end up with (actually, expect to end up with is a better way of putting it): $\$1(1+i_{01})(1+i_{11})$

Equilibrium.

The Expectation hypothesis claims that since these options are perfect substitutes, then in equilibrium we should see:

$(1+i_{02})^2 = (1+i_{01})(1+i_{11})$ sometimes written as $1+i_{02} = [(1+i_{01})(1+i_{11})]^{0.5}$ (the righthand side is called the geometric mean)

Logic of Equilibrium.

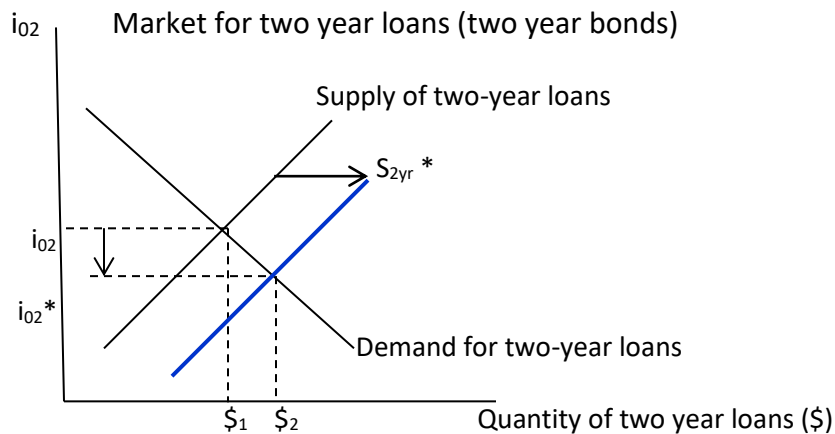
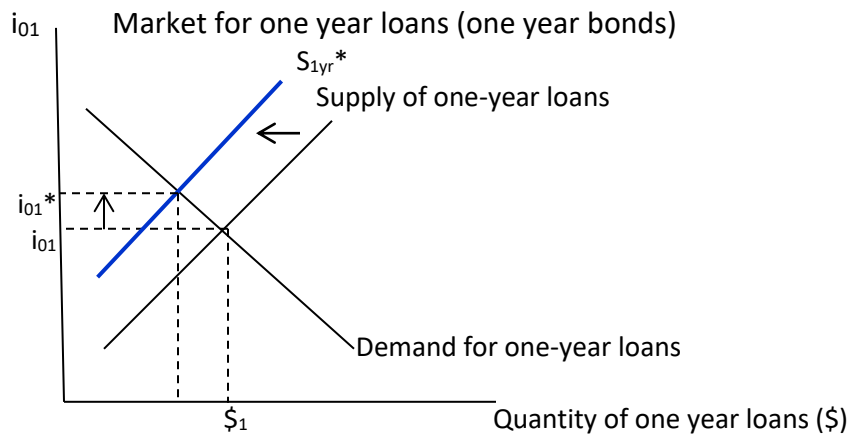
Why should this be considered an equilibrium? Suppose $i_{02} = 10\%$, $i_{01} = 8\%$ and $i_{11} = 11\%$.

If choose option A: $\$1(1+i_{02})^2 = \$1(1.1)^2 = \$1.21$ at end of 2 years.

If choose option B: $\$1(1+i_{01})(1+i_{11}) = \$1(1.08)(1.11) = \$1.19$ at end of 2 years.

So you get a better deal by choosing option A where we see $(1+i_{02})^2 > (1+i_{01})(1+i_{11})$

However, this will result in the following change in behavior ("market forces") in the various loanable funds markets. The supply of loans in the one-year loan market would decrease (i.e., the demand for one year bonds would decrease) and the supply of loans in the two year loan market would increase (i.e. the demand for 2-year bonds would rise). The resulting shifts in supply would in turn drive up the return on one-year loans and drive down the return on two-year loans. $(1+i_{02})^2 \downarrow > [(1+i_{01})(1+i_{11})] \uparrow$ until $(1+i_{02})^2 = (1+i_{01})(1+i_{11})$. Notice that this is just a variation on the Law-of-One-Price story.



Estimating Forward Rate.

Notice that information that can be collected currently on i_{02} and i_{01} can be used to infer **the forward return** (assuming the Expectations Theory is correct). For example, suppose that $i_{02} = 3\%$ and $i_{01} = 1\%$. We want to find i_{11} .

$(1 + i_{02})^2 = (1 + i_{01})(1 + i_{11}) \rightarrow (1 + i_{02})^2 / (1 + i_{01}) = (1 + i_{11})$
 $(1.03)^2 / (1.01) = 1.0609 / 1.01 = 1.0504 = (1 + i_{11})$ or $i_{11} = 5.04\%$. Thus, we expect the short term rate on one year bonds to rise from 1% today to 5.04% in one year.

We can solve for various forward rates. As another example, we can find the expected one year yield on a bond purchased two years from now by:

$(1 + i_{03})^3 = (1 + i_{01})(1 + i_{11})(1 + i_{21})$ divided by $(1 + i_{02})^2 = (1 + i_{01})(1 + i_{11})$ equals:

$(1 + i_{03})^3 / (1 + i_{02})^2 = (1 + i_{21})$. The terms on the left-hand side are all observable (check out the financial section).

$(1 + i_{04})^4 / (1 + i_{03})^3 = (1 + i_{31})$ etc. Generally, $(1 + i_{0n})^n / (1 + i_{0,n-1})^{n-1} = (1 + i_{n-1,1})$

$(1 + i_{03})^3 / (1 + i_{02})^2 = (1 + i_{21})$

Expectations Theory combined with Fisher equation always to deduce expected inflation rates: Recall that the Fisher equation is $i = r + \pi^e$. If we can assume the real rate is relatively constant, then this expectation of a higher one year yield may reflect people's expectation of higher inflation in the future. For example: $i_{01} = r + \pi_{01} \rightarrow 1\% = 1\% + 0\%$ (assumed r was 1% and expected inflation was 0%).

If r is constant, then $i_{11} = r + \pi_{11} \rightarrow 5\% = 1\% + \pi_{11} \rightarrow \pi_{11} = 4\%$.

In short, the Expectations Theory allows us to estimate then i_{11} , i_{21} , i_{31} etc. from the observable date of i_{01} , i_{02} , i_{03} etc. Then, once we have estimated then i_{11} , i_{21} , i_{31} ... we can use the Fisher equation to estimate π_{11} , π_{21} , π_{31} ASSUMING we know r (or r today and assuming r does not change)

Suppose we want to compare purchase of a 3-year bond purchased today with three one-year bonds purchased consecutively. In equilibrium:

$$(1 + i_{03})^3 = (1 + i_{01})(1 + i_{11})(1 + i_{21}) \text{ or } 1 + i_{03} = [(1 + i_{01})(1 + i_{11})(1 + i_{21})]^{1/3}.$$

Stated more generally

$$(1 + i_{0n})^n = (1 + i_{01})(1 + i_{11})(1 + i_{21}) \dots (1 + i_{n-1,1}) \text{ or } 1 + i_{03} = [(1 + i_{01})(1 + i_{11})(1 + i_{21}) \dots (1 + i_{n-1,1})]^{1/n}.$$

Approximate version of the model:

Return to $(1 + i_{02})^2 = (1 + i_{01})(1 + i_{11})$ which can be written as

$$1 + 2i_{02} + i_{02}^2 = 1 + i_{01} + i_{11} + i_{01} \cdot i_{11}$$

The cross product terms on the end of each side are likely to be small (especially nowadays). So they can be dropped leaving us with the following approximate version of the theory:

$$1 + 2i_{02} = 1 + i_{01} + i_{11} \rightarrow 2i_{02} = i_{01} + i_{11} \rightarrow i_{02} = (i_{01} + i_{11})/2$$

More generally, the approximation version is

$$i_{0n} = (i_{01} + \dots + i_{n-1,1})/n$$

Liquidity Premium Theory

Options A and B are NOT perfect substitutes.

Why these options may not be perfect substitutes.

Consider two bonds that pay interest annually. Each has coupon rate of 12% and a face value of \$1000. One bond is 5 years from maturity and the other is 10 year to maturity. Also consider a perpetuity that pays \$120 per year forever.

For each bond, $CPN = c \cdot F = .12 \cdot 1000 = \120 .

Suppose current **interest rate is 10%**.

For Bond 1: $P_{\text{bond}} = 120/1.1 + 120/(1.1)^2 + 120/(1.1)^3 + 120/(1.1)^4 + 120/(1.1)^5 + 1000/(1.1)^5 = \$1,075.81$.

For Bond 2: $P_{\text{bond}} = 120/1.1 + 120/(1.1)^2 + 120/(1.1)^3 + 120/(1.1)^4 + \dots + 120/(1.1)^{10} + 1000/(1.1)^{10} = \$1,122.89$

For Bond 3 (perpetuity) = $120/.10 = 1200$.

Now suppose the **interest rate rises to 15%**. What happens to each price?

For Bond 1 (5 year bond): $P_{\text{bond}} = 120/1.15 + 120/(1.15)^2 + 120/(1.15)^3 + 120/(1.15)^4 + 120/(1.15)^5 + 1000/(1.15)^5 = \899.44 .

For Bond 2 (10 year bond): $P_{\text{bond}} = 120/1.15 + 120/(1.15)^2 + 120/(1.15)^3 + 120/(1.15)^4 + \dots + 120/(1.15)^{10} + 1000/(1.15)^{10} = 849.44$

For Bond 3 (perpetuity) = $120/.15 = \$800$.

For bond 1, the decline in price from 1075 to 899 represents a 16.4% decline.

For bond 2, the decline in price from 1122 to 849 represents a 24% decline.

For bond 3, the decline in price from 1200 to 800 represents a 33% decline.

FV=1000,c=12%	At i= 10%	At i= 15%	% drop in price	
5 year bond	\$1,075.81	\$899.44	16.4% drop	
10 year bond	\$1,122.89	\$849.44	24% drop	
Perpetuity	\$1,200	\$800	33% drop	

This fluctuation in bond prices due to interest changes is known as **interest rate risk**.

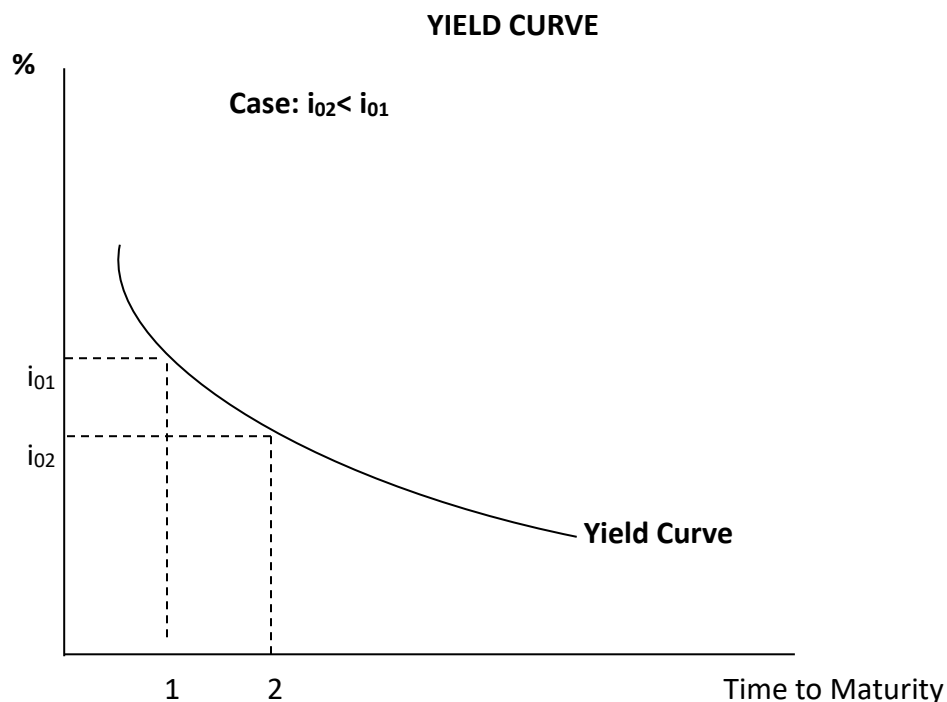
Conclusion: longer maturity bonds are more price sensitive to a given change in the interest rate compared to the bonds with short maturities. That is, long maturity bonds have greater interest rate risk. This has implications for our theory of the term structure of interest rates. Under the expectations hypothesis, we assumed option A and option B were perfect substitutes. But this analysis on interest rate risk implies that they are not. For example, suppose this was initially the case:

$(1+i_{02})^2 = (1+i_{01})(1+i_{11})$. But if the first option (left hand side) is considered riskier, then lenders may move out that option, so the supply of two-year loans drops, causing i_{02} to rise. These lenders would have an incentive to move into the option b (right hand side), thus increasing the supply of one-year loans, causing i_{02} to fall. This implies we end up with an equilibrium in which:

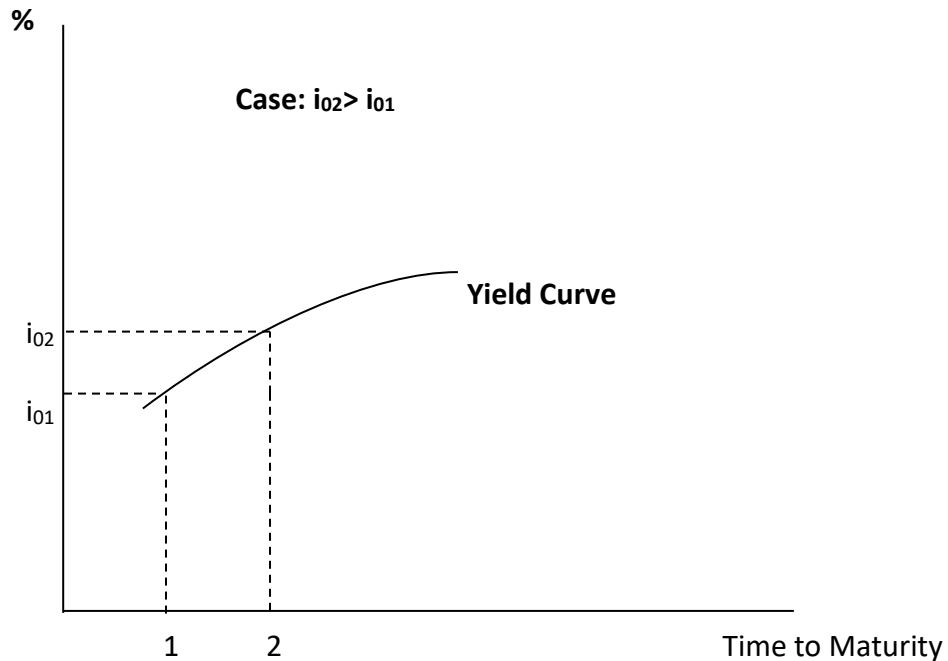
$(1+i_{02})^2 > (1+i_{01})(1+i_{11})$. Suppose the risk premium (basically the gap between these two alternatives) is given by η with this being a positive number. Therefore in equilibrium:

$(1+i_{02})^2 = (1+i_{01})(1+i_{11}) + \eta$. This is known as the Liquidity Premium Theory.

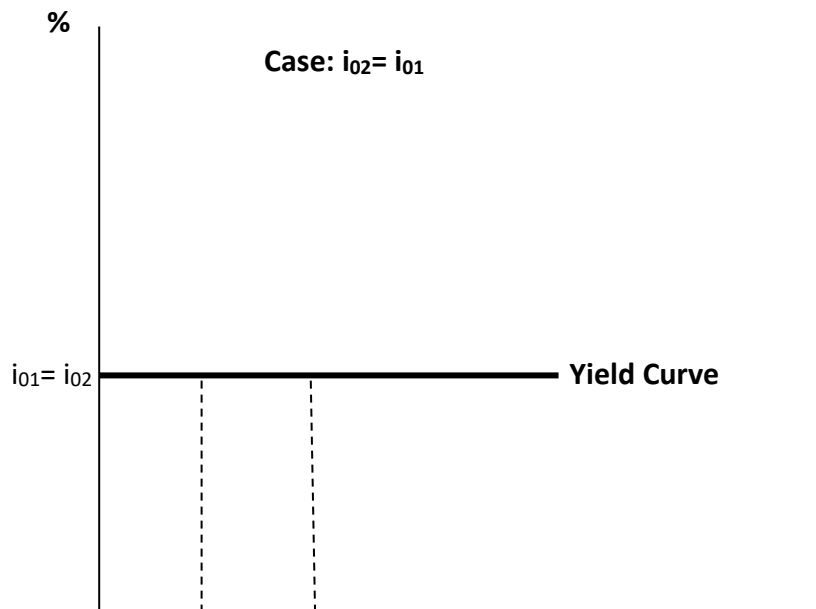
The approximate version of the model is $i_{02} = [(i_{01} + i_{11})/2] + \eta$.



Under Expectations Theory (approx. version) $\rightarrow i_{02} = (i_{01} + i_{11})/2$ but given $i_{02} < i_{01}$ we can deduce that $i_{11} < i_{01}$ (short term rates are expected to fall next year). If recessions tend to correspond to lower nominal S-T rates then the negative slope Y.C. might be evidence of an impending recession.



Under Expectations Theory (approx. version) $\rightarrow i_{02} = (i_{01} + i_{11})/2$ but given $i_{02} > i_{01}$ we can deduce that $i_{11} > i_{01}$ (short term rates are expected to rise next year).

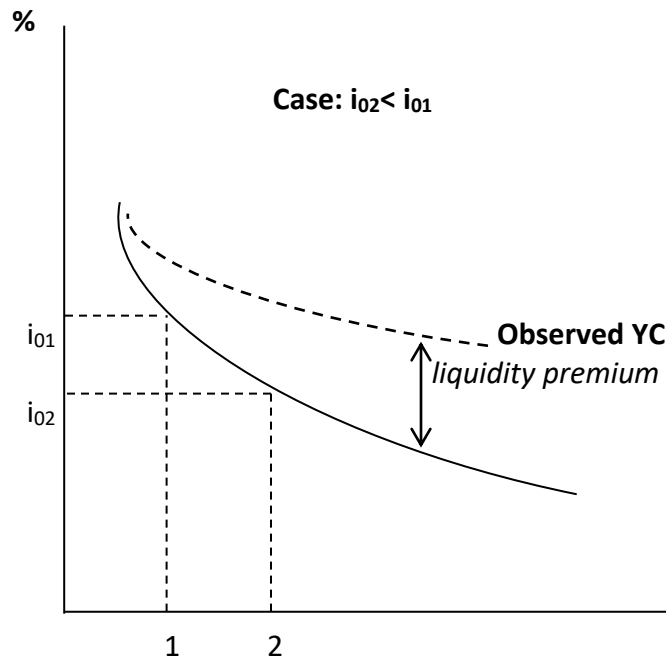


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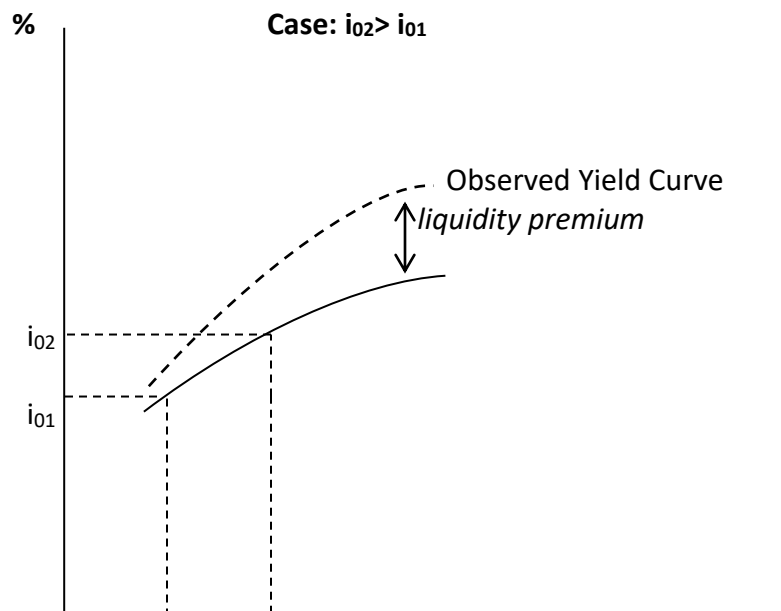
Time to Maturity

Under Expectations Theory (approx. version) $\rightarrow i_{02} = (i_{01} + i_{11})/2$ but given $i_{02} = i_{01}$ we can deduce that $i_{11} = i_{01}$ (short term rates are not expected to change).

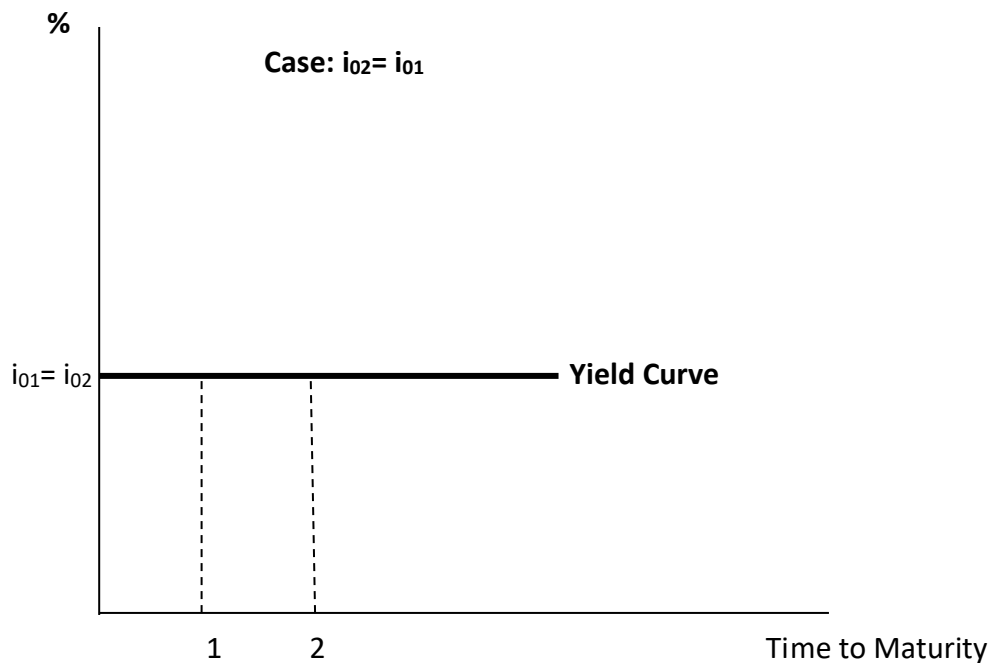
YIELD CURVE & LIQUIDITY PREMIUM CURVE



If observed is downward sloping, then the “unbiased” curve must also be downward sloping. The approximate version of the LPT is $i_{02} = [(i_{01} + i_{11})/2] + \eta$. It is given (that is, it is observable from the YC) that $i_{02} < i_{01}$, so for example, suppose $i_{02} = 6\% < i_{01} = 7\% \rightarrow 6 = [(7 + i_{11})/2] + \eta$, or rearranged, $6 - (+) = [(7 + i_{11})/2] \rightarrow$ left-hand side is less than 6, thus the righthand side must be less than 6. The only way for that to be true is if i_{11} is less than 7 $\rightarrow$ so must be case $i_{11} < i_{01}$.



If observed is upward sloping, then “unbiased” curve could be either upward, horizontal or downward sloping. The approximate version of the LPT is $i_{02} = [(i_{01} + i_{11})/2] + \eta$. It is given (that is, it is observable from the YC) that $i_{02} > i_{01}$, so for example, suppose $i_{02} = 8\% > i_{01} = 7\% \rightarrow 8 = [(7 + i_{11})/2] + \eta$, or rearranged, $8 - (+) = [(7 + i_{11})/2] \rightarrow$ lefthand side is less than 8, thus the righthand side must be less than 8. But this is *consistent with either $i_{11} < i_{01}$ or $i_{11} > i_{01}$ or $i_{11} = i_{01}$* . [see if you can provide numerical examples of each.] Thus, if the YC slopes upward, then under the LPT (assuming we do not know the size of the risk premium) we cannot deduce what will happen to the forward rate.

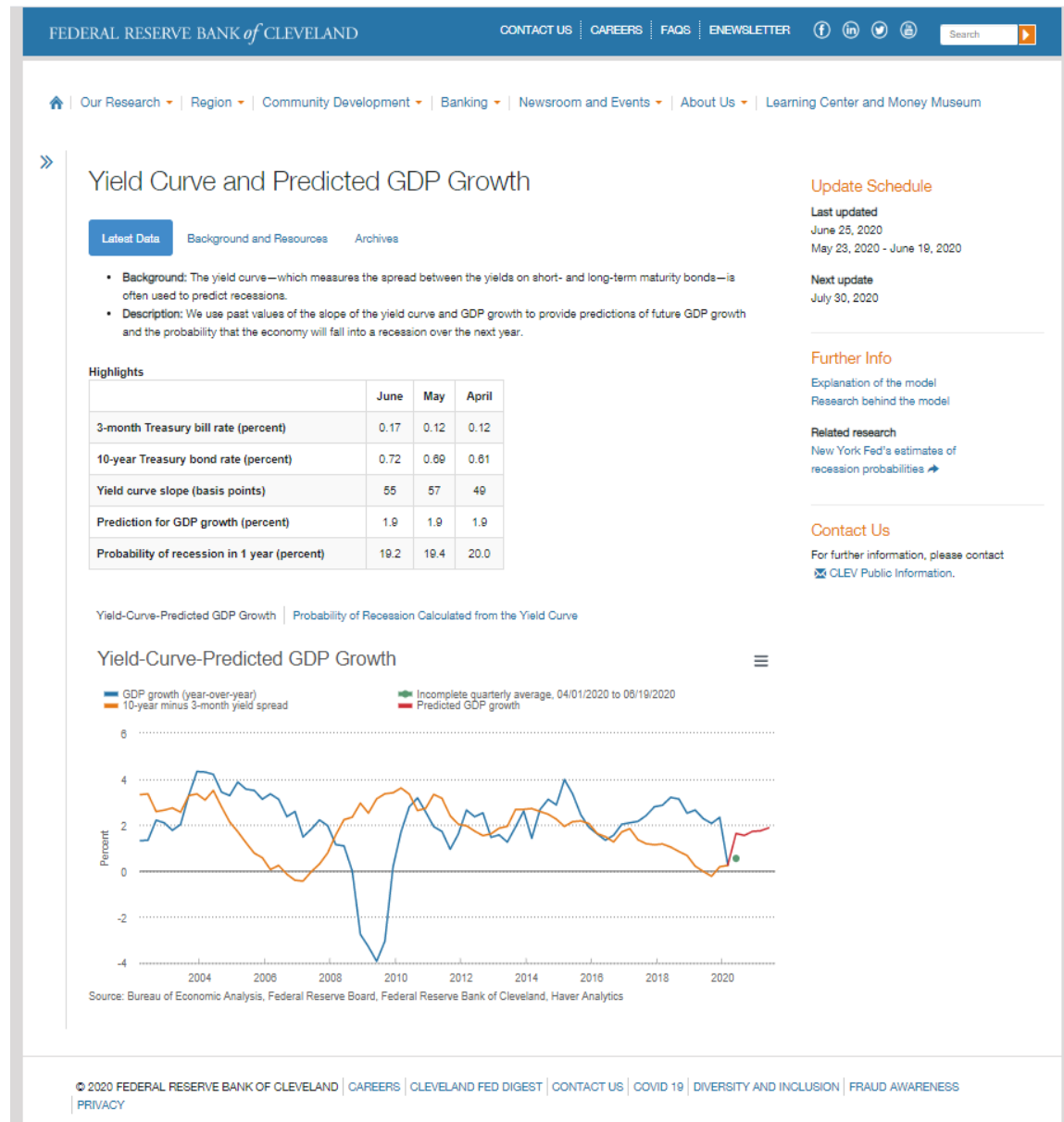


The approximate version of the LPT is $i_{02} = [(i_{01} + i_{11})/2] + \eta$. It is given (that is, it is observable from the YC) that $i_{02} = i_{01}$, so for example, suppose $i_{02} = 7\% = i_{01} = 7\% \rightarrow 7 = [(7 + i_{11})/2] + \eta$, or rearranged, $7 - (+) = [(7 + i_{11})/2] \rightarrow$ lefthand side is less than 7, thus the righthand side must be less than 7. The only way for that to be true is if i_{11} is less than 7 $\rightarrow$ so must be case $i_{11} < i_{01}$.

Using Yield Curve to predict recessions

The yellow line below is, in our notation $i_{010} - i_{01}$. So if $i_{010} - i_{01} > 0 \rightarrow i_{010} > i_{01}$, an upward sloping yield curve. If $i_{010} - i_{01} < 0 \rightarrow i_{010} < i_{01}$, a downward sloping yield curve. If $i_{010} - i_{01} = 0 \rightarrow i_{010} = i_{01}$, a horizontal yield curve.

NOTE: 100 basis points is 1%, 50 basis points is 0.5%, etc.



Money supply and the Fed

1. Definitions and data
2. Institutional details
3. Simplified banking system model
4. Advanced model
5. Components of model
6. Tool of monetary management

1. Definitions

Money is defined by its functions (“money is what money does”).

The functions of money are:

- a. **medium of exchange**—refers to thing *commonly* used to make purchases and pays debts.
- b. **unit of account**—the item we quote prices in. Goods and services are quoted in terms of dollars per unit, not, say, apples per unit. If there are **n goods**, the number of independent relative prices is given by $[n(n-1)]/2$ whereas in a money economy there are n money prices.
- c. **store of value-the unit(s)** we store our wealth. Since receipts of income and purchases of income are typically not perfectly synchronized, we need a means by which store our purchasing power. Money is one asset that enables us to do so (During periods of high inflation rate, money is not a good store of value).

Operational definitions of money

M1 consists of (1) currency outside the U.S. Treasury, Federal Reserve Banks, and the vaults of depository institutions; (2) traveler's checks of nonbank issuers; (3) demand deposits at commercial banks (excluding those amounts held by depository institutions, the U.S. government, and foreign banks and official institutions) less cash items in the process of collection and Federal Reserve float; and (4) other checkable deposits (OCDs), consisting of negotiable order of withdrawal (NOW) and automatic transfer service (ATS) accounts at depository institutions, credit union share draft accounts, and demand deposits at thrift institutions. Seasonally adjusted M1 is constructed by summing currency, traveler's checks, demand deposits, and OCDs, each seasonally adjusted separately.

More simply, $M1 = C + D$, where C is currency in the hands of non-bank public and D is demand deposits (“checkable deposits”).

M2 consists of M1 plus (1) **savings deposits** (including money market deposit accounts); (2) small-denomination **time deposits** (time deposits in amounts of less than \$100,000), less individual retirement account (IRA) and Keogh balances at depository institutions; and (3) balances in retail money market mutual funds, less IRA and Keogh balances at money market mutual funds. Seasonally adjusted M2 is constructed by summing savings deposits, small-denomination time deposits, and retail money funds, each seasonally adjusted separately, and adding this result to seasonally adjusted M1.

Data can be found at <https://www.federalreserve.gov/releases/h6/current/default.htm>

Date	Seasonally adjusted	
	M1 ¹	M2 ²
June 2018	3,657.5	14,120.0
July 2018	3,677.0	14,153.0
Aug. 2018	3,686.3	14,197.0
Sept. 2018	3,703.9	14,228.5
Oct. 2018	3,719.1	14,235.4
Nov. 2018	3,698.0	14,245.4
Dec. 2018	3,746.4	14,351.8
Jan. 2019	3,740.5	14,434.6
Feb. 2019	3,759.7	14,464.4
Mar. 2019	3,729.9	14,511.8
Apr. 2019	3,781.0	14,558.3
May 2019	3,792.4	14,653.2
June 2019	3,832.8	14,780.7
July 2019	3,858.2	14,859.9
Aug. 2019	3,853.3	14,931.2
Sept. 2019	3,903.2	15,020.8
Oct. 2019	3,922.8	15,147.1
Nov. 2019	3,947.3	15,247.5
Dec. 2019	3,976.8	15,302.4
Jan. 2020	3,976.9	15,417.9
Feb. 2020	4,009.8	15,511.5
Mar. 2020	4,268.2	16,103.7

Date	Seasonally adjusted	
	M1 ¹	M2 ²
Apr. 2020	4,819.0	17,176.9
May 2020	5,062.6	18,044.0

[Make Full Screen](#)

Percent change at seasonally adjusted annual rates
3 Months from Feb. 2020 TO May 2020
6 Months from Nov. 2019 TO May 2020
12 Months from May 2019 TO May 2020

2.NOTE: Some of the institutional detail about the Fed that follows is optional reading, I am mainly concerned with what we covered in class (see yellow highlights).

Structure and Functions of the Federal Reserve System (the Fed). The Federal Reserve System is the *central bank* of the United States. It was founded by Congress in 1913. Over the years, its role in banking and the economy has greatly expanded.

The Federal Reserve has several functions:

- a. Monetary Policy (setting the money supply, the most important function),
- b. Banking Regulation (other institutions also regulate banks—more on this later),
- c. helps to clear checks,
- d. serves as the bank for the U.S. Treasury, and
- e. other.

Control of the Fed resides with the **Board of Governors**, located in Washington, D.C. The Board of Governors, also known as the Federal Reserve Board, is the national component of the Federal Reserve System. The board consists of the seven governors, appointed by the president and confirmed by the Senate. Governors serve 14-year, staggered terms to ensure stability and continuity over time. The chair (currently, Jerome Powell) and vice-chairman are appointed to four-year terms and may be reappointed subject to term limitations.

Other info on Fed: The Board also exercises broad supervisory control over the financial services industry, administers certain consumer protection regulations, and oversees the nation's payments system (such as clearing checks). The Board oversees the activities of District Reserve Banks, approving the appointments of their presidents and some members of their boards of directors. The Board sets reserve requirements for depository institutions and approves changes in discount rates recommended by Reserve Banks. The Board's most important responsibility is participating in the **Federal Open Market Committee (FOMC)**, which conducts our nation's monetary policy; the seven governors comprise the voting majority of the FOMC with the other five votes coming from Reserve Bank presidents. Board members are called to testify before Congress, and they maintain regular contact with other government

organizations as well. The chairman reports twice a year to Congress on the Fed's monetary policy objectives, testifies on numerous other issues, and meets periodically with the Secretary of the Treasury.

Federal District Reserve Banks: A network of 12 Federal Reserve Banks and 24 branches make up the Federal Reserve System under the general oversight of the Board of Governors. Reserve Banks are the operating arms of the central bank. Each of the 12 Reserve Banks serves its region of the country, and all but three have other offices within their Districts to help provide services to depository institutions and the public. The Banks are named after the locations of their headquarters - Boston, New York, Philadelphia, Cleveland, Richmond, Atlanta, Chicago, St. Louis, Minneapolis, Kansas City, Dallas and San Francisco. The Reserve Banks serve banks, the U.S. Treasury, and, indirectly, the public. A Reserve Bank is often called a "banker's bank," storing currency and coin, and processing checks and electronic payments. Reserve Banks also supervise commercial banks in their regions. As the bank for the U.S. government, Reserve Banks handle the Treasury's payments, sell government securities and assist with the Treasury's cash management and investment activities. Reserve Banks conduct research on regional, national and international economic issues. Research plays a critical role in bringing broad economic perspectives to the national policymaking arena and supports Reserve Bank presidents who all attend meetings of the Federal Open Market Committee (FOMC). Each Reserve Bank's board of directors oversees the management and activities of the District bank. Each board appoints the president and first vice president of the Reserve Bank, subject to the approval of the Board of Governors. All member banks hold stock in Reserve Banks and receive dividends. Unlike stockholders in a public company, banks cannot sell or trade their Fed stock. Reserve Banks interact directly with banks in their Districts through examinations and financial services and bring important regional perspectives that help the entire Federal Reserve System do its job more effectively. **"Member" Banks:** Approximately 38 percent of the 8,039 commercial banks in the United States are members of the Federal Reserve System. National banks must be members; state-chartered banks may join if they meet certain requirements. The member banks are stockholders of the Reserve Bank in their District and as such, are required to hold 3 percent of their capital as stock in their Reserve Banks.

Other Depository Institutions: In addition to the approximately 3,000 member banks, about 17,000 other depository institutions provide the American people checkable deposits and other banking services. These depository institutions include nonmember commercial banks, savings banks, savings and loan associations, and credit unions. Although not formally part of the Federal Reserve System, these institutions are subject to System regulations, including reserve requirements, and have access to System payments services.

Federal Open Market Committee: The Federal Open Market Committee, or FOMC, is the Fed's monetary policymaking body. It is responsible for formulation of a policy designed to promote stable prices and economic growth. Simply put, the FOMC manages the nation's money supply. The voting members of the FOMC are the Board of Governors, the president of the Federal Reserve Bank of New York and presidents of four other Reserve Banks, who serve on a rotating basis. All Reserve Bank presidents participate in FOMC policy discussions. The chairman of the Board of Governors chairs the FOMC. The FOMC typically meets eight times a year in Washington, D.C. At each meeting, the committee discusses the outlook for the U.S. economy and monetary policy options. The FOMC is an example of the interdependence built into the Fed's structure. It combines the expertise of the Board of Governors and the 12 Reserve Banks.

Advisory Councils: Three statutory advisory councils - the Federal Advisory Council, the Consumer Advisory Council, and the Thrift Institutions Advisory Council - advise the Board on matters of current interest. These councils, whose members are drawn from each of the 12 Federal Reserve Districts, meet two to four times a year. The individual Reserve Banks have advisory committees as well, including thrift

institutions advisory committees, small business and agricultural advisory committees. Moreover, officials from all Reserve Banks meet periodically in various committees.

Stylized Balances Sheets

Note: Assets $\equiv$ Liabilities + Net Worth $\rightarrow$ NW = assets – liab.

Liabilities—claims on the firm or individual by outside members or non-owners.

Fed

Assets		Liabilities and Net Worth	
U.S. Treasury Bonds	\$4.2 Trillion	Federal Reserve Notes	\$2.0 Trillion
Mortgage-backed Securities	\$1.9 Trillion	Reserve Deposits	\$2.0 Trillion
Other		Other	

(Reserve Deposits: deposits that banks have at the Fed)

Fed and US Treasury are NOT the same institution

Commercial Bank

Assets		Liabilities and Net Worth	
Reserves	\$1.5 Trillion	Deposits	\$15.7 Trillion (apprx 20% checkable dep.)
Securities and Loans	\$16 Trillion	Other (Borrowings)	\$2.3 Trillion
other	\$2.5T	New Worth	\$2 Trillion
Total	20 Trillion	Liab + NW	\$20 Trillion

Reserves = vault cash + reserve deposits

<https://www.federalreserve.gov/releases/efa/consolidated-bank-balance-sheet-data.htm>

3. Assumptions of Simplified Banking System.

1. a. Banks only assets are reserves (see discussion below) and loans; only liability is demand deposits (checking accounts); NW is zero.
- b. For the Fed, only asset is US Treasury bonds and only liability is reserve deposits. Net worth is zero.
2. All purchases are made by checks; Any check received is deposited in bank.
3. Banks do not desire to hold excess reserves. Banks reduced excess reserves by expanding loans.
4. Required Reserve ratio (*rr*) is 10% on all Demand Deposits.

Discussion on actual reserves, required reserves and excess reserves.

Actual reserves = vault cash + reserve deposits. Reserve deposits are deposits banks have at the Fed (the Fed serves as a banker's bank in this sense). The Fed requires that banks hold a certain minimum level of reserves. Required reserves are determined by the required reserve ratio (*rr*, set by the Fed) and the level of bank deposits (*D*). Required reserves = (*rr*)(*D*) where *D* is demand deposits.

So, for example, if *rr* = 20% and *D* = \$1 Billion, then required reserves are equal to .20(\$1 Billion) = \$200 million. Excess reserves (*ER*) = actual reserves minus required reserves.

EXAMPLE of the money creation process (traditional approach)

Step 1. Now assume the Fed purchases \$100,000 worth of government bonds from the non-bank public. It does so by writing a check on itself. This check, in turn, is deposited by the member of the non-bank public in Bank A. Bank A returns the check to the Fed for payment. The Fed payoffs on the check by transferring reserves to Bank A.

Immediately after Step 1.

Bank A		Fed	
<u>Assets</u>	<u>Liabilities</u>	<u>Assets</u>	<u>Liabilities</u>
Reserves (R)+100,000 (1)	Deposits (D)+100,000 (1)	US T-Bonds 100,000	Reserves 100,000
Loans (L) 0			

Step 2. How do these changes impact Bank A? Need to calculate excess reserves to answer this.

$R = 100,000$, $RR = rr * D = .10 * 100,000 = 10,000$, $ER = 100,000 - 10,000 = 90,000$. Bank A expands loans by 90,000 by creating checking account for non-bank public member (person taking out the loan). This leads to

Bank A	
<u>Assets</u>	<u>Liabilities</u>
Reserves (R)+100,000	Deposits (D)+100,000
	+90,000 (2)
Loans (L) +90,000 (2)	

Step 3. Suppose this person X uses the proceeds of the loan to purchase \$90,000 worth of goods from firm Y. Firm Y deposits the check at Bank B. Bank B returns check to Bank A for payment. Bank A pays off by transferring reserves to Bank B. Thus, we know have:

Bank A		Fed	
<u>Assets</u>	<u>Liabilities</u>	<u>Assets</u>	<u>Liabilities</u>
Reserves (R)+100,000	Deposits (D)+100,000	US T-Bonds 100,000	Reserves +100,000
-90,000 (3)	+90,000		
	-90,000 (3)		
Loans (L) 0			
+90,000			

Immediately after these transactions for Bank A: $R = 10,000$, $rr * D = .1 * 100,000 = 10,000$, and $ER = 0$. Bank A can no longer expand loans. Done with A.

Bank B	
<u>Assets</u>	<u>Liabilities</u>
Reserves (R)+90,000 (3)	Deposits (D)+90,000 (3)
Loans (L) 0	

For Bank B: $R = 90,000$, $rr * D = .10 * 90,000 = 9,000$ and $ER = 81,000$. Expands loans by \$81,000.

Story repeats itself, although with smaller numbers at each step (for example, 81K now rather than 90K).

Creation of money (demand deposits) = 100,000 (step 1) + 90,000 (step 2) + 81,000 (step 4, not shown, but just repeats step 2, only now for B instead of A, and for 81K instead of 90K) +.....

= **total increase in D.**

$$= 100,000 + 100,000 (1 - rr) + 100,000 (1 - rr)^2 + 100,000 (1 - rr)^3 + \dots = \text{total increase in D}$$

$$= 100,000 [1 + (1 - rr) + (1 - rr)^2 + (1 - rr)^3 + \dots] = \text{total increase in D.}$$

$$= 100,000 [1/rr] = \text{total increase in D} \rightarrow 100,000 [1/.10] = \mathbf{1,000,000.}$$

More generally, $\Delta R * (1/rr) = \Delta D$ where $(1/rr)$ is known as the simple money multiplier. With $rr = 10\%$, then $100K * (1/.10) = 1 \text{ million.}$

4. Advanced Model of money creation.

We need to start with a definition: Monetary Base (MB) = C + R (not to be confused with the money supply as measured by M1, recall that $M1 = C + D$). The variable C is currency in the hands of the non-bank public. For our next step, need to modify the money multiplier.

Here are the steps:

a. By substitution for R: $MB = C + rr * D + ER$. Now, letting $c = C/D$ where c is the non-bank public's desired holding of currency expressed as a % of D and $e = ER/D$ where e represents the banking system's desired holdings of excess reserves expressed as a % of D, leads to the next step.

b. By rearranging and by substitution:

$$C = c * D \text{ and } ER = e * D \text{ and } MB = c * D + rr * D + e * D = D (c + rr + e).$$

c. Solving for D: $D = MB * [1/(c + rr + e)]$.

d. However, the money supply is $M1 = C + D$. This can be written as $M1 = c * D + D = D (1 + c)$.

e. Finally, by substitution we get

$$\mathbf{M1 = MB * (1 + c) / (c + rr + e), \text{ with mm = money multiplier} = (1 + c) / (c + rr + e).}$$

Currently, $rr = 0\%$ (as of March 2020), $c = 0.58$, $e = 0.9$ so $mm = 1.07$ (apprx)

5. Looking at each term in more detail:

Specifics on the currency ratio

The variable c is determined by the non-bank public's preferences to hold currency. Hence, this makes the job of the Fed more difficult. If its goal is to target M1, then factors in the mm that they do not control (such as c and e) will make this job more difficult.

Factors that influence c:

- i. *bank panics* (sudden rush to withdraw currency if lose confidence in banking system; this is less common nowadays since bank deposits are insured by the FDIC—see below for optional reading on the FDIC). Bank panics or bank runs increase c.
- ii. *level of illegal transactions* (tend to use cash not checks). As the level of illegal transactions rises, this will tend to cause c to rise. A testable hypothesis relies on the observation countries with high tax rates tend to witness a greater level of illegal (or unreported) transactions to avoid paying taxes. Would expect a higher proportion of cash transactions.

- iii. *return on alternative assets* (if the return on other assets rises, tend to hold less cash). As the return on other assets rises, would expect c to fall.
- iv. *payments technology* that impact the cost of obtaining currency (ATMs was such a change or introduction and diffusion of credit cards),
- v. *foreign demand for currency* (estimated that 50% of all US dollars held outside the US).

How a change in c impacts mm and therefore $M1$ will depend on if $mm > 1$ or $mm < 1$ to begin with. If $mm > 1$, then an increase in c will lower mm and therefore lower $M1$. If $mm < 1$, then get the opposite impact of an increase in c .

Specifics on the excess reserve ratio

The variable e is determined by banks.

Factors that influence e :

- i. *relative return on other assets compared to reserves*. As of 2008, the Fed is allowed to pay interest rate on reserves (i_{or}). Since the Federal Funds market represents the borrowing/lending of reserves between banks, it makes sense that the best alternative to holding reserves and receiving i_{or} would be to lend your reserves in the FF market and receive i_{FF} . Assuming $(i_{FF} - i_{or}) > 0$, this would represent the opportunity cost of holding reserves and therefore any increase in this term would presumably lower the demand to hold reserves. Thus, a rise in $(i_{FF} - i_{or})$ would cause e to fall.
- ii. *expectations of banks on future deposit withdrawals* (likely to hold more reserves if expect an increase in deposit withdrawals)
- iii. *expectations regarding the riskiness of loans* (hence during a financial crisis the expectation that risk on loans has increased is likely to increase e). Look at recent data below.

How a change in e impacts mm and therefore $M1$ is straightforward since this variable only shows up in the denominator of the mm . An increase in e results in a fall in mm and $M1$.

Specifics of the required reserve ratio (rr)

This is directly set by the Fed. An increase in rr reduces mm and $M1$.

As announced on March 15, 2020, the Board reduced reserve requirement ratios to zero percent effective March 26, 2020. This action eliminated reserve requirements for all depository institutions. The following content explains the Board's authority to impose reserve requirements and how reserve requirements were administered prior to the change in reserve requirement ratios to zero.

The Federal Reserve Act authorizes the Board to establish reserve requirements within specified ranges for purposes of implementing monetary policy on certain types of deposits and other liabilities of depository institutions.

Prior to the change effective March 26, 2020, reserve requirement ratios on net transactions accounts differed based on the amount of net transactions accounts at the depository institution. A certain amount of net transaction accounts, known as the "reserve requirement exemption amount," was subject to a reserve requirement ratio of zero percent. Net transaction account balances above the reserve requirement exemption amount and up to a specified amount, known as the "low reserve tranche," were subject to a reserve requirement ratio of 3 percent. Net transaction account balances above the low reserve tranche were subject to a reserve requirement ratio of 10 percent. The reserve

requirement exemption amount and the low reserve tranche are indexed each year pursuant to formulas specified in the Federal Reserve Act

Reserve Requirements

Liability Type	Requirement	
	% of liabilities	Effective date
Net transaction accounts ¹		
\$0 to \$16.9 million ²	0	1-16-20
More than \$16.9 million to \$127.5 million ³	0	3-26-20
More than \$127.5 million	0	3-26-20
Nonpersonal time deposits	0	12-27-90
Eurocurrency liabilities	0	12-27-90

Monetary Base (MB) and non-borrowed monetary base

The MB is not perfectly controlled by the Fed. Due to borrowing at the Discount Window (discussed below) some of the reserves injected into the banking system by the Fed are borrowed (BR = borrowed reserves) and although the Fed controls the Discount Rate, it cannot always determine exactly how many reserves banks will desire to borrow at any given Discount Rate. On the other hand, non-borrowed reserves are directly put into the banking system by Fed's purchase of bonds (OMO) and can be directly determined. Thus, the MB is broken down as follows for purposes of money supply creation analysis:

$MB_n = MB - BR$ where MB_n is referred to as the non-borrowed monetary base. However, in practice BR tends to be small relative to MB. Might also write as $MB = MB_n + BR$.

An increase in MB, other things constant, causes M1 rise.

Problems does the Fed encounter in its attempts to control the money supply.

The Fed has to be able to correctly anticipate changes in the variables it does not directly control (such as e , c , and BR) in order to undertake the proper offsetting policies. For example, if the Fed expects e to rise, and therefore the money multiplier and money supply to fall, then the Fed could take offsetting action by, say, engaging in an open market purchase (discussed below), or cutting the discount rate relative to other rates (discussed below), or reducing rr . Likewise, if it expects c to rise (assuming $m > 1$), and thus expects mm and $M1$ to fall it would likewise take offsetting action by again engaging in an open market purchase, or cutting the discount rate relative to other rates, or reducing rr . If the BR increased, *by itself* this would increase MB ($MB = MB_n + BR$) and thus $M1$. The Fed could take offsetting action by an open market sale (which lowers the non-borrowed monetary base component and thus leaves MB

unchanged), raise rr or raising the discount rate relative to other rates to discourage borrowing of reserves (this would cause BR to fall offsetting the initial rise in BR).

Summary

Variable	Controlled by	Impact on MB or mm?	Impact on M1
MB	Fed*	MB	MB rises $\rightarrow$ M1 Rises
rr	Fed	mm	cut in $rr \rightarrow$ mm rises $\rightarrow$ M1 rises
e	banks	mm	rise in $e \rightarrow$ mm falls $\rightarrow$ M1 falls
c	non-bank pub	mm	see note below

Note: Since c is in the numerator and denominator of the mm ratio, its impact on mm will depend on whether mm is initially greater than 1, less than 1, or equal to 1

If $mm > 1 \rightarrow$ then, say, increase in c will have a proportionately larger impact on the denominator, thus mm will fall and M1 will fall.

If $mm < 1$, then opposite result.

If $mm = 1$, then no change in mm . (in recent years, mm has been around 1 or little above 1)

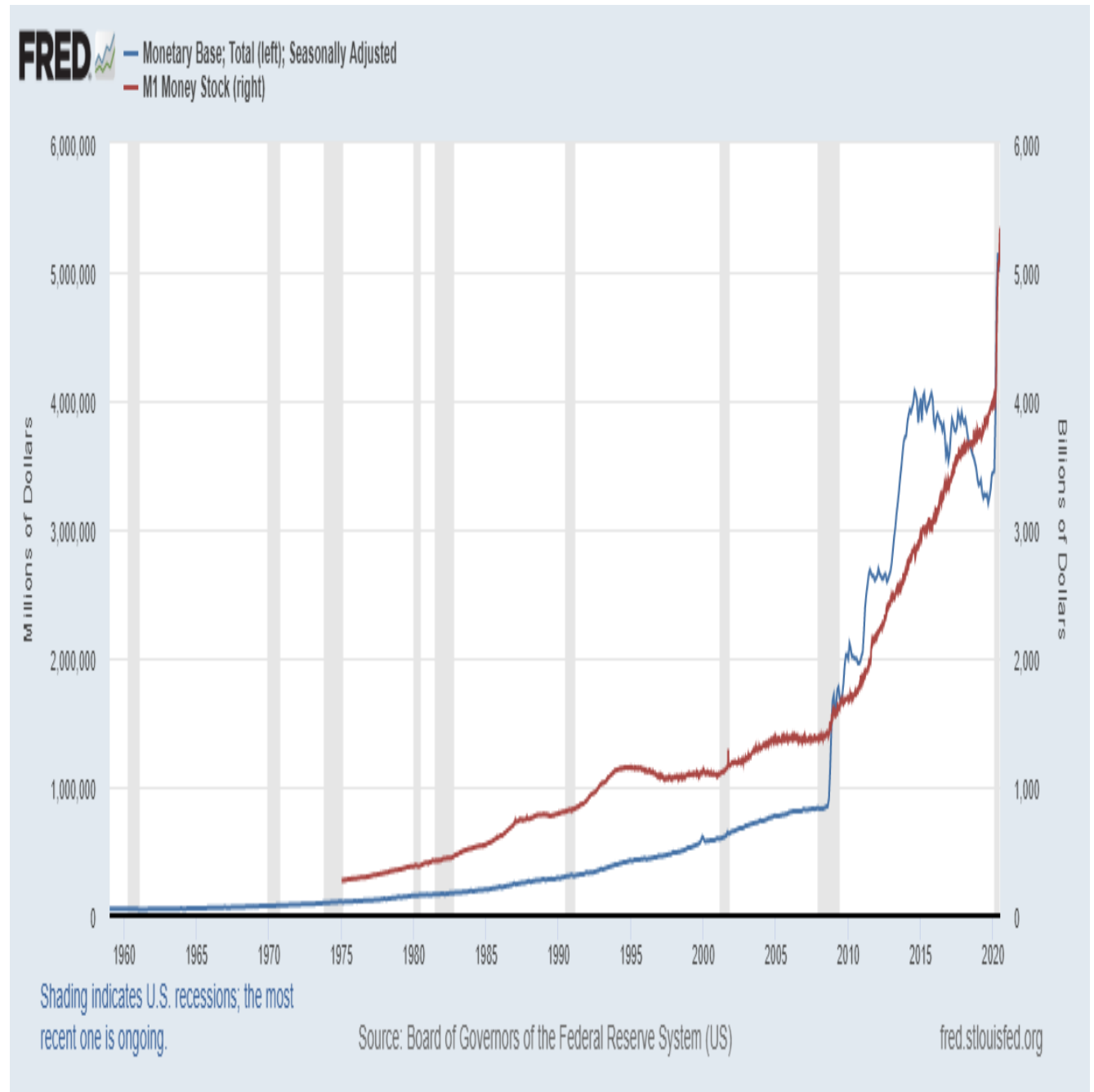
*we will later modify this statement

Data

Monetary Base. Notice the large increase in 2020.



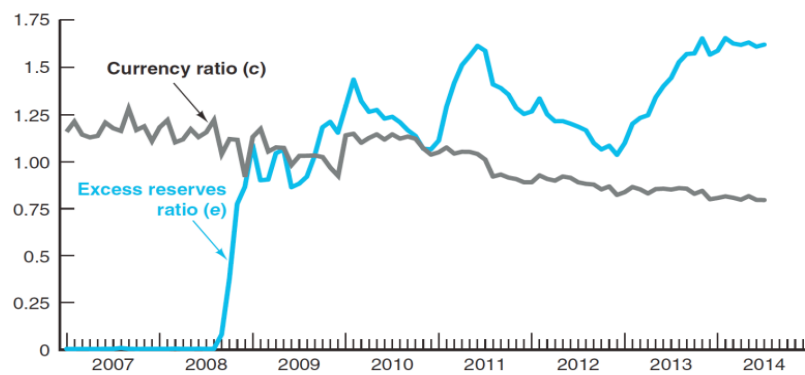
M1 and MB



Money multiplier (mm) = M1/MB. Large decline in 2020. Money multiplier is a bit above 1.

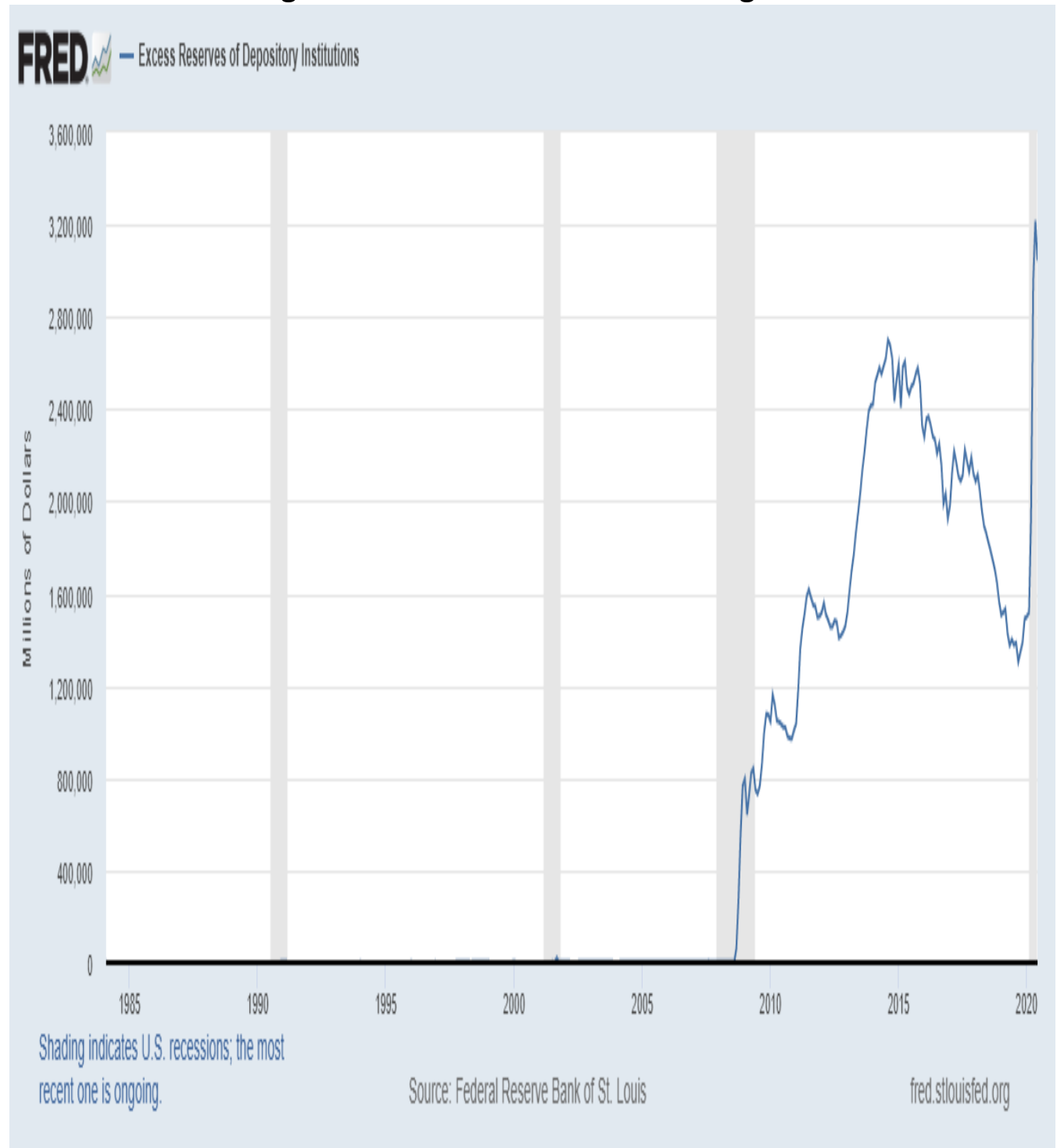


Historical data on c and e .



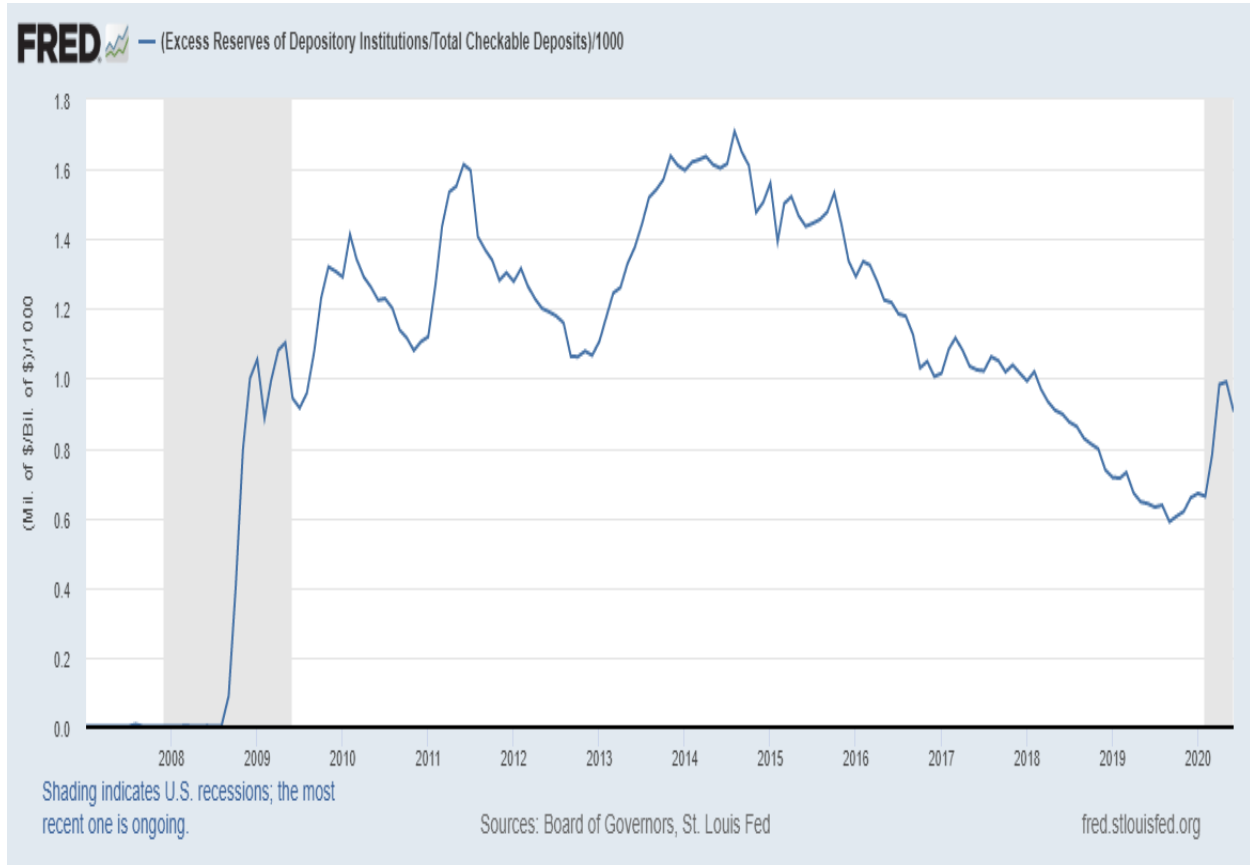
- No shift to currency (Difference to Great Depression: FDIC insurance.)
- Huge increase in excess reserves.

Excess reserves. Large increase excess reserve holdings in 2020.

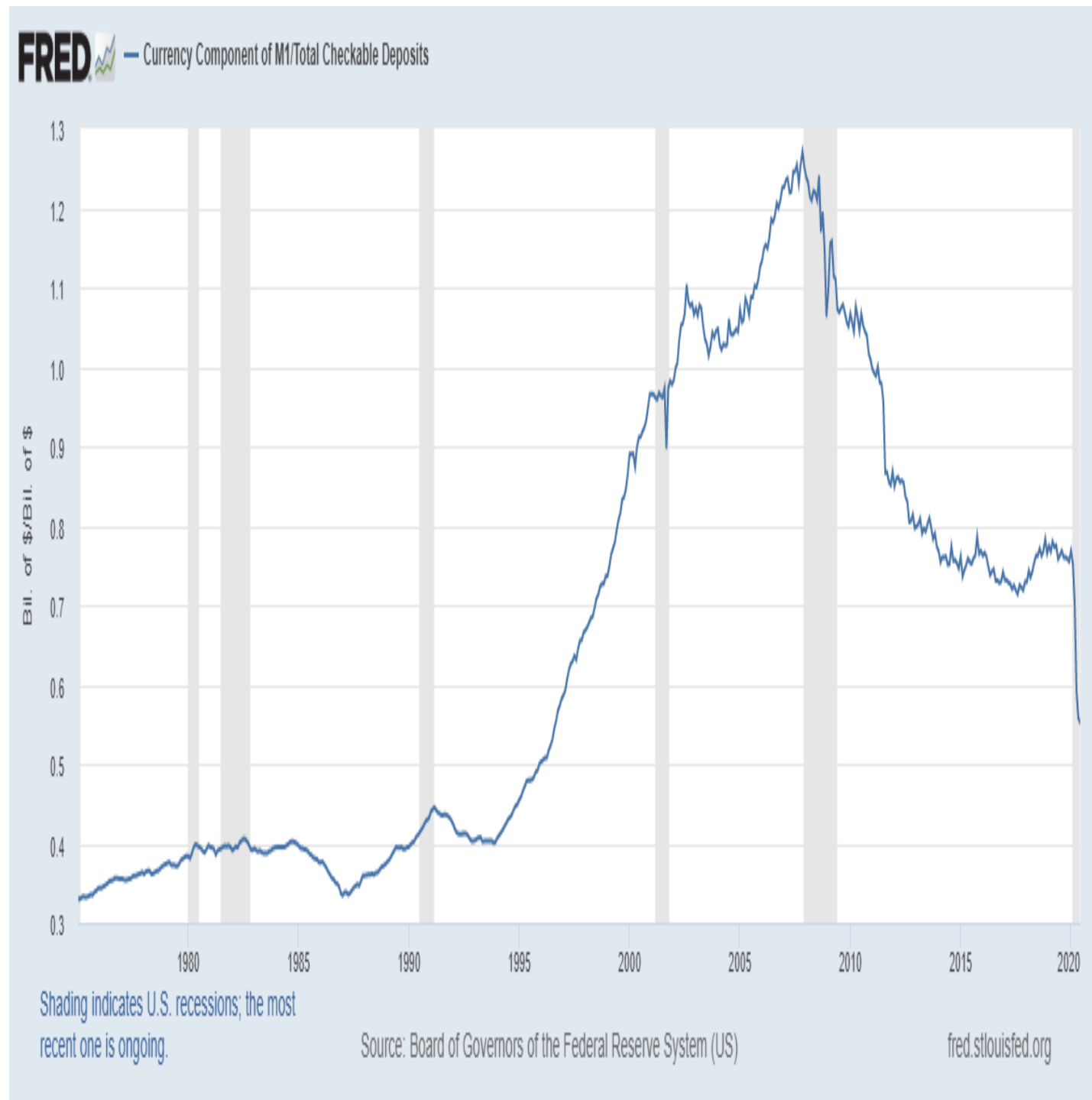


Excess reserves have increased dramatically in 2020. Now around \$3 Trillion.
e is roughly 0.90

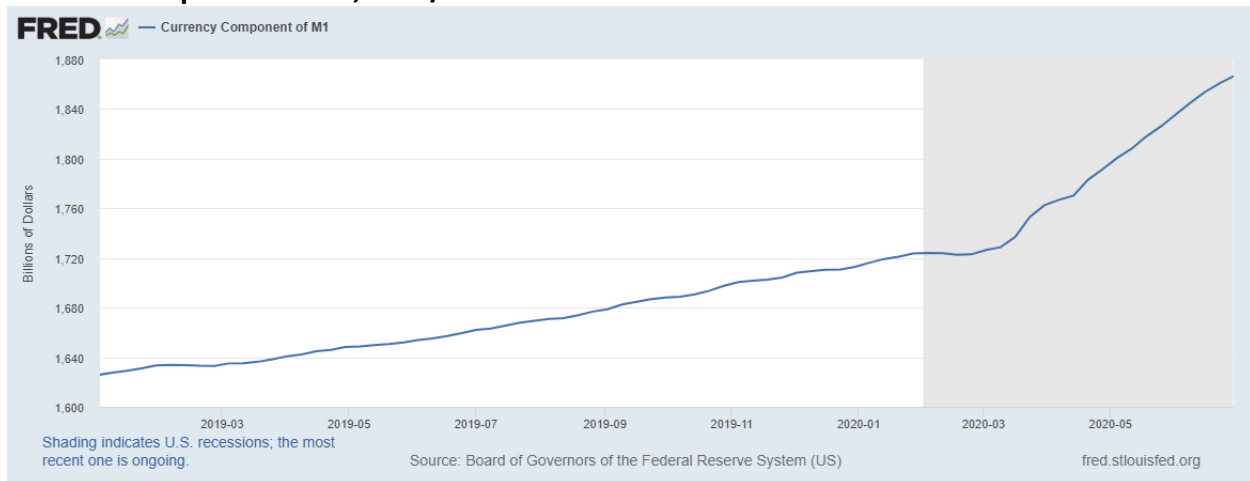
excess reserve ratio (e). Notice large increase in e in 2020. This helps explain the large drop in mm in 2020.



Currency as a ratio to D (this is our variable c)



Notice that although the ratio c has fallen, C (*total currency* holdings have risen in recent months). This sharp fall in c while C has risen implies a proportionately greater increase in checkable deposits. Recall, $c = C/D$



OPTIONAL READING ON THE FDIC (FROM FDIC WEBSITE)

What is the FDIC? The FDIC—short for the Federal Deposit Insurance Corporation—is an independent agency of the United States government. The FDIC protects depositors of insured banks located in the United States against the loss of their deposits if an insured bank fails. Any person or entity can have FDIC insurance coverage in an insured bank. A person does not have to be a U.S. citizen or resident to have his or her deposits insured by the FDIC. FDIC insurance is backed by the full faith and credit of the United States government. Since the FDIC began operations in 1934, no depositor has ever lost a penny of FDIC-insured deposits.

FDIC Coverage Basics: FDIC insurance covers depositors' accounts at each insured bank, dollar-for-dollar, including principal and any accrued interest through the date of the insured bank's closing, up to the insurance limit.

FDIC insurance covers all types of deposits received at an insured bank but does not cover investments, even if they were purchased at an insured bank.

What the FDIC Covers

- Checking accounts
- Negotiable Order of Withdrawal (NOW) accounts
- Savings accounts
- Money market deposit accounts (MMDA)
- Time deposits such as certificates of deposit (CDs)
- Cashier's checks, money orders, and other official items issued by a bank

What the FDIC Does Not Cover

- Stock investments
- Bond investments
- Mutual funds
- Life insurance policies
- Annuities
- Municipal securities
- Safe deposit boxes or their contents
- U.S. Treasury bills, bonds or notes*

*These investments are backed by the full faith and credit of the U.S. government.

The standard deposit insurance amount is \$250,000 per depositor, per insured bank, for each account ownership category. The FDIC insures deposits that a person holds in one insured bank separately from any deposits that the person owns in another separately chartered insured bank. For example, if a person has a certificate of deposit at Bank A and has a certificate of deposit at Bank B, the amounts would each be insured separately up to \$250,000. Funds deposited in separate branches of the same insured bank are not separately insured. The FDIC provides separate insurance coverage for funds depositors may have in different categories of legal ownership. The FDIC refers to these different categories as "ownership categories." This means that a bank customer who has multiple accounts may qualify for more than \$250,000 in insurance coverage if the customer's funds are deposited in different ownership categories and the requirements for each ownership category are met.

Ownership Categories

This section describes the following FDIC ownership categories and the requirements a depositor must meet to qualify for insurance coverage above \$250,000 at one insured bank.

- Single Accounts
- Certain Retirement Accounts
- Joint Accounts
- Revocable Trust Accounts
- Irrevocable Trust Accounts
- Employee Benefit Plan Accounts
- Corporation/Partnership/Unincorporated Association Accounts
- Government Accounts

Single Accounts

A single account is a deposit owned by one person. This ownership category includes:

- An account held in one person's name only, provided the owner has not designated any beneficiary (ies) who are entitled to receive the funds when the account owner dies
- An account established for one person by an agent, nominee, guardian, custodian, or conservator, including Uniform Transfers to Minors Act accounts, escrow accounts and brokered deposit accounts
- An account held in the name of a business that is a sole proprietorship (for example, a "Doing Business As" or DBA account)
- An account established for or representing a deceased person's funds—commonly known as a decedent's estate account
- An account that fails to qualify for separate coverage under another ownership category

If an account title identifies only one owner, but another person has the right to withdraw funds from the account (e.g., as Power of Attorney or custodian), the FDIC will insure the account as a single ownership account. The FDIC adds together all single accounts owned by the same person at the same bank and insures the total up to \$250,000.

Tools of monetary management-these are the main tools the Fed uses to control the money supply, especially the first tool.

1. Open market operations. Open market operations are the buying and selling of government securities (usually Treasury Bills) by the Fed. Recall in our simplified banking system, that when the Fed engaged in an open market purchase of US T-Bills, it caused reserves to be injected into the banking system. The opposite, an Open Market Sale (the Fed sells US T-Bills), would have the opposite impact—it would take reserves out of the banking system and cause the MB and M1 to fall. These transactions directly impact MB_n mentioned above. Suppose the Fed engages in an open market purchases of securities (bonds) by directly buying bonds from a bank. The Fed pays for the bonds by injecting reserves into the banking system. So the immediate impact is this:

Bank A		Fed	
Assets	Liabilities	Assets	Liabilities
Reserves (R)+	Deposits (D)	US Sec. +	Reserve Deposits +
US Securities -			

This increase in reserves raises the MB and thus increases M1. An open market sale of securities by the Fed has the opposite impact.

2. Discount Rate. Discount rate—this is the rate of interest the Fed charges banks for discount loans (typically loans of reserves—this is the BR term mentioned above). This policy needs to be discussed in conjunction with the Federal funds rate (the rate of interest charged on overnight inter-bank loans of reserve deposits) and federal funds market. We do this below in our discussion on the market for reserves. Suffice it to say for now, that a sufficient cut in the discount rate relative to the federal funds rate will cause banks to want to borrow more reserves. The immediate impact is:

Bank A		Fed	
Assets	Liabilities	Assets	Liabilities
Reserves (R)+	Deposits (D)	US Sec.	Reserve Deposits +
US Securities	Borrowings from Fed +	Loans to Banks +	

This increase in reserves raises the MB and thus increases M1.

3. Change in rr. This was discussed above. A change in rr directly impacts mm and thus M1.

Problems the Fed encounters in its attempts to control the money supply.

The Fed has to be able to correctly anticipate changes in the variables it does not directly control (such as e, c, and BR, note: BR is relatively small and is typically not much of a problem) in order to undertake the proper offsetting policies.

For example:

a. If the Fed expects e to rise, and therefore the money multiplier and money supply to fall, then the Fed could take offsetting action by, say, engaging in an open market purchase (which raises MB), or cutting the discount rate sufficiently relative to other rates (which raises MB), or reducing rr (which increases mm).

- b. Likewise, if it expects $\underline{c}$ to rise (and assuming $mm > 1$), and thus expects mm and $M1$ to fall it would likewise take offsetting action by again engaging in an open market purchase, or cutting the discount rate sufficiently relative to other rates, or reducing rr .
- c. If the BR increased, by itself this would increase MB (recall that $MB = MB_n + BR$) and thus $M1$. The Fed could take offsetting action by an open market sale (which lowers the non-borrowed monetary base and thus leaves MB unchanged), or raise rr or raising the discount rate relative to other rates to discourage borrowing of reserves (this would cause BR to fall offsetting the initial rise in BR).

Summary of model.

$$M1 = MB * (1+c)/(c+rr+e)$$

Variable	Controlled by:	If Controlled by Fed, tool it uses	Impact on MB or mm?	Impact on M1?
MB	Fed*	OMO & discount rate	MB	MB up, M1 up
rr	Fed	rr	mm	rr up, mm down, M1 down
e	Banks**	-----	mm	e up, mm down, M1 down
$\underline{c}$	Non-bank Public***	-----	mm	Depends on if $mm > 1$, or < 1 , or $= 1$.****

* $MB = MB_n + BR$. The Fed directly controls MB_n via OMO. It indirectly (and therefore imperfectly) controls BR through discount rate policy

**e (rate on paid on reserves relative to other rates, expected future deposit withdrawals, expected riskiness on loans)

*** $\underline{c}$ (bank panics, level of illegal activities, relative return on other assets, payments technology, foreign demand for currency)

****If $mm > 1 \rightarrow$ then, say, increase in c will have a proportionately larger impact on the denominator, thus mm will fall and $M1$ will fall. If $mm < 1$, then opposite result. If $mm = 1$, then no change in mm . (in recent years, mm has been around 1 or little above 1)

Numerical Example

$$M1 = MB * (1+c)/(c+rr+e)$$

Assume MB is \$5 (in trillions), $\underline{c} = 0.50$, $rr = 0$, and $e = 1$. The Fed desires to keep $M1$ constant but expects e to rise to 1.1. What type and of what amount of OMO should the Fed engage in? To answer this:

Step 1: Find the initial value of $M1$. Find mm by $mm = (1.5/1.5) = 1$, and then plug in the givens to get the following So $M1 = \$5 (1.0) = \5 Trillion

Step 2: Now suppose e rises to 1.1, what kind of OMO to hold $M1$ constant?

Recalculate mm by $mm = (1.5/1.6) = 0.94$, following that set it up as $5 = MB(0.94)$ with MB as the unknown $\rightarrow MB = 5/0.94 = \5.3 Trillion

Step 3: This would require $\Delta MB = 5.3 - 5.0 = \$0.30 \text{ Trillion} > 0$. That is an Open Market Purchase of Securities of \$0.30 Trillion.

Non-numerical Example. The current COVID and recession events have lead firms to perceive that the riskiness of loans has increased. To prevent any change $M1$ what should the Fed do in terms of OMO? Here is how you approach this:

First step. Perceived increase in riskiness on loans $\rightarrow e$ rises $\rightarrow mm$ falls $\rightarrow M1$ falls, other things constant.

Second step. To prevent the fall in $M1$, the Fed has to do something to raise $M1$ offsetting this. This would be an open market purchase of government bonds $\rightarrow MB$ rises $\rightarrow M1$ rises.

The market for reserves and the determination of the federal funds rate

We frequently hear about the Fed “setting interest rates”. This can be a bit confusing because interest rates are market determined variables—determined by supply and demand (except for the discount rate and the rate paid on by reserves by the Fed—those rates are directly set by the Fed). To understand how the Fed does impact interest rates, specifically the federal funds rate (rate charged for interbank borrowing/loans of reserves) we need to develop the market for reserves which determines the federal funds rate (i_{ff}). We do that next.

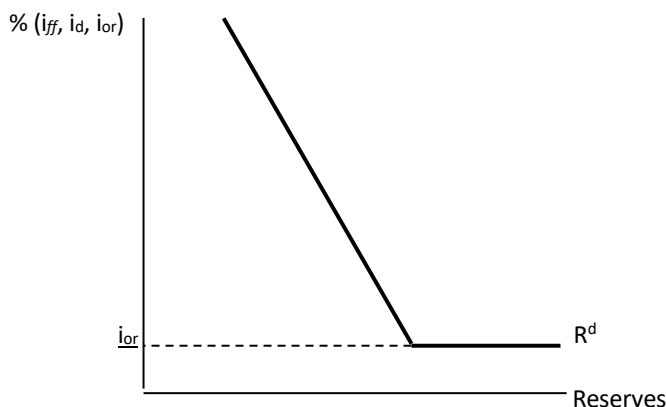
The Demand for reserves curve (R^d). See graph below. Two aspects of this curve need to be explained.

First, why does it have a negative slope and why is horizontal (perfectly elastic at i_{or})?

--Justification of negative slope: The *opportunity cost* of holding reserves is $i_{ff} - i_{or}$ where i_{ff} = Federal Funds rate (again, this is the rate banks charge each other for the borrowing of reserves) and i_{or} is the rate of interest the Fed pays on reserves (as of 2008). Assuming $i_{ff} - i_{or} > 0$, then an increase in $i_{ff} - i_{or}$ would raise the opportunity cost of HOLDING reserves versus lending them out. This would lower the demand to hold reserves. So other things constant, as i_{ff} rises R^d falls → a negative slope (notice: this relationship between i_{ff} & R^d is represented as a movement along the R^d curve).

-- Why does the demand for reserves become perfectly elastic (horizontal) at i_{or} ? To understand the horizontal stretch of the demand curve, consider the (temporary) case of $i_{ff} - i_{or} < 0$ or $i_{ff} < i_{or}$. If this was the case, banks could make infinite profit by continually borrowing reserves in the Federal Funds market at the rate i_{ff} and then holding these reserves to receive the higher i_{or} from the Fed. Obviously, this cannot remain this way—demand for reserves would get infinitely large (hence the perfectly elastic stretch of the demand curve) leading to a shortage of reserves in the FF market which, in turn, would cause the federal funds rate to be bid up. Effectively, i_{or} acts as a floor on how low the federal funds rate can fall. (in reality, it might fall below i_{or} due to transactions costs of using the federal funds market—we will assume zero transactions costs to rule this possibility out; we shall also assume all financial institutions that hold reserves qualify for the rate i_{or} , otherwise our story above may fail to hold).

Demand for Reserves

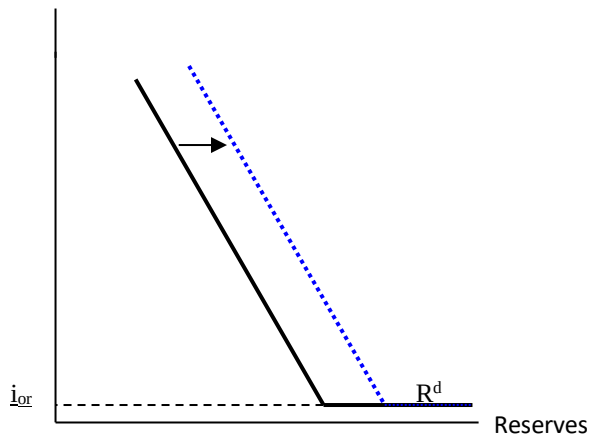


The next important step is to determine what are the **shift variables** for R^d and HOW do they shift the demand curve?

a. **Bank preferences for holding reserves.** This would be influenced by various factors in the economy. One that have already discussed, is if banks thought there would be a sudden surge in deposit withdrawals by its customers, it would very likely want to increase their holdings of reserves to satisfy their customers. Banks would probably not survive too long if depositors come into withdraw cash and find out the bank has none.

Another circumstance we have discussed would be the bank's perception of the riskiness of holding other assets, particularly loans. If they view loans have becoming riskier they want to hold onto more reserves (this seems to have been what happened in the wake of the recession of 2008-09). It would cause a rightward shift.

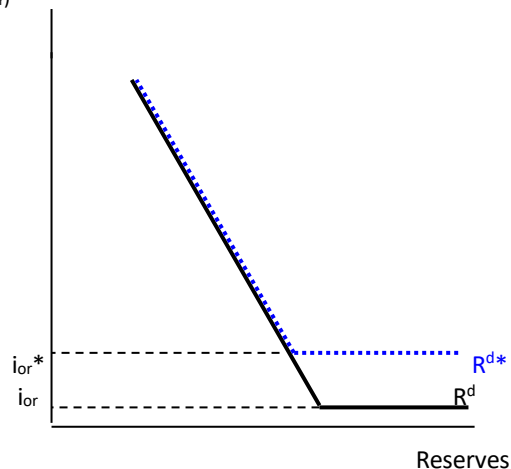
% (i_{ff} , i_d , i_{or})



b. A **change in the required reserve ratio (rr)** . If the Fed raised rr , this would increase demand for reserves. See graph for previous question.

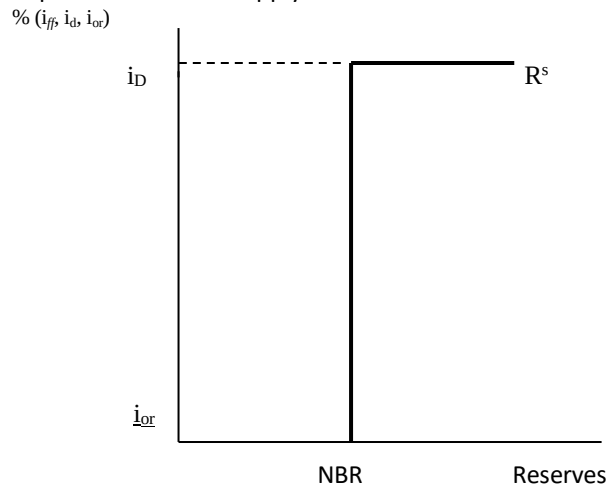
c. A **change in i_{or}** . If the Fed was to raise this rate, it would lead to the following change (see graph below).

% (i_{ff} , i_d , i_{or})



Supply of Reserves (R^s).

The **supply curve for reserves (R^s)** is vertical up to the discount rate (i_D) and starts at NBR. Why? The amount of reserves available in the banking system (the supply of reserves) is determined by the Fed. However, there are different ways of the Fed injecting reserves into the banking system (at least in the conventional sense). Non-borrowed reserves (NBR) refer to the reserves the Fed injects into the banking system via open market operations. Because this amount is completely determined by the Fed, it is NOT a function of the federal funds rate. That is, if the federal funds rate was to change the Fed does not have to supply more or less as a consequence. This has the implication that the supply curve is vertical at whatever level of NBR that Fed settles on. See below.

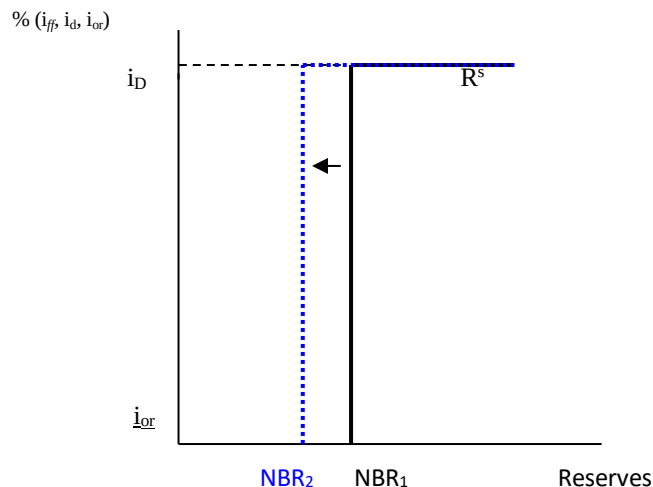


NBR is the SAME as MB_{net}

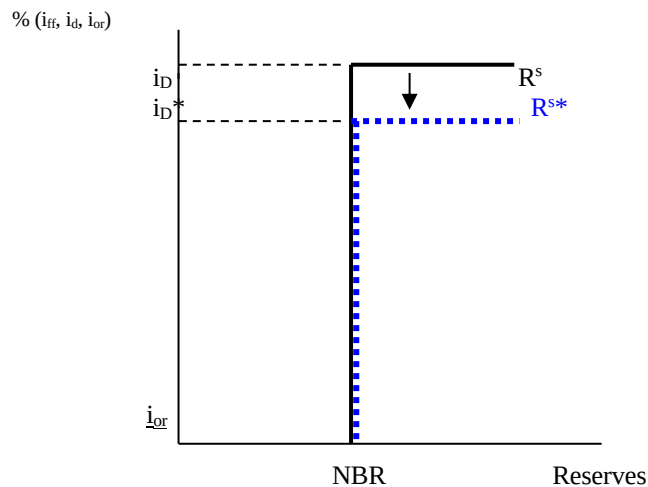
Why does it become horizontal at the discount rate? The discount rate is the rate the Fed charges banks for the borrowings of reserves (BR). This is another way the Fed can inject reserves into the banking system. To understand the horizontal stretch, assume for now that initially $i_{ff} > i_D$. This would create the following incentives: borrow from the Fed at the lower discount rate and then *supply* these funds in the FF market at the higher federal funds rate. You want to keep doing this to make endless profit! Hence the perfectly elastic section of R^s . Naturally, this cannot be an equilibrium as the massive surplus of funds in the FF market would drive the federal funds rate down. Effectively, the discount rate acts as a ceiling on how high the federal funds rate could climb.

Shift variables for R^s .

a. Open market operations would change the amount of NBR in the banking system. For example, an open market sale of bonds by the Fed would decrease NBR causing the following shift.

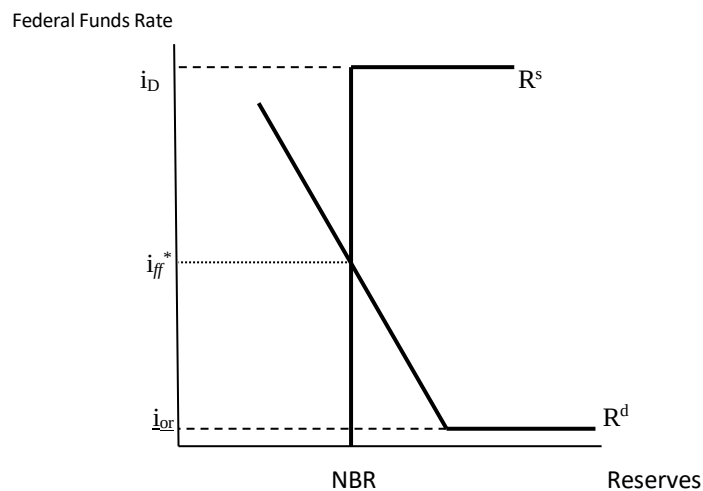


b. A change in the discount rate can also cause a shift in this curve. For example, a cut in the discount rate would do this:



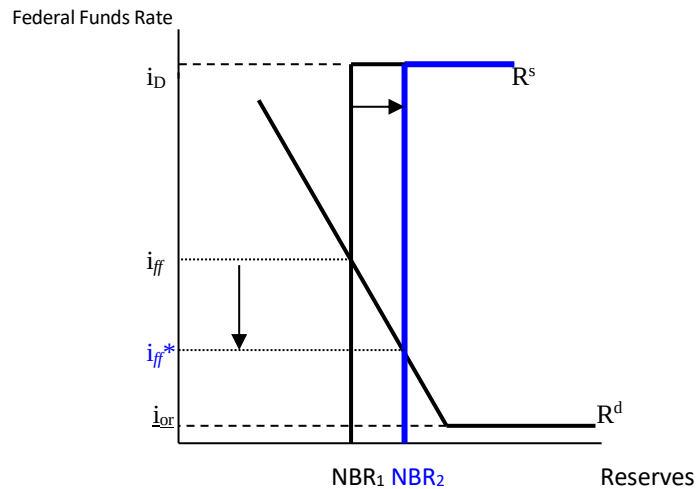
Determination of the federal funds rate

The federal funds rate is determined by the supply and demand for reserves but the Fed controls the supply of reserves and partially controls the demand side. So the Fed can target a particular rate by shifting supply and/or demand to hit the target rate. So there is nothing inconsistent with saying both that the federal funds rate is market determined (S&D) and yet it is controlled by the Fed. Any rate above i_{ff}^* in the graph below would lead to a surplus of reserves ($R^s > R^d$), causing the federal funds rate to fall. Any rate below i_{ff}^* would lead to a shortage of reserves ($R^s < R^d$), causing the federal funds rate to be bid up.

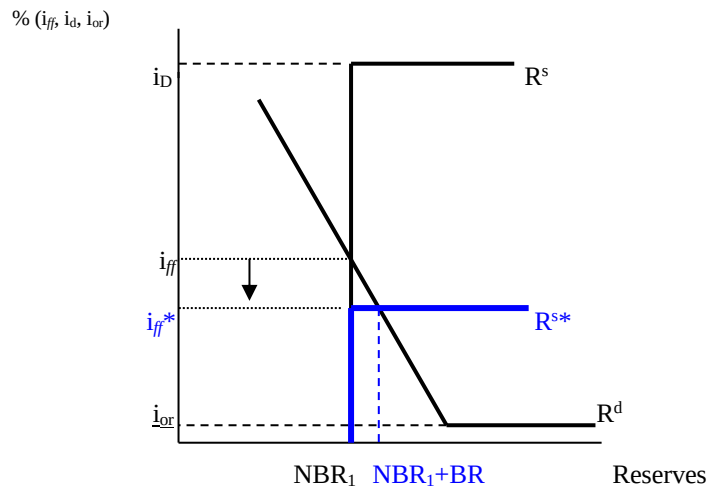


Comparative Statics: changes in equilibrium values.

Example one: The impact of an Open Market Purchase.

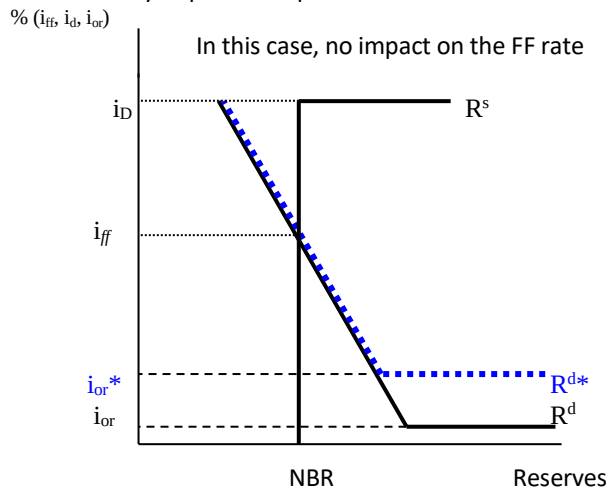


Example 2: A (sufficient) cut in the discount rate (notice that if discount rate is not cut by enough—that is, below i_{ff} , it will not have an impact). As represented in the graph, this cut is below i_{ff} and so INITIALLY banks would find it profitable to borrow at the discount rate and lend out at the federal funds rate. This, however, will create a massive surplus of federal funds causing the federal funds rate to be driven down to its new equilibrium value of i_{ff}^* . Notice that if i_D is not cut sufficiently (that is, it is still left above the original i_{ff}) then it will have not impact.

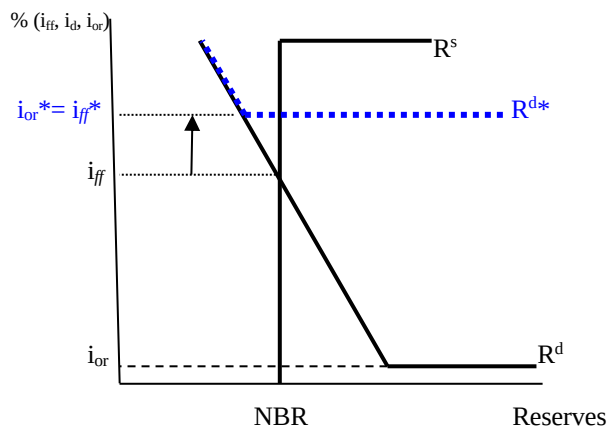


Example 3: An increase in i_{or} . Two possibilities exist here.

Case A. If the increase is not sufficient enough, so that even after the increase we still have original $i_{ff} > i_{or}$, then this will not have any impact on equilibrium federal funds rate (see graph).



Case B. In this case, i_{ff} rises. To see why, notice that the IMMEDIATE impact of setting i_{or}^* above the initial i_{ff} will be to cause a shortage of federal funds. It would be profitable to borrow in the federal funds market and hold the reserves and earn i_{or} on them. This shortage of federal funds will cause i_{ff} to be bid up to the new equilibrium value of i_{ff}^* .



Example 4. Suppose there is an increase in e . This would increase the demand for reserves and shift out the R^d curve. If the Fed has goal of maintaining a constant federal funds rate, then it might respond to this by shifting out the R^s curve via Open Market Purchases. See graph below. Notice that the demand curve shifts out first, then the Fed responds by increasing the supply curve to keep the federal funds rate constant.

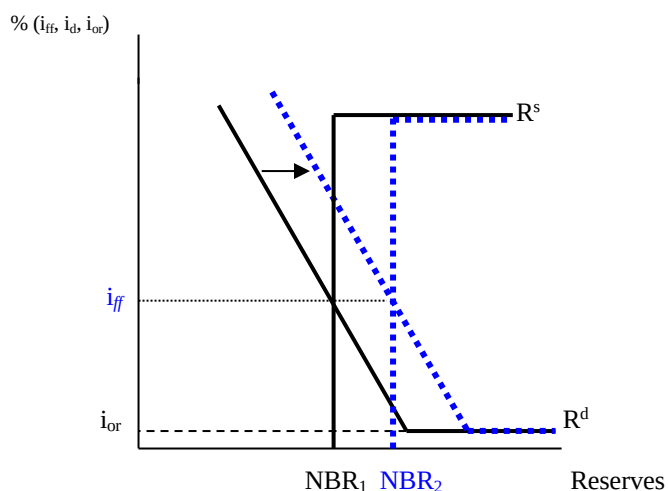
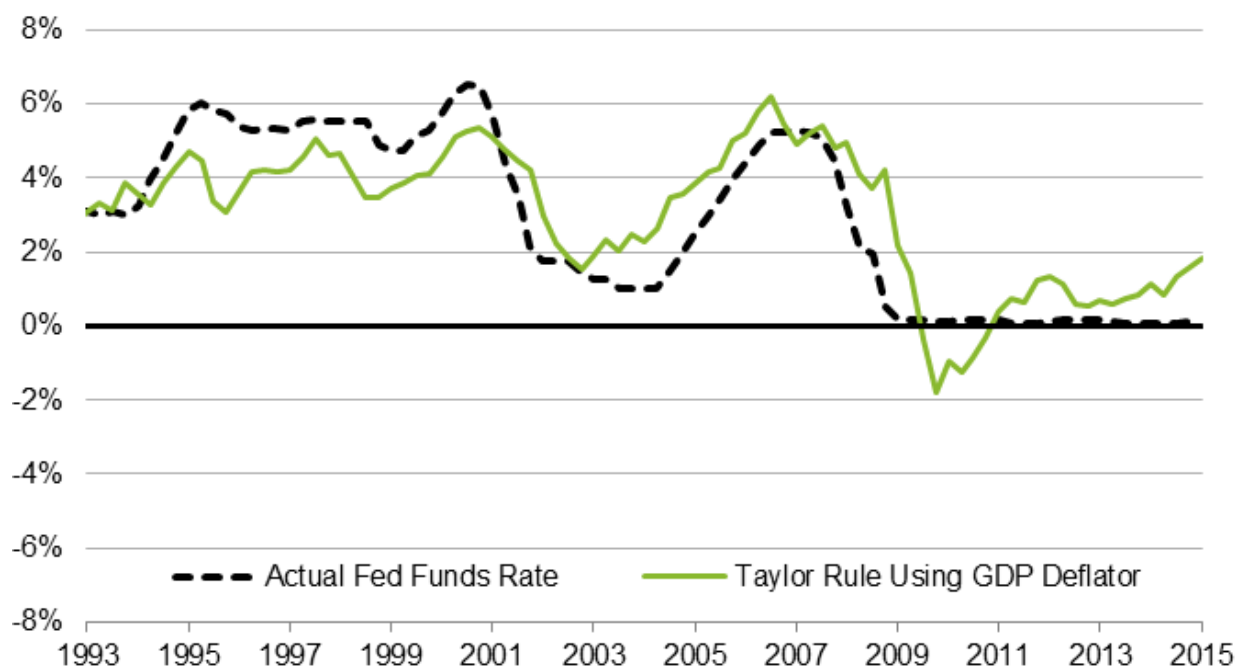


Figure 1: The Original Taylor Rule, 1993-Present



Now that we know how the Fed can target a particular value of federal funds rate, the next question is how does it chooses its target value?

An empirical rule some follow in the US to make these predictions of target rate is known as the Taylor Rule.

1. In general the Taylor rule is $i_{ff} = \pi + \text{equilibrium real federal funds rate}^* + .5(\text{output gap}) + .5(\pi - \pi_{\text{target}})$

a. i_{ff} = federal funds rate predicted by the Taylor Rule as the Fed's target rate.

b. The term π is the inflation rate. Taylor used **2%** as the inflation rate target (π_{target}).

c. The equilibrium real federal funds rate is defined as the real FF rate that is consistent with full employment in the long run (Taylor used **2%** for this rate).

d. Output gap is the % deviation of real GDP from a target. For example, suppose we index target GDP to be 100. Actual GDP is actually, say, 96. So the deviation from the target is -4 and the percentage deviation is $-4/100 = -0.04$ (-4%). Target GDP typically means trend or full employment GDP when used in practice (trend GDP or GDP in long run would be full employment GDP in that we expect full employment in the long run). Formally, the part of the equation is done by the log differences (see the table below for an explanation).

2. Original Taylor Rule: $i_{ff} = \pi + .5(\text{output gap}) + .5(\pi - 2) + 2$

Example: Assume -4% output gap, and an inflation rate of 1%. Then by Taylor Rule predicts the Fed will:

$$i_{ff} = \pi + .5(\text{output gap}) + .5(\pi - 2) + 2 = 1 + 0.5(-4) + 0.5(1-2) + 2 = 1 - 2 - 0.5 + 2 = \mathbf{0.5\%}$$

Notice that the logic of including things like output and inflation targets in this equation is to recognize that the Fed uses interest rate policies as an intermediate goal to achieve some larger goal such as low inflation and full employment output. If output is below its full employment level, then this term (output gap) shows up as a negative value and thus implies, other things constant, that the Fed should lower the federal funds rate so as to stimulate the economy (or at least try to stimulate the economy). On the other hand, looking at the term $“.5(\pi - \pi_{\text{target}})“$ in the equation predicts that if actual inflation is above the target rate, and thus this term takes on a positive value, then the Fed should raise the FF rate ----to try to bring down inflation. In principle, the relative weights (that is, the coefficients) on the output gap and inflation gap should depend on, among other things, the extent to which policymakers are willing to accept greater variability in inflation in exchange for greater stability in output. Lower federal funds rates are typically associated with stimulative money supply expansions (notice that an Open Mkt purchase both lowers the federal funds rate and raises M1, other things constant) . However, the higher money supply growth which might stimulate the economy (that is, increase real GDP) also runs the risk of higher inflation. Some research subsequent to John Taylor's original publication of this rule, summarized by Taylor (1999), found a case for allowing a larger response of the funds rate to the output gap (specifically, a coefficient of 1.0 rather than 0.5). The FOMC paid closer attention to variants of the Taylor rule that include the higher output gap coefficient. For example, Janet Yellen has suggested that the FOMC's "balanced approach" in responding to inflation and unemployment is more consistent with a coefficient on the output gap of 1.0, rather than 0.5. In the modified Taylor rule it was assumed the higher coefficient on the output gap.

Footnote on output gap. Output gap in equation is natural logarithm of real GDP minus natural log of target or full employment real GDP. Example:

	real GDP (index)	ln(real GDP)	ln(x _{actual})-ln(x _{full employ})
full employment GDP	100	4.6052	
actual GDP1	96	4.5643	-4.1% (or roughly 4%)
actual GDP2	103	4.6347	3.0%
actual GDP3	100	4.6052	0.0%

You might find this interesting. You can set your own weights for the Taylor Rule.

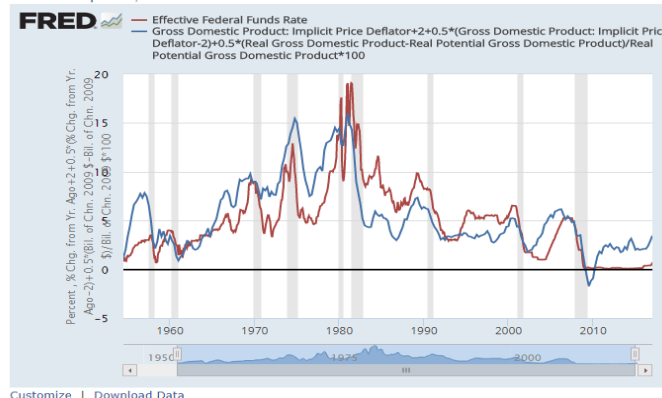
<https://www.frbatlanta.org/cqer/research/taylor-rule.aspx?panel=1>

The FRED® Blog

The Taylor Rule



Posted on April 24, 2014



This graph shows in blue the Taylor Rule, which is a simple formula that John Taylor devised to guide policymakers. It calculates what the federal funds rate should be, as a function of the output gap and current inflation. Here, we measure the output gap as the difference between potential output (published by the Congressional Budget Office) and real GDP. Inflation is measured by changes in the CPI, and we use a target inflation rate of 2%. We also assume a steady-state real interest rate of 2%. These are a lot of assumptions, and you are welcome to change them on the graph by playing around with the formula to see how the Taylor Rule matches up with the effective federal funds rate. To read up on the Taylor Rule, see the [original article](#) or an article by former St. Louis Fed president William Poole.

How this graph was created: To create a new series from several series, first add the series by modifying the existing series in the "Graph" tab. Once you have assembled them all, expand the series section in the same tab and "create your own transformation." Finally, as the axis legend has become unwieldy, remove it by checking off the mark in the graph tab.

Suggested by: [Christian Zimmermann](#)

Optional Reading

For a more detailed discussion on monetary rules, see below.

Monetary policymakers often use simple monetary policy rules, like the Taylor rule, as an input into their decision-making. However, there are many different simple rules, and there is no agreement on a single "best" rule. We look at the federal funds rates coming from seven simple rules and three economic forecasts to investigate the range of results that can be produced. While there are some commonalities, we document that the differences in the federal funds rates suggested by the rules can be quite pronounced.

Simple monetary policy rules provide a relationship between the central bank's policy rate and a relatively small number of indicators on real economic activity and inflation. Since the early 1990s, a large academic research literature has developed around the creation and testing of simple monetary policy rules. At the same time, policymakers have come to often look at a variety of monetary policy rules as an input into their decision-making and as a means of communicating the rationale of those decisions to the public (see, e.g., Yellen 2012, Meyer 2002, Plosser 2014, and Asso et al. 2010).

In this *Commentary*, we look at a range of simple monetary policy rules and their implications for the federal funds rate based on multiple forecasts. Examining a variety of rules is helpful because there is no agreement on a single "best" rule. At the same time, it is useful to consider multiple forecasts for the current state of the economy and how the economy is likely to evolve going forward, because the future is inherently uncertain and our ability to measure present economic circumstances is imperfect.

This exercise reveals some commonalities but also quite pronounced differences across the simple policy rules we consider. In a 2012 speech, then-Vice Chair Janet Yellen noted that "a wide variety of simple rules have been proposed in the academic literature, and their policy implications can differ significantly depending on the particular specification." In looking across the rules and forecasts over the next two years, the differences between

the implied federal funds rates at the 75th percentile and the 25th percentile range from 0.9 to 1.4 percentage points. Even for a single forecast, some of the differences among the rules are as large as three percentage points.

The General Form of Many Simple Monetary Policy Rules

Simple monetary policy rules often take the general form:

$$i_t = \rho i_{t-1} + (1 - \rho)[r^* + \pi_t + \alpha(\text{inflation gap}) + \beta(\text{activity gap})].$$

In this rule, i_t is the central bank's policy rate, such as the federal funds rate in the United States, at time t . The variable r^* refers to the equilibrium real interest rate, which within the above rule can be thought of as the long-run federal funds rate adjusted for inflation. The inflation rate is π_t , and the inflation gap is the difference between inflation and target inflation, denoted π^* . The activity gap measures the extent to which economic activity exceeds "normal" levels. Commonly, activity is measured using the output gap (the percentage difference between actual output and potential output) or the negative of the unemployment gap (the percentage point difference between the unemployment rate and the natural rate of unemployment). The coefficient ρ captures the amount of inertia in the policy rule, α determines the responsiveness of policy to the inflation gap, and β determines the responsiveness of policy to the activity gap.

In practice, the simplicity of this general monetary policy rule belies a host of complexities. Academic researchers and policymakers have not settled on the choice of the coefficients ρ , α , and β . While some policy rules use the contemporaneous values of the inflation gap and the activity gap, others assume that the current policy rate is affected by forecasts of the inflation gap and the activity gap, given that it typically takes monetary policy some time to affect inflation and real economic activity. There are also many different approaches to estimating concepts such as r^* , the activity gap, and the inflation gap, and considerable uncertainty surrounds such estimates.¹

Seven Simple Policy Rules

Partly as a result of the complexities laid out above, there is a large and ever-growing literature related to simple policy rules.² Faced with a multitude of rules, we take a pragmatic approach: We select seven simple rules that have broadly representative features from this literature, and we compare and contrast the federal funds rates coming from those rules, based on current economic conditions and based on forecasts for how economic conditions may develop in the future. We lay out the seven rules we will use in this section, with formulae in table 1, and we detail the sources of the forecasts in the next section.

Table 1

Taylor (1993) rule:	$i_t = r^* + \pi_t + 0.5(\pi_t - \pi^*) + 0.5(\text{output gap}_t)$
Core inflation in Taylor (1999) rule:	$i_t = r^* + \pi_t^{\text{Core}} + 0.5(\pi_t^{\text{Core}} - \pi^*) + (\text{output gap}_t)$
Inertial rule:	$i_t = \rho i_{t-1} + (1 - \rho)[r^* + \pi_t^{\text{Core}} + 0.5(\pi_t^{\text{Core}} - \pi^*) + (\text{output gap}_t)]$
Alternative r^* rule:	$i_t = \rho i_{t-1} + (1 - \rho)[r_{\text{alt}}^* + \pi_t^{\text{Core}} + 0.5(\pi_t^{\text{Core}} - \pi^*) + (\text{output gap}_t)]$
Forward-looking rule:	$i_t = r^* + \pi_{t+3}^F + 0.5(\pi_{t+3}^F - \pi^*) + 0.5(\text{output gap}_t)$
First-difference rule:	$i_t = i_{t-1} + 1.74(\pi_{t+3}^F - \pi^*) - 1.19(u_{t-1} - u_{t-2})$
Low weight on output gap rule:	$i_t = 0.91i_{t-1} + (1 - 0.91)[r^* + \pi^* + 1.58(\pi_{t+1}^{Q,F} - \pi^*) + 0.14(\text{output gap}_{t+1}^F)]$

The first rule is based on the policy rule suggested by Taylor (1993). Taylor (1993) fixed r^* to 2 percent and used the GDP deflator as the measure of inflation. To update the rule, we make two modifications. First, given the Federal Open Market Committee's (FOMC) [Statement on Longer-Run Goals and Monetary Policy Strategy](#), we measure π_t as inflation in the price index for personal consumption expenditures (PCE inflation) over the previous

four quarters, and we set π^* to 2 percent.³ Second, we account for potential changes to r^* by using the median implied long-run real federal funds rate from the most recently available FOMC Summary of Economic Projections (SEP), which was 1 percent in June 2016.⁴

The second rule is based on a parameterization considered by Taylor (1999), in which the coefficient on the output gap is 1.0 instead of 0.5 as in the Taylor (1993) rule. Thus, policymakers are assumed to place twice the amount of weight on slack in this rule compared with the earlier rule.⁵ In an effort to span the range of policy rules and provide additional contrast to the first rule above, for inflation we use the four-quarter PCE inflation rate excluding food and energy, or core PCE inflation, π_t^{Core} . Core inflation measures are often used to smooth through volatility coming from food and energy price movements, in order to identify the inflation trend. Bernanke (2015) provides one perspective on the inclusion of core PCE inflation in this rule. Monetary policymakers often use versions of the Taylor (1993) and Taylor (1999) rules as benchmarks in assessing the stance of policy; see, e.g., Yellen (2012).

The third rule adds inertia, such that the funds rate does not adjust immediately to the value of the Taylor (1999) rule with core PCE inflation, but rather moves there slowly over time. Empirical estimates consistently find such inertial responses by central banks around the world (see, e.g., Goodhart 1999, Coibion and Gorodnichenko 2012), and theoretical and simulation-based studies also typically find that policy inertia improves economic outcomes (see, e.g., Levin et al. 2003, Woodford 2003b, Taylor and Williams 2011). As in Canzoneri et al. (2015), we set $\rho = 0.8$, a coefficient that is “typical of estimates that are found in the empirical literature” (p. 385).

The fourth rule captures uncertainty around r^* by using an alternative estimate of it.⁶ Laubach and Williams (2003, 2015) estimate the natural rate of interest and find that those estimates have varied significantly through U.S. history. Their most recent estimate, which we denote r_{alt}^* , was 0.2 percent using data available on May 31, 2016. The fifth rule incorporates forward-looking elements by replacing the realized inflation rate over the last four quarters with a forecast of PCE inflation three quarters ahead, π_{t+3}^F , as suggested by Bernanke (2010).⁷ Forward-looking rules begin to account for the lag between monetary policy actions and their impact on the economy, and they are also able to distinguish between transitory and persistent inflationary pressures.⁸ Because forecasts will encompass information that may not be available in current readings on inflation or the activity gap, forecast-based policy rules inherently bring more information to bear in the policy rule, but in an easy-to-understand and compact way (see Orphanides and Wieland 2008).

The sixth rule is a forward-looking, first-difference rule, based on Orphanides and Williams (2008, 2013). In this rule, the change—or first-difference—in the policy rate is related to a forward-looking inflation gap and the (backward-looking) change in the unemployment rate in the previous quarter, $u_{t-1} - u_{t-2}$. This rule is notable because it omits variables that are unobservable in practice and difficult to estimate. The omission of both a long-run real funds rate r^* and an activity “gap” term precludes the federal funds rates coming from this rule from being contaminated by mismeasurement of unobservable concepts.⁹

The seventh rule departs from the others by placing a relatively low weight on the output gap, along the lines of the literature on pure inflation-targeting.¹⁰ We base the parameters of this rule on estimates reported in Clarida et al. (2000) for a forward-looking rule with inertia, which captured U.S. monetary policy actions over the period 1982–1996. In this case, we use the forecast for quarterly annualized inflation in the next quarter, $\pi_{t+1}^{Q,F}$, and the forecast for the output gap in the next quarter, $output_{t+1}^F$, in the policy rule.¹¹

Data and Forecasts

To generate the federal funds rates based on these simple policy rules, both for the current quarter and going forward, we use data and forecasts from a variety of sources. One well-known and publicly available source for economic forecasts is the quarterly Survey of Professional Forecasters (SPF) from the Federal Reserve Bank of Philadelphia; we use the median responses to ensure robustness to potential outliers. The Congressional Budget Office (CBO) is another source of publicly available forecasts, which are usually updated twice per year, in January and August. Finally, we generate forecasts from a small statistical Bayesian vector autoregression model used in previous *Economic Commentaries* (FRBC BVAR).¹²

The CBO provides detailed forecasts for PCE inflation, core PCE inflation, the unemployment rate, and the output gap over a multiyear horizon, which is all the information needed to calculate the federal funds rates implied by the seven rules, for the current quarter and for the next several years. While the SPF and the FRBC BVAR forecast PCE inflation and core PCE inflation, they only report an unemployment rate and not an output gap. However, we can construct unemployment gaps for the SPF and the FRBC BVAR.¹³ When working with these two forecasts, we replace the output gap in all the policy rules with the product of the unemployment gap times the inverse Okun’s coefficient as the proxy for the output gap.¹⁴

Crucially, in all cases, we take the forecast as given and then calculate the implied average federal funds rate for each quarter based on the seven simple policy rules (see also, e.g., Plosser 2014). That is, we do not model the manner in which different paths for the federal funds rate would affect the forecast—even though theory and empirical evidence suggest that higher paths for the federal funds rate would tend to slow economic activity and inflation, while lower paths for the federal funds rate would tend to increase economic activity and inflation.¹⁵

Federal Funds Rates Based on Simple Policy Rules, Today and Going Forward

With seven rules, three forecasts, and quarterly data, there are a large number of federal funds rates to share. Figure 1 provides a summary of the results using quartiles. We present the median (50th percentile) federal funds rate across all the policy rules and forecasts, along with the 25th and 75th percentiles and the maximum and minimum funds rate at each point in time. Based on data and forecasts available as of June 23, 2016, the median federal funds rate from the simple policy rules we consider rises from 0.7 percent in 2016:Q2 to 2.1 percent in 2018:Q2.¹⁶

Figure 1. Summary of Federal Funds Rates
Based on Seven Simple Policy Rules

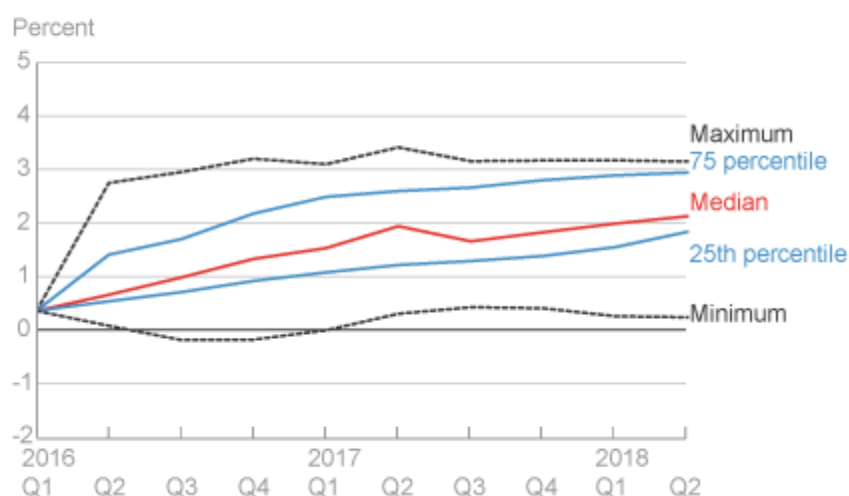
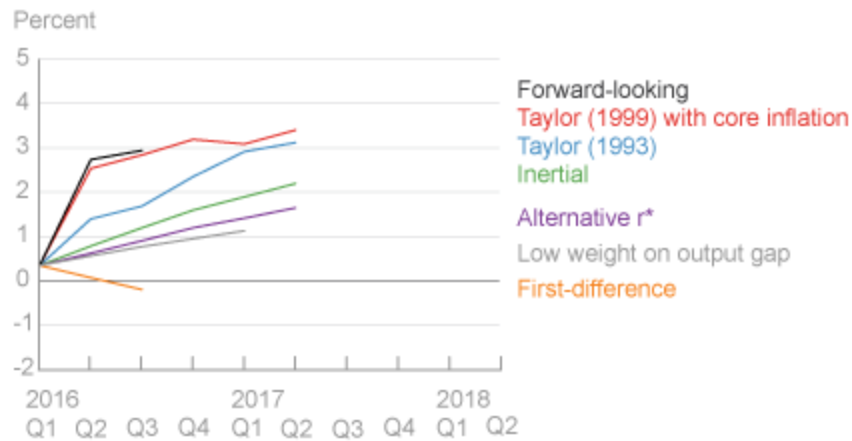


Figure 1 highlights pronounced differences among the federal funds rates coming from the various policy rules and forecasts. In the summary figure, the distance between the 75th and 25th percentiles ranges from 0.9 to 1.4 percentage points over the two-year horizon shown. Based on forecasts for the first quarter considered, 2016:Q2, the difference between the 75th and 25th percentiles is 0.9 percentage point, and the difference between the maximum and minimum federal funds rates is 2.7 percentage points.

Of course, some of this variation is attributable to differences in forecasts across the three sources. Figures 2 through 4 show the federal funds rates from each simple monetary policy rule conditional on forecasts from the SPF, CBO, and FRBC BVAR, respectively. (Federal funds rates based on the SPF are limited by the SPF's forecast horizon, which extends only one year.) Even conditional on a single forecast, we still see a wide range of implied federal funds rates: In each figure, the differences between the maximum funds rate and the minimum funds rate at a given point in time range from 1.7 to 3.1 percentage points.

Figure 2. Federal Funds Rates Based on Forecasts from the Survey of Professional Forecasters and Seven Simple Policy Rules



Note: The short forecast horizon for the SPF, combined especially with forward-looking rules, limits the number of observations in some cases.

Figure 3. Federal Funds Rates Based on Forecasts from the Congressional Budget Office and Seven Simple Policy Rules

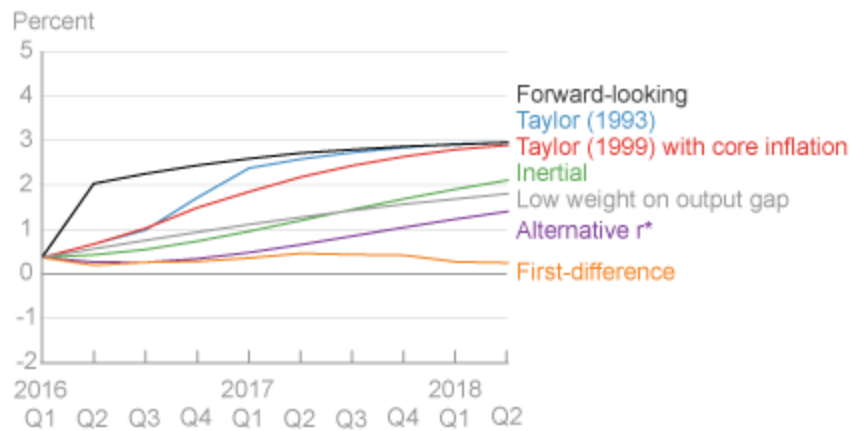
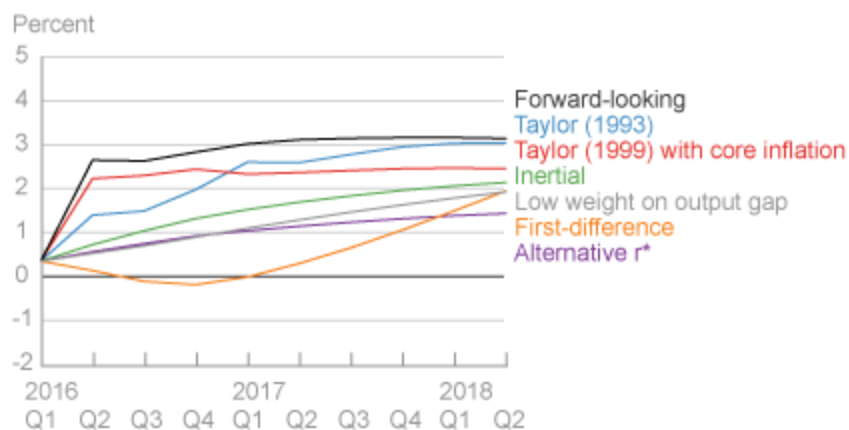


Figure 4. Federal Funds Rates Based on Forecasts from the Cleveland Fed Staff Small BVAR Model and Seven Simple Policy Rules



Finally, while we consider multiple forecasts for inflation and real economic activity to address some of the uncertainties inherent in predicting the future and assessing the current state of the economy, it is worth noting that there is considerable uncertainty surrounding all forecasts. Thus, the summary figure likely understates the range of potential federal funds rates implied by these simple policy rules going forward, as the forecast may evolve in unexpected ways based on new incoming data.

Conclusion

This *Commentary* presents a relatively straightforward look at the federal funds rates coming from a range of simple monetary policy rules and forecasts. Considering a range of rules and forecasts is important, because there is not a single agreed upon “best” rule, and because there are always differing views on the economy and its future prospects. This analysis highlights some commonalities but also quite pronounced differences across simple policy rules. Even for a single forecast, some of the differences among the rules are as large as three percentage points.

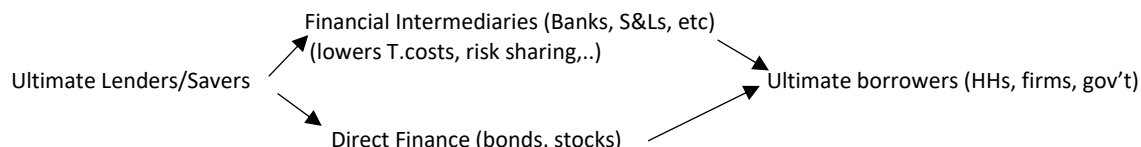
Footnotes

1. Uncertainty around the inflation gap typically relates to the choice of the inflation measure to include in the rule. Regarding r^* and the activity gap, in addition to considerable estimation uncertainty given that these concepts are unobservable, there are multiple alternative theoretical constructs for each concept (see, e.g., Carlstrom and Stehulak 2015, and Tasci and Verbrugge 2014). [Return](#)
2. For one survey, see Taylor and Williams (2011). [Return](#)
3. With the sum of coefficients on π_t of 1.5, which is greater than 1, this rule satisfies the Taylor principle, in that the real interest rate ($i_t - \pi_t$) rises in response to increases in the inflation rate, which eventually helps to stabilize inflation. In many macroeconomic models, the policy rule coefficient on π_t needs to exceed 1 in order for a unique stable equilibrium to exist (see, e.g., Woodford 2003a for a discussion). [Return](#)
4. The median implied long-run real federal funds rate is the median long-run federal funds rate (from figure 2 of the SEP) minus the median long-run PCE inflation rate. Carlstrom and Stehulak (2015) provide evidence of a recent decline in the long-run natural rate of interest. [Return](#)
5. The implications of the difference in the coefficients on the output gap have been studied elsewhere—e.g., Taylor (1999a), Meyer (2002), and Taylor and Williams (2011)—and are typically model-dependent. [Return](#)

6. See Clark and Kozicki (2005). Carlstrom and Fuerst (2016) find that monetary policy may achieve better outcomes if policy rules incorporate fluctuations in long-term productivity growth that proxy for a time-varying natural rate of interest. [Return](#)
7. Technically, we use the four-quarter rate of PCE inflation expected three quarters in the future: Within quarter t , the last observation of the price level comes from quarter $t - 1$, and we look at inflation four quarters ahead from that point, which is $t + 3$. [Return](#)
8. See also Ashley et al. (2014). [Return](#)
9. While the parameterization of the first-difference rule we consider is optimal in the context of a single model (see Orphanides and Williams 2008), first-difference rules are often robust to various forms of model uncertainty, misspecification, and learning, and can outperform optimal control policies; see, e.g., Walsh (2003), Orphanides and Williams (2008, 2013), and Tetlow (2015). [Return](#)
10. In fact, the coefficient on the output gap is not statistically different from zero for this case in Clarida et al. (2000). [Return](#)
11. While Clarida et al. (2000) estimate a relatively high value for π^* , to maintain consistency with the other rules, we set $\pi^* = 2$ and $r^* = 1$ based on the FOMC's most recent SEP. To conform to the Clarida et al. (2000) specification, the right-hand side of this rule includes π^* in two locations. [Return](#)
12. See, e.g., Knotek et al. (2015). The Cleveland Fed staff consult a variety of forecasting models, thus this forecast does not necessarily represent the official forecast of Cleveland Fed staff or the president of the Cleveland Fed. [Return](#)
13. For the SPF, we take the difference between the unemployment rate and the estimate of the natural rate of unemployment (commonly called the nonaccelerating inflation rate of unemployment, or NAIRU) that the SPF updates once per year. For the FRBC BVAR, we take the difference between the unemployment rate and the long-run unemployment rate from the most recently released FOMC SEP to compute the unemployment gap. [Return](#)
14. We estimate Okun's coefficient for the gap version of Okun's law, $\text{unemployment gap}_t = b(\text{output gap}_t)$, using CBO estimates for the output gap and the unemployment gap and rolling 13-year windows, as in Knotek (2007). [Return](#)
15. The extent to which the forecasts would endogenously differ based on alternative policy rules depends on the particular modeling assumptions employed, as well as the differences between the funds rate path implied by the policy rule and the funds rate path in the original forecast. Given the typical lags from monetary policy to real economic activity and inflation, the assumption of exogeneity of the forecast is less problematic in the near term and more problematic as the forecast horizon lengthens. [Return](#)
16. The three forecasts put inflation on a rising trajectory toward the FOMC's 2 percent objective. The unemployment rate is near estimates of the natural rate, suggesting little slack in the unemployment gap, while the CBO's output gap is negative but forecasted to gradually close by mid-2018. [Return](#)

Credit risk management, asymmetric information, adverse selection, moral hazard in the market for credit (loanable funds market).

Overview



Under financial intermediation, the claim of savers is against the FI and not against the ultimate borrowers. **Primary claim:** claim against the ultimate borrower. **Secondary claim:** claim against FI and is not directly related to claim held by FI. Some advantages of FI: allows banks to pool the funds of many small savers and make relatively large loans. Also, allows the depositors to effectively get diversification (effectively have claims on proportions of many assets, not just one or a few assets), FI specialize in the production of information.

Asymmetric Information and Financial Markets

Definition of Asymmetric Information: buyers and sellers have different information on the characteristics of the good, service or asset being traded. Bonds (probability of default, how the funds will be used, etc.), health care etc. Information is costly to acquire.

Problems that arise from asymmetric information:

Adverse selection a problem created by asymmetric information *before* a transaction occurs: the people who are the most undesirable from the other party's point of view (say, high risk borrowers) are the ones most likely to want to engage in the financial transaction.

Adverse Selection Examples

Example #1: Auto insurance—sold in units of \$10; each person in each group knows its own probability. Suppose initially there is perfect information—that is, the insurance company also knows the probabilities of each person in each group. Assuming also perfect competition (the implication being in long-run equilibrium economic profit is driven to zero), then we have the following set up:

Group	Probability of filing Claim	Under perfect information, price charged	Revenue	Expected Cost
1 (100)	1/10	\$1 for \$10 of coverage	\$100	\$100
2 (100)	2/10	\$2 for \$10 of coverage	\$200	\$200
3 (100)	4/10	\$4 for \$10 of coverage	\$400	\$400
4 (100)	5/10	\$5 for \$10 of coverage	\$500	\$500

Expected profit = 0 in each category.

Now we change one of our assumptions. Suppose the insurance company does not know who is who. It reasons "I will charge \$3 since that would, on average, cover costs". This, however, creates an adverse selection problem. To see this, look at groups 1 & 2:

For group one, the expected benefit is $(1/10)(\$10) = \$1 < \$3$ They would drop out of market.

For group two, the expected benefit is $(2/10)(\$10) = \$2 < \$3$ They also drop out.

We market is now left with highest probability groups--- 3 & 4. An average premium of \$4.5 would apparently solve the problem, but in fact it would not → group 3 drops out ($4 < 4.5$). Left with only the bad risk group.

Example #2: Loan Example

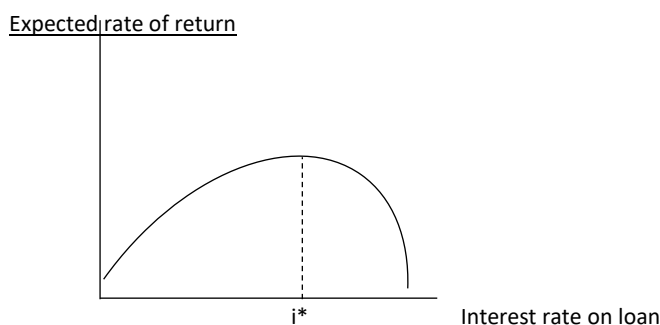
I. Assume a 94% chance borrower pays back a lender and 6% chance does not. Assume charge 15% to cover opportunity cost.

Expected payoff to lender = $(.94)(15\%) + (.06)(-100\%) = 8.1\%$ or (for example, if made 5000 one dollar loans, expected back \$5405 dollars at 15% (this is from $.94 * 5000 * 1.15$) → thus, $\$5405 / \$5000 = 1.081$) So charging 15% on each loan does NOT give the lender an expected return of 15%, but only an 8.1% expected return.

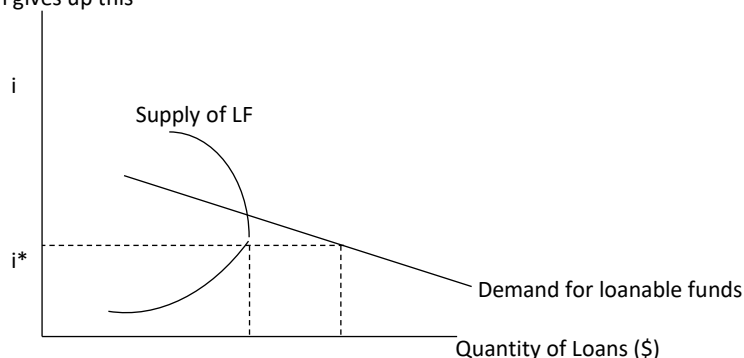
II. Suppose the lender reasons: “what rate x should I charge to get an expected return of 15%?”.

$15\% = (.94)(x) + (.06)(-100\%) \rightarrow x = 22.3\%$. This would seem to allow the lender enough to cover opportunity costs, but this would be incorrect. Why? It ignores the adverse selection problem. At the higher rate 22%, some of the low risk people would drop out, thus we can no longer assume a probability of .94. Suppose instead with the new pool of borrowers it is now 0.88. Then: Expected payoff = $(.88)(22.3\%) + (.12)(-100\%) = 19.6 - 12 = 7.6\%$.

You actually get a DROP in the expected return by raising the rate charged from 15% to 22.3%! This is the adverse selection problem at work. This leads us to conclude that we would at some i^* expect to see the following relationship:



Which in turn gives up this



There is no longer any reason for i to be bid up even though Q demanded exceeds Q supplied. Thus, the limited funds are rationed—that is, borrowers will not be able to borrow as much as they are able and willing to at i^* . This is called Credit Rationing. Lenders have no incentive to raise i above i^* since, as explained above, this would lower the **expected** return on the loan.

Some **solutions** to some of the asymmetric info problems (adverse selection).

1. Private production and sale of information—Standard and Poor's; Moody's; etc. This is not perfect solution since the information tends to be a "public good"—one it is out there anyone can use it without paying for it. This leads to free rider issues.
2. Government regulations to increase information. SEC, audits, standardized accounting principles. Not perfect solution because leaves open the question of government incentives to do this properly---who monitors the monitor?
3. **Financial intermediation** (banks for example)—experts in production of information. They specialize in information acquisition and can take advantage of economies of scale.
4. Collateral (borrower pledges an asset that is forfeited if default on loan)
5. **Reputation and repeated transactions.** A borrower is less likely to default if it engages in repeated loans with particular banks. Defaulting would make it harder to get future loans.

Moral hazard is said to exist in the context of a financial market if, after a purchase agreement has been concluded between a buyer and seller of a financial asset: the seller changes his or her behavior in such a way that the probabilities (risk calculations) used by the buyer to assess the quality of the financial asset are no longer accurate; the buyer of the financial asset is only imperfectly able to monitor (observe) this change in the seller's behavior. For example, a moral hazard problem arises if, after a lender has purchased a debt security from a borrower, the borrower increases the risks originally associated with the debt security by investing his borrowed funds in more risky projects than he originally reported to the lender. Here is an example.

Moral Hazard, an example.

Loan out \$1000 with expectation get back \$1100. Firm has two possible projects.

Project one has 2 possible outcomes, probab. of each outcome is ½.

- | | | | |
|----|----------------------------|------------------------------|---------------------------------|
| A. | <u>firm gets back 1200</u> | <u>bank gets back \$1100</u> | <u>firm: 1200- 1100 = \$100</u> |
| B. | <u>Firm gets back 900</u> | <u>bank gets \$900</u> | <u>firm: 0</u> |

Project two (has more risk):

- | | | | |
|----|------------------------------|-------------------------------|----------------------------------|
| A. | <u>Firm gets back \$2100</u> | <u>banks gets back \$1100</u> | <u>firm gets: 2100-1100=1000</u> |
| B. | <u>Firm gets back 0</u> | <u>bank gets zero</u> | <u>firm gets zero</u> |

The incentive of the borrower is to use funds for the higher risk project since outcomes are clearly better (using borrowed money, "things go well I get the big upside gain, things go poorly get zero either way---not my money, borrowed money" . The lender however prefers the low risk project as can be seen from the payoffs.

Partial solutions to moral hazard problem

- a. Monitor behavior of borrowers—can be very costly, possible free rider problem
- b. Bond covenants (restrictions on how can use funds; or cannot issue any more bonds..)
- c. FI --as explained above
- d. Net worth requirements (not allowed to use more than X amount of debt—make the borrower put up some of their own funds)
- e. Collateral requirements
- f. Reputation and repeated transactions. A borrower is less likely to default if it engages in repeated loans with particular banks. Defaulting would make it harder to get future loans.

Interest rate risk management: duration analysis, financial futures market

Interest rate risk and Duration Gap Analysis

Duration Analysis: measures sensitivity of the market value of the FI net worth to changes in interest rates.

Macaulay's Duration (DUR) measures the average lifetime of a security's stream of payments.

DUR for individual assets

How to calculate DUR. Consider the following stream of payments for an asset: \$1 per year for four years (first payment one year from today). Assume an interest rate of 10%. Find Macaulay's Duration.

Period	Dollar payment	PV at 10%	PV_i/PV (weight)	weight	yr to payment * weight
1	\$1	0.91	$PV_1/PV=0.91/3.17=.29$	0.29	$1*0.29=0.29$
2	\$1	0.83	$PV_2/PV=.83/3.17=.26$	0.26	$2*0.26=0.52$
3	\$1	0.75	$PV_3/PV=.75/3.17=.24$	0.24	$3*0.24=0.72$
4	\$1	0.68	$PV_4/PV=.68/3.17=.21$	0.21	$4*0.21=0.84$
PV=\$3.17				1.00	DUR= 2.37

The PV of asset is \$3.17.

First, find PV for each future dollar amount. Add up to get PV

$\$1/1.1 = \0.91 ; $\$1/(1.1)^2 = \0.83 ; $\$1/(1.1)^3 = \0.75 ; $\$1/(1.1)^4 = \0.68 ; PV is sum of these, 3.17

Second, find weights for each period, PV_i/PV

Third, multiply respective periods by weights, add up to get

Macaulay's Duration = $1(.29) + 2(.26) + 3(.24) + 4(.21) = 2.37$ years

What is the intuition of this formula? To see the intuition first notice that

Macaulay's $D = (1)(PV_1/PV) + (2)(PV_2/PV) + (3)(PV_3/PV) + \dots + (N)(PV_N/PV)$ with N being the last year in the series of payments, PV_1 is present value of payment received one year from today, PV_2 is the present value of payment received two years from today, etc. on to PV_N is the present value of payment received N years from today. The denominator is PV which is the present value of the entire series of payments (the price of the asset). The formula weights the length (in years) of each payment by (PV_i/PV) . This particular ratio is the proportion of the PV of the asset accounted for by present value of the payment made one year from now, and other ratios are interpreted accordingly. Thus, it places more weight on those years which account for a larger share of the overall PV.

It can be shown that sensitivity of an asset's market value (price) to a change in its interest rate can be approximated by: $\% \Delta P = - \text{DURATION} * (\Delta i / 1+i)$. To illustrate this, suppose the interest rate increases to 11%, using our "rule of thumb" equation determine the percentage change in the price (PV) of this asset. First, for purposes of comparison, let's do it the "long way":

$PV = 1/1.11 + 1/(1.11)^2 + 1/(1.11)^3 + 1/(1.11)^4 = \3.10

The absolute change in price is $\$3.17 - \$3.10 = \$0.07$; the % change is $0.07/3.17 =$ (negative) 2.2% decrease.

Now, compare with our “short cut”, we use $\% \Delta P = - \text{DURATION} * (\Delta i / 1+i) = -2.37 (0.01/1.1) = -2.2\%$ (rounding)

Example #2. Consider the following stream of payments for an asset: \$1 per year for three years and \$10 in year four. (first payment one year from today). Assume an interest rate of 10%. Find Macaulay's Duration.

Period	Dollar payment	PV at 10%	PV _i /PV (weight)	PV _i /PV (weight)	yr to payment * weight
1	\$1	0.91	PV ₁ /PV=0.91/9.34	0.10	0.10
2	\$1	0.83	PV ₂ /PV	0.09	0.18
3	\$1	0.75	PV ₃ /PV	0.08	0.24
4	\$10	6.85	PV ₄ /PV	0.73	2.92
\$9.34				1.00	3.44

PV of asset = \$9.34.

Macaulay's Duration = $1(.10) + 2(.09) + 3(.08) + 4(.73) = 3.44$

Relative to the first example the duration is greater. That is not surprising since the amount received 4 years ago is larger and accounts for a larger proportion of the overall PV, thus more weight is placed on 4. If interest rate rises to 11%:

$\% \Delta P = - \text{DURATION} * (\Delta i / 1+i) = -3.44 * (.11-.10)/1.10 = -3.40 * (0.01/1.1) = -0.031 (-3.1\%)$

[Long way, taking the PV at 11% is \$9.03 which is approximately a 3.1% drop (closer to 3.3% but I am rounding)].

Collection of Assets and Liabilities

Having calculated the duration of each asset and liability, how do we find the duration of all assets and all liabilities?

$\text{DUR}_{\text{assets}} = (A_1/A) * \text{DUR}_{A1} + (A_2/A) * \text{DUR}_{A2} + (A_3/A) * \text{DUR}_{A3} + \dots$

That is, the duration of all assets is the weighted average of the duration of each individual asset where the “weight” is the proportion of total asset value accounted for by that particular asset.

$\text{DUR}_{\text{liabilities}} = (L_1/L) * \text{DUR}_{L1} + (L_2/L) * \text{DUR}_{L2} + (L_3/L) * \text{DUR}_{L3} + \dots$

In other words, the duration of all liabilities is the weighted average of the duration of each individual liability where the “weight” is the proportion of total liability value accounted for by that particular liability

One last equation is $\text{DUR}_{\text{gap}} = [\text{DUR}_{\text{assets}} - (L/A) * \text{DUR}_{\text{liabilities}}]$

Example #1. Now suppose we assume that $\text{DUR}_L = 2$ and that the ratio of Liabilities to Assets is 0.85. The bank holds two assets with the following durations and values: $\text{DUR}_{A1} = 1$, $\text{DUR}_{A2} = 4$ and $A_1 = \$100$, $A_2 = \$100$.

We desire to know how a bank's net worth (NW = assets-liabilities) will be affected by a change in the interest rate. Suppose the interest rate changes from 10% to 11%. What is the impact on NW expressed as a percentage of Assets and as a percentage of NW?

To Solve, we can use the following equation to do this: $\Delta NW/A = - [(\Delta i / 1+i)] [DUR_{gap}]$

Step 1. Find DUR_{gap} and DUR_A . By definition, $DUR_{gap} = [DUR_{assets} - (L/A) * DUR_{liabilities}]$ and $DUR_{assets} = (A_1/A) * DUR_{A1} + (A_2/A) * DUR_{A2} \rightarrow DUR_{assets} = (100/200) * (1) + (100/200) * (4) = 2.5$
Plugging in, $DUR_{gap} = [DUR_{assets} - (L/A) * DUR_{liabilities}] = [2.5 - (0.85)(2)] = 2.5 - 1.7 = \mathbf{0.80}$.

Step 2. $\Delta NW/A = - [(\Delta i / 1+i)] [DUR_{gap}] = - [(0.11-0.10)/1.1][0.80] = - .0091 * .80 = -0.0073$ (or a drop of **0.7%** in NW expressed as a percentage of assets.)

Intuition: an increase in the interest rate will lower the PV of both assets and liabilities, but assets on average are of longer duration and are therefore are more interest rate sensitive.

Step 3. We know that $\Delta NW/A = -0.0073$ or rearranged, $\Delta NW = -0.0073 (A) = -0.0073 (\$200) = -\$1.46$ (drop in NW in absolute dollar terms). Since $NW = A - L$, then the initial value of NW is $NW = A - L = \$200 - (0.85)(200) = 200 - 170 = \30 . We can express the change in terms of NW change $\rightarrow \Delta NW/NW = -1.46/30 = \mathbf{-0.049}$ (4.9% drop in NW expressed as a percentage of NW).

Example #2. How can bank eliminate interest rate risk? Suppose the bank wanted to eliminate interest rate risk, but did not want to change the proportion of its asset holdings, it would have to change DUR_L to what value?

Step 1: find $DUR_A = (A_1/A) * DUR_{A1} + (A_2/A) * DUR_{A2} = (100/200) * (1) + (100/200) * (4) = 2.5$

Step 2: Set up $DUR_{gap} = [DUR_{assets} - (L/A) * DUR_{liabilities}]$ with DUR_{gap} set equal to zero (to eliminate interest rate risk). So $[DUR_{assets} - (L/A) * DUR_{liabilities}] = 0$.

Step 3: Plug in values and solve for the unknown $D_L \rightarrow [2.5 - (0.85) * DUR_{liabilities}] = 0 \rightarrow$

$2.5 = 0.85(DUR_L) \rightarrow (DUR_L) = 2.5/.85 = \mathbf{2.94}$. Thus, the bank would have to greatly increase the duration of its liabilities to eliminate interest rate risk. Notice this question could be asked in different ways, holding some variables constant and solving for the remaining unknown.

Footnote (not required to know derivation for exam)

$NW = A - L$ where A = assets (or the market value of assets), L = liabilities (market value of liabilities).

$\Delta NW = \Delta A - \Delta L$

Divide both sides by A : $\Delta NW/A = \Delta A/A - \Delta L/A$

$\Delta NW/A = \Delta A/A - (L/L)(\Delta L/A) = \Delta A/A - (L/A)(\Delta L/L)$

$\Delta A/A = DUR_{assets} * (\Delta i / 1+i)$

$\Delta L/L = DUR_{liabilities} * (\Delta i / 1+i)$

By substitution $\Delta NW/A = DUR_{assets} * (\Delta i / 1+i) - (L/A)[DUR_{liabilities} * (\Delta i / 1+i)]$

$\Delta NW/A = [(\Delta i / 1+i)] [DUR_{assets} - (L/A) * DUR_{liabilities}]$

By definition, $-DUR_{gap} = [DUR_{assets} - (L/A) * DUR_{liabilities}]$

So we are left with $\Delta NW/A = - [(\Delta i / 1+i)] [DUR_{gap}]$

NOTES on Financial Futures

A futures market is an auction market in which participants buy and sell commodity and futures contracts for delivery on a specified future date. Examples of futures markets are the New York Mercantile Exchange, the Kansas City Board of Trade, the Chicago Mercantile Exchange, the Chicago Board of Options Exchange and the Minneapolis Grain Exchange. More specifically, a *futures contract* is an agreement between buyer and seller to trade some well-defined item (wheat, corn, oil, Treasury bonds, etc.) at some specified future date, some specified price agreed upon today but paid/received on the future date of delivery. This specified price is called the futures price. A futures contract is similar but not the same as a forward contract in that futures markets have formal and standardized contract sizes (say, units of \$10,000 for a financial future), delivery dates. There are formal, centralized futures exchanges (for example, the Chicago Mercantile Exchange is the largest in the world).

Some terminology:

Going long in the futures market means you are *buying* a futures contract, that is, you are agreeing to buy the item on the established delivery date.

Going short means the opposite: *selling* a futures contract. Agreeing to deliver the specified item at the agreed upon future date.

FFM and Speculation

Assume you are speculating in the financial futures market (that is, you are hoping to gain by either the price of the good in question going up or down by the time of delivery date), when would you go long? ...same question for going short.

---a. You would profit by going long when you expect the market price of the good on the date of delivery (spot price) to be greater than the price you agreed to pay upon delivery on the delivery date. For a financial futures contract, that is the equivalent you betting that the interest rate will fall and thus the price of bond rises.

---b. You would profit by going short when you expect the market price of the good on the date of delivery to be less than the price you agreed to pay upon delivery on the delivery date. For a financial futures contract, that is the equivalent of betting that the interest rate will rise and thus the price of bonds falls. If that happens, you can buy the bond at the lower price (in the so-called spot market) and turn around to deliver the item you promised at the higher agreed price.

Example.

Assume a face value for a one year, zero coupon, T-bond is \$100. You can take delivery of such a bond on September 1st. The current listed futures price of such a delivery is, say, **\$83.33** (that is, a yield of 20%). As a speculator, if you thought the interest rate on September 1st was going to be 22% (implicitly you are assuming the market is wrong), would you go long or go short? Explain.

If you thought the interest would be higher than 20% on that date for a one-year bond, that is equivalent to assuming you expect one-year bond prices to be lower than \$83.33. Thus, you would go short.

Here is the story:

a. Sell a futures contract today at the current futures price of \$83.33 (that is, you agree to deliver a one-year bond on September 1st at the agreed upon price of \$83.33. Notice: you do not yet own such a bond! You are gambling that the spot price (price for immediate delivery) on September 1 will be lower than this, thus, you can buy it at the low price (if you guess correctly) and sell at the agreed price of \$83.33.

b. Suppose that on September 1st the interest rate is in fact 22% as you had guessed. Then the price of a one year bond on that date is $\$100/1.22 = \81.97 . You proceed as follows: buy a one-year bond on the spot market for \$81.97, then make good on your futures contract by delivering the one year bond at the agree upon price of \$83.33. You profit by the difference of 83.33 minus 81.97. Of course, if you are wrong and the interest rate fell below 20%, then the bond price would be greater than \$83.33 and you would suffer a loss. That is why it is called speculation.

Using the financial futures be used by banks to reduce interest rate risk (to “hedge”).

Case A. **Suppose the duration gap is positive**, that is, $DUR_{gap} = [DUR_{assets} - (L/A) * DUR_{liabilities}] > 0$ or $DUR_{assets} > (L/A) * DUR_{liabilities}$. In this case, an increase in interest rates would cause a drop in NW as discussed above in the questions on duration analysis. So, again, if $DUR_{gap} > 0$, and if interest rates rise, NW falls.

--So if you combine such a duration position with going short in the financial futures market this would mean the following:

- If interest rates do go up (price of security falls), the loss that results from a position of $DUR_A > (L/A) * DUR_L$ can be partially or potentially completely offset by the gains in the financial futures market.
- if interest rates fall (price of security rises), then you would suffer a loss in the financial futures market but this would be offset by the gain from having a position of $DUR_A > (L/A) * DUR_L$. Thus, no matter which direction interest rates change, a loss in one area is offset by a gain in another area—you thus reduce fluctuations in net worth and profit.

Case B. **Suppose the duration gap is negative**, that is, $DUR_{gap} = [DUR_{assets} - (L/A) * DUR_{liabilities}] < 0$ or $DUR_A < (L/A) * DUR_L$. In this case, an increase in interest rates would cause a rise in net worth.

--So if you combine such duration position with going long in the financial futures market this would mean: a. If interest rates do go up (bond prices down), the gain that results from $DUR_A < (L/A) * DUR_L$ will be partially or potentially completely offset by the loss in the financial futures market. b. if interest rates fall, then you would capture a gain in the financial futures market but this would be offset partially or completely by the loss from having a position of $DUR_A < (L/A) * DUR_L$. Thus, once again, no matter which direction interest rates change, a loss in one area is offset by a gain in another area.

Using the futures market to forecast the federal funds rate (optional reading).

Federal funds futures are contracts with payouts at maturity based on the average effective federal funds rate during the month of expiration. These securities have been trading on the Chicago Board of Trade (CBOT) since late 1988. The value of the contract at expiration is $100 - r$, where r is the average effective federal funds rate over the expiry month. Contracts with expiration maturities out to two years are offered, but most of the trading takes place in contracts with expiry dates within six months. Notice how investors expectations about the future federal funds rate translates into expectations about the future price of this contract. So looking at market prices of these contracts we can infer the average belief about the future federal funds rate.

Fed Watch Tool's Assumption and Interpretations:

- Probability of a rate hike is calculated by adding the probabilities of all target rate levels above the current target rate.
- Probabilities of possible Fed Funds target rates are based on Fed Fund futures contract prices assuming that the rate hike is 0.25% (25 basis points) and that the Fed Funds Effective Rate (FFER) will react by a like amount.
- FOMC meetings probabilities are determined from the corresponding CME Group Fed Fund futures contracts.

Methodology:

The FedWatch tool calculates unconditional probabilities of Federal Open Market Committee (FOMC) meeting outcomes to generate a binary probability tree. CME Group lists 30-Day Federal Funds Futures (FF) futures, prices of which incorporate market expectations of average daily Federal Funds Effective Rate (FFER) levels during futures contract months. (E.g., the market price of FFU5 reflects the market consensus expectation of the average FFER level during the month of September 2015.) The FFER is published by the Federal Reserve Bank of New York each day, and is calculated as a transaction-volume weighted average of the previous day's rates on trades arranged by major brokers in the market for overnight unsecured loans between depository institutions.

In the FedWatch tool's probability analysis, the implementation assumes that the size of a rate change is always 25 basis points and that for a given FOMC meeting month, prior or post FF futures contract prices contain information that either is independent of the outcome of that meeting or is solely dependent on that meeting's outcome. Additionally, the FedWatch tool incorporates the assumption that FFER is bounded below by zero. Because the price of each FF futures contract represents the expected average daily FFER for the contract month, if one were in a FOMC meeting month where there was no FOMC meeting in the prior month, then the FF futures price of the previous month contains information independent of the current month's meeting. Likewise, if one were in a FOMC meeting month such that there was no FOMC meeting scheduled for the next following month, then the FF futures price of the following contains only information about the outcome of the current month's meeting. If one assumes that in its current month meeting the FOMC will decide either to raise its daily FFER target or to maintain the status quo, then the probabilities of a rate hike versus no rate hike would be calculated as:

$$P(\text{Hike}) = [\text{FFER}(\text{end of month}) - \text{FFER}(\text{start of month})] / 25 \text{ basis points}$$

$$P(\text{NoHike}) = 1 - P(\text{Hike})$$

Whether the FOMC sets its target for daily FFER as a level or as a range should not affect either the pricing of FF futures or the calculation of implied probabilities of FOMC meeting outcomes, because calculation is based on a comparison of FFER (end of month) versus FFER (start of month). Provided that changes in the FOMC target levels are of the magnitude of 25 basis points (whether as the change in a given target level or in the location of a target range), the probability of a rate change is relative to the expected End-of-month target versus the expected Start-of-month target.

To calculate unconditional probability of a change in the target at the current month FOMC meeting, the primary consideration is whether there is an FOMC meeting in the month immediately before "meeting" month. If there was no meeting in the month prior, it is categorized as a "Type 2 meeting." Otherwise, it is categorized as a "Type 1 meeting." To see this, consider the following table:

	Type 1	Type 2
N	Days in Meeting Month	Days in Meeting Month
M	Day(MeetingDate) – 1	Day(MeetingDate) – 1
FFER.Start	(N/M) * [Implied Rate – FFER(end)*((N-M)/N)]	(1 – FF.MonthBefore)
Implied Rate	100 – FF.MeetingMonth	
FFER.End	(1 – FF.MonthAfter)	(N/(N-M)) * [Implied Rate – (M/N)*FFER(start)]

Example, September 17, 2015 FOMC – Type #2

FFQ5 = 99.8675
 FFU5 = 99.805
 N = 30
 M = 16
 FFER(start) = 0.1325 (100-99.8675)
 ImpliedRate = 0.195 (100-99.805)
 FFER(end) = 30/14 * [0.195 – (16/30)*0.1325]
 = 0.26643
 P(Hike) = (0.26643 – 0.1325) / 0.25 = 53.6%
 P(NoHike) = 46.4%

After the FedWatch tool computes the unconditional probability for each known meeting date (as published by the Federal Reserve Board of Governors website), it calculates a binary policy decision tree.

For the first node of the tree, there are probabilities for two outcomes: (1) Maintenance of current target or (2) a change to a different target (25 bps higher or 25 bps lower). In the current example and in subsequent examples, there will only be two outcomes, i.e. hike or no hike, cut or no cut.

For the second node, assuming that the expectation is for the target rate to be raised or not, then at the second meeting we have the following probabilities: probability of a decreased target rate at the second meeting, probability of an unchanged target rate FFTR at the second meeting, probability of an increased target rate at the second meeting.

The equations are as follows:

$$P(\text{FFER decreased}) = \text{Probability}(\text{FFER Decrease previous}) * (1 - \text{Probability of a rate change})$$

$$P(\text{FFER unchanged}) = \text{Probability}(\text{FFTR increase previous}) * (1 - \text{Probability of a rate change}) + (\text{Probability of a FFTR decrease previous}) * (\text{Probability of a rate change})$$

$$P(\text{FFER increased}) = (\text{Probability FFTR Increase previous}) * (\text{Probability of a rate change})$$

Note 1: Recall that one of the assumptions of this methodology is that the FOMC will always move rates in 25 basis point increments. It is possible for the expected Federal Funds Rate (as implied from futures prices) to be more than 25 basis points above the current effective rate – in this case the market is implying some chance of a rate hike of rate hike greater than 25 basis points.

In such cases, the calculated probability will exceed 100%. To simplify the interpretation, the calculated probability is re-distributed between the probability of a single hike and the probability of two hikes

The amount that is in excess of 100% will be defined as the probability of a 50 basis point rate hike, to the next target increment, defined as P50bp hike

The probability of a single hike is now defined as (1- P50bp hike)

Example: Due to high implied rate in futures contract, calculated probability is 104% probability of a 25 bps rate hike, with a -4% probability of no change. Using a formula, this will be redistributed and shown as P(NoHike)=0%, P(25bpHike) = 96%, and P(50bpHike) = 4%

Note 2: Based on market commentary and market assumptions the FFER is bounded by zero. As such the scenarios for the second node are as follows: Probability of unchanged FFER at the second meeting, Probability of an increased FFER in the second meeting (or probability of a decrease in the second meeting after an increase in the first), Probability of an increased FFER in the first and second meeting.

The formulas for these probabilities are as follows:

$$P(\text{FFER unchanged}) = \text{Probability}(\text{FFER NoHike previous meeting}) * (1 - \text{Probability of a rate change})$$

$$P(\text{FFER first increased on this meeting date, or decreased at second meeting if hike in the first meeting}) = \text{Probability}(\text{FFTR hike previous meeting}) * (1 - \text{Probability of a rate change}) + (\text{Probability of a FFTR NoHike previous meeting}) * (\text{Probability of a rate change})$$

$$P(\text{FFER increased this meeting date as well as previous meeting date}) = (\text{Probability FFTR Increase previous meeting}) * (\text{Probability of a rate change})$$

Our Fed rate monitor calculator is based on CME Group 30-Day Fed Fund futures prices, which tend to signal the markets' expectations regarding the possibility of changes to US interest rates based on Fed monetary policy. The tool allows users to calculate the likelihood of an upcoming Fed rate hike or cut. The prices of Option contracts on fed funds futures (traded on the Chicago Board of Trade) can be used to infer the market's expectations of future Fed policy changes. Based on CME Group 30-Day Fed Fund futures prices, which have long been used to express the market's views on the likelihood of changes in U.S. monetary policy, the CME Group FedWatch tool allows market participants to view the probability of an upcoming Fed Rate hike. One set of such implied probabilities is published by the Cleveland Fed.

Fed Fund Futures and Options

30-Day Fed Fund futures and options are one of the most widely used tools for hedging short-term interest rate risk. Fed Fund futures are a direct reflection of collective marketplace insight regarding the future course of the Federal Reserve's monetary policy.

Sep 20, 2017



Meeting Time: **Sep 20, 2017 02:00PM ET**

Future Price: **98.840**

1.00 - 1.25

94.5%

1.25 - 1.50

5.5%

Target Rate	Current Probability%	Previous Day Probability%	Previous Week Probability%
1.00 - 1.25	94.5%	100.0%	92.2%
1.25 - 1.50	5.5%	—	7.7%
1.50 - 1.75	—	—	0.1%

Updated: Jul 28, 2017 05:19PM ED

Interest rate Swaps

Interest rate swap: A financial contract that allows one party to exchange (swap) a set of interest payments for another set of interest payments owned by another party. Allows for efficient allocation of risk.

Can be used by banks to reduce interest rate risk. By increasing or decreasing interest rate exposure in various parts of the yield curve using swaps, managers can either ramp-up or neutralize their exposure to changes in the shape of the curve, and can also express views on credit spreads. Swaps can also act as substitutes for other, less liquid fixed income instruments. Moreover, long-dated interest rate swaps can increase the duration of a portfolio, making them an effective tool in Liability Driven Investing, where managers aim to match the duration of assets with that of long-term liabilities.

LIBOR = London Interbank Offer Rate; an interest rate offered by London banks on loans made by other well established banks. This rate could fluctuate over the time period of the loan and is famous for being used as a base for setting the rate on other borrowers on so-called floating loans and is the most often the used rate as a base for interest rate swaps.

[Though the London Interbank Offered Rate (LIBOR) and the federal funds rate are concerned with the same action, i.e. interbank loans, they are distinct from one another, as follows: The target federal funds rate is a target interest rate that is set by the FOMC for implementing U.S. monetary policies. The (effective) federal funds rate is achieved through open market operations at the Domestic Trading Desk at the Federal Reserve Bank of New York which deals primarily in domestic securities (U.S. Treasury and federal agencies' securities).

LIBOR is based on a questionnaire where a selection of banks guess the rates at which they could borrow money from other banks. LIBOR may or may not be used to derive business terms. It is not fixed beforehand and is not meant to have macroeconomic ramifications.]

--Assume **Bank A** has a \$100 asset that pays LIBOR + 1.5%. As LIBOR rises, and so LIBOR +1.5 rises, then the amount Bank A gets on this asset goes up and if LIBOR falls, and so LIBOR +1.5 falls, then Bank A receives less on this asset.

--Assume that **Bank B** holds a \$100 asset that pays a fixed interest rate of 7%.

--Bank A's risk preferences are such that it would like to trade (swap) its floating stream of payments for a fixed stream of payments. Bank B's risk preferences are such that it would like the opportunity for a large payoff and desires to trade its fixed payment stream for a floating payment stream.

a. How can A and B engage in mutually beneficial interest rate swap?

b. What happens between A and B when LIBOR = 4.5%? When LIBOR is 6.5%?

a. Bank A agrees to pay Bank B LIBOR + 1.5% on the \$100 principal and Bank B in return agrees to pay Bank A a fixed 7% on a principal of \$100. They are not formally trading assets (they are still each on the bank's books) but of course implicitly that is what they are doing.

b. If LIBOR is 4.5%.

Bank A: receives 4.5% + 1.5% on its asset of \$100. So:

Revenue = $(0.06)(100)$ + payments from Bank B under swap agreement of $(0.07)(100)$ = \$6 + \$7 = \$13

Expenses = payments made to B under swap agreement = $(0.06)(100)$ = \$6

Net = \$7 (7% return on \$100)

Bank B: receives 7% on its asset of \$100. So:

Revenue = $(0.07)(100)$ + payments from Bank A under swap agreement of $(0.06)(100)$ = \$7 + \$6 = \$13.

Expenses = payments made to A under swap agreement = $(0.07)(100)$ = \$7

Net = \$6 (or 6% on \$100).

b. If LIBOR is 6.5%.

Bank A: receives 6.5% + 1.5% on its asset of \$100. So:

Revenue = $(0.08)(100)$ + payments from Bank B under swap agreement of $(0.07)(100)$ = \$8 + \$7 = \$15.

Expenses = payments made to B under swap agreement = $(0.08)(100)$ = \$8

Net = \$7 (7% return on \$100)

Bank B: receives 7% on its asset of \$100. So:

Revenue = $(0.07)(100)$ + payments from Bank A under swap agreement of $(0.08)(100)$ = \$7 + \$8 = \$15.

Expenses = payments made to A under swap agreement = $(0.07)(100)$ = \$7

Net = \$8 (or 8% on \$100).

Thus, it can be seen that Bank A has converted an uncertain situation (payments depended on which way LIBOR went) into a fixed payment stream—Bank A gets 7% no matter which way LIBOR changes.

On the other hand, Bank B now has an opportunity to get more than 7% (but of course a chance of getting less than this!). So Bank B's fixed payment stream has been converted to a floating payment stream—6% in one case but 8% in the other case. The numerical example is a bit of overkill since all the banks are really doing is trading interest rate payment streams. However, the assets are still listed on their books.

General Description

An interest rate swap's (IRS's) effective description is a derivative contract, agreed between two counterparties, which specifies the nature of an exchange of payments benchmarked against an interest rate index. The most common IRS is a fixed for floating swap, whereby one party will make payments to the other based on an initially agreed fixed rate of interest, to receive back payments based on a floating interest rate index. Each of these series of payments is termed a 'leg', so a typical IRS has both a fixed and a floating leg. The floating index is commonly an interbank offered rate (IBOR) of specific tenor in

the appropriate currency of the IRS, for example LIBOR in USD, GBP, EURIBOR in EUR or STIBOR in SEK. To completely determine any IRS a number of parameters must be specified for each leg; the notional principal amount (or varying notional schedule), the start and end dates and date scheduling, the fixed rate, the chosen floating interest rate index tenor,

Discussion on options to be added.